Quantum Subspace Expansion: Revolutionizing Molecular Energy Calculations for Drug Discovery

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Abstract

This article explores Quantum Subspace Expansion (QSE) as a transformative computational technique for calculating molecular energies, a critical task in drug discovery and development. We provide a comprehensive analysis for researchers and pharmaceutical professionals, covering the foundational principles of QSE, its methodological implementation in simulating molecular systems, strategies for troubleshooting and optimizing performance on noisy hardware, and a comparative validation against other quantum algorithms. The discussion synthesizes recent breakthroughs and practical insights, highlighting QSE's potential to drastically reduce the time and cost associated with bringing new therapeutics to market.

What is Quantum Subspace Expansion? The Foundation for Advanced Molecular Simulation

Defining Quantum Subspace Expansion (QSE) and its Core Principles

Quantum Subspace Expansion (QSE) is a **post-processing algorithm** for quantum computers that enhances the accuracy of ground and excited state energy calculations. It operates by constructing and diagonalizing an effective Hamiltonian within a small, classically-tractable subspace built upon a initial quantum state, known as a "root state" [1] [2]. This method has emerged as a powerful alternative to purely variational approaches, such as the Variational Quantum Eigensolver (VQE), offering a pathway to mitigate errors and perform spectral calculations on noisy intermediate-scale quantum (NISQ) devices without significantly increasing quantum circuit depth [1] [3] [2].

The core value of QSE lies in its hybrid quantum-classical nature. A quantum processor is used to prepare a starting state and measure the matrix elements needed to define the subspace, while a classical computer solves a generalized eigenvalue problem to find the best energy eigenvalues and eigenvectors within that subspace [4]. This makes QSE particularly promising for near-term quantum hardware, as it exchanges increased circuit depth for additional measurements, a resource that is often more readily available than perfect, long-coherence quantum gates [3].

Theoretical Foundation and Mathematical Framework

Core Mathematical Principle

The foundational principle of QSE is to expand a prepared quantum root state, (\rho0), *into a subspace spanned by applying a set of (L) Hermitian expansion operators ({\sigma*i}). The states within this subspace are expressed as: [\rho*{\text{SE}}(\vec{c}) = \frac{W^\dagger \rho*0 W}{\Tr[W^\dagger \rho*0 W]}, \quad \text{with} \quad W = \sum*{i=1}^{L} ci \sigma*i] where (\vec{c} = (c1, \dots, cL)) is a vector of complex coefficients that parametrize the subspace [1]. The expectation value of an observable (O) within this subspace is given by the Rayleigh-Ritz quotient: [\Tr[O\rho*{\text{SE}}(\vec{c})] = \frac{\sum*{i,j=1}^{L} ci^* cj \mathcal{O}*{ij}}{\sum*{i,j=1}^{L} ci^* cj \mathcal{S}*{ij}}] where (\mathcal{O}*{ij} = \Tr[\sigma*i^\dagger \rho*0 \sigma*j O]) and (\mathcal{S}*{ij} = \Tr[\sigma*i^\dagger \rho*0 \sigma*j]) [1]. The matrices (\mathcal{H}*{ij} = \Tr[\sigma*i^\dagger \rho*0 \sigma*j H]) (subspace-projected Hamiltonian) and (\mathcal{S}*{ij}) (overlap matrix) are measured on the quantum computer.

The Generalized Eigenvalue Problem

To find the ground or excited states, QSE solves the **generalized eigenvalue problem**: [\mathcal{H} \vec{c} = \lambda \mathcal{S} \vec{c}] Here, (\mathcal{H}) is the matrix representation of the Hamiltonian in the subspace, (\mathcal{S}) is the overlap matrix, and the eigenvalues (\lambda) correspond to the energy estimates for the states within the subspace [4]. The lowest eigenvalue (\lambda_0) provides an improved estimate of the ground state energy. Solving this equation is a classical computation step, making it tractable for reasonable subspace dimensions (L).

The Krylov Subspace

A particularly effective choice for the expansion operators is using powers of the Hamiltonian, (H^p). This generates a **Krylov subspace** [1]. The overlap of the states (H^p \rho*0 H^{p\dagger}) *with the true ground state increases exponentially with (p), provided the root state (\rho*0) has

non-zero overlap with the true ground state [1]. This connects QSE to well-established classical Krylov subspace diagonalization methods, known for their rapid convergence for ground state problems [1] [3].

Key Protocols and Experimental Methodologies

Protocol 1: Standard QSE for Molecular Excited States

This protocol outlines the standard procedure for performing a QSE calculation to compute ground and excited states of a molecular system on a quantum computer, following the example of a minimal basis hydrogen molecule [4].

- **Step 1: Define the Problem and Generate the Initial State**
  - Construct the molecular Hamiltonian (H) in the fermionic space and map it to a qubit operator using a mapping such as Jordan-Wigner or Bravyi-Kitaev [4].
  - Prepare a root state (\rho_0) close to the ground state. This is often the optimized state from a VQE calculation using an ansatz like UCCSD [4].
- **Step 2: Define the Expansion Subspace**
  - Select a set of expansion operators ({\sigma*i}). For molecular excited states, these are often chosen as fermionic excitation operators. For example, to generate spin-adapted states, one can use singlet single excitation operators (E*{ij}) [4].
  - Map these operators to the qubit space.
- **Step 3: Quantum Measurement of Matrix Elements**
  - On the quantum computer, measure all matrix elements of the projected Hamiltonian (\mathcal{H}*{ij} = \langle \psi*{ij} | H | \psi*{kl} \rangle) and the overlap matrix (\mathcal{S}*{ij} = \langle \psi*{ij} | \psi*{kl} \rangle), where (|\psi_{ij}\rangle) are the states generated by applying the expansion operators to the root state [4].
  - This can be done using shot-based protocols like Pauli Averaging [4].
- **Step 4: Classical Post-Processing**
  - On a classical computer, solve the generalized eigenvalue problem (\mathcal{H}\vec{c} = \lambda \mathcal{S}\vec{c}) [4].
  - Handle numerical instabilities, often caused by near-linear dependencies in the basis (ill-conditioning of (\mathcal{S})), by removing dimensions corresponding to small singular values of (\mathcal{S}) via truncation [4].
- **Step 5: Analysis**
  - The eigenvalues (\lambda) correspond to the energies of the ground and excited states within the subspace.
  - The eigenvectors (\vec{c}) define the expanded state within the subspace.

Protocol 2: Large-Scale QSE with Classical Shadows

A major bottleneck in QSE is the measurement overhead required to estimate all matrix elements (\mathcal{H}*{ij}) and (\mathcal{S}*{ij}). This protocol leverages **informationally complete (IC) measurements**, specifically **classical shadows**, to overcome this bottleneck and enable large-scale implementations [1].

- **Step 1: Prepare the Root State**
  - Prepare the root state (\rho_0) on the quantum processor.
- **Step 2: Perform Informationally Complete Measurements**
  - Instead of measuring each observable individually, perform randomized measurements on the state (\rho_0). A classical shadow is a classical snapshot of the quantum state, built from these randomized measurements [1].
  - For each of many measurement basis randomizations (e.g., over 32,000 per circuit), collect measurement samples [1] [5].
- **Step 3: Construct Classical Shadows and Estimate Matrices**
  - From the collected samples, reconstruct unbiased estimators for the matrix entries of (\mathcal{H}) and (\mathcal{S}) by processing the classical shadows [1].
  - This allows for the simultaneous estimation of a vast number of Pauli observables from the same set of measurements. One implementation evaluated over (10^{14}) Pauli traces for an 80-qubit system [1] [5].
- **Step 4: Reformulate as a Constrained Optimization**

- To avoid numerical ill-conditioning from direct matrix inversion and to obtain rigorous statistical error bars, reformulate the QSE problem as a constrained optimization problem [1].
- This involves finding the minimal expectation value (\min{\vec{c}} \Tr[H\rho{\text{GSE}}(\vec{c})]) given a maximally tolerated statistical error, which provides an external tuning knob to control the trade-off between potential bias and variance [1].

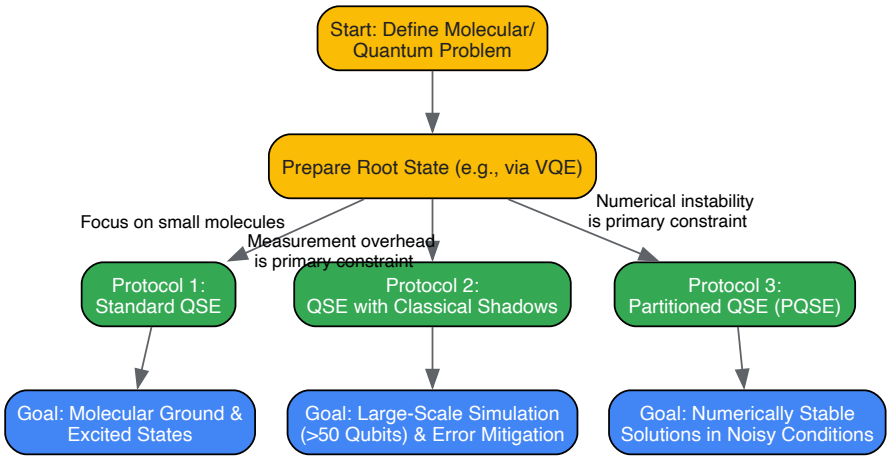
- **Key Application:** This protocol has been successfully demonstrated for probing quantum phase transitions in spin models with three-body interactions, achieving accurate ground state energy recovery for systems of up to **80 qubits** [1] [5].

Protocol 3: Partitioned QSE (PQSE) for Numerical Stability

The Partitioned Quantum Subspace Expansion (PQSE) is an iterative generalization of QSE designed to improve numerical stability in the presence of finite sampling noise [3].

- **Step 1: Build a Krylov Subspace**
  - Follow the standard Krylov-based QSE approach to build an initial subspace using powers of the Hamiltonian.
- **Step 2: Iterative Partitioning and Recombination**
  - Instead of diagonalizing the entire Krylov subspace at once, the algorithm breaks it down into a sequence of smaller subspaces [3].
  - These subspaces are connected sequentially via their lowest-energy states. The solution from one subspace informs the construction of the next.
- **Step 3: Variance-Based Sequencing**
  - A variance-based criterion is used to determine a "good" iterative sequence. This criterion helps ensure stability against statistical noise [3].
- **Step 4: Classical Diagonalization**
  - Diagonalize the Hamiltonian in each of the smaller, connected subspaces. This requires additional classical processing with polynomial overhead but uses the same quantum resources as the single-step approach [3].
- **Key Advantage:** PQSE substantially alleviates the numerical instability that limits the accuracy of standard QSE in a parameter-free way, without requiring additional quantum measurements [3].

The workflow below illustrates the logical relationship and decision points between these three primary QSE protocols.



Critical Reagents and Computational Tools

Successful implementation of QSE relies on a suite of theoretical constructs and software tools. The table below details the key "research reagents" essential for experiments in this field.

Table 1: Essential Research Reagents and Tools for Quantum Subspace Expansion

Category	Reagent / Tool	Function in QSE Protocol
Theoretical Constructs	Krylov Subspace [1] [3]	A subspace spanned by powers of the Hamiltonian ((H^p)) applied to a root state. Provides rapid convergence for ground state calculations.

Category	Reagent / Tool	Function in QSE Protocol
	Expansion Operators [4]	A set of operators ({\sigma_i}) (e.g., fermionic excitations, Pauli strings) used to define the search subspace beyond the root state.
	Classical Shadows [1] [5]	A framework for informationally complete (IC) measurements that dramatically reduces the measurement overhead for estimating many observables.
Software & Libraries	InQuanto [4]	A quantum computational chemistry toolkit that provides high-level interfaces (e.g., `AlgorithmQSE`) to run QSE calculations.
	Qiskit / SQD Addon [6]	An open-source SDK for quantum computing. The SQD (Sample-Based Quantum Diagonalization) addon implements related subspace methods.
	Tangelo [7]	An open-source quantum chemistry package, used in hybrid workflows like DMET-SQD for fragment-based simulations.
Hardware & Backends	Pauli Averaging [4]	A shot-based protocol for estimating expectation values of Pauli operators on quantum hardware or simulators.
	Quantum Processing Units (QPUs)	Physical quantum devices (e.g., IBM's Eagle processor [7] [6]) used to execute quantum circuits and sample measurement outcomes.

Quantitative Performance and Applications

QSE and related subspace methods have been demonstrated on a variety of problems, from small molecules to large spin models. The following table summarizes key quantitative results from recent experimental implementations.

Table 2: Performance Metrics of QSE and Related Subspace Methods in Recent Implementations

System / Application	Method	System Size (Qubits)	Key Performance Metric	Reference
Hydrogen Molecule (H_2)	Standard QSE	N/A	Successfully obtained ground and spin-adapted excited states beyond VQE solution.	[4]
Spin Model with Three-Body Interactions	QSE + Classical Shadows	16 to 80	Accurate ground state recovery and mitigation of local order parameters; over 32,768 measurement basis randomizations per circuit.	[1] [5]
Cyclohexane Conformers	DMET-SQD (Related method)	27 to 32	Energy differences between conformers within 1 kcal/mol of classical benchmarks (chemical accuracy).	[7]
Water & Methane Dimers (Non-covalent interactions)	SQD (Related method)	27 to 54	Binding energy deviations within 1.000 kcal/mol from leading classical methods (CCSD(T)).	[6]
General Quantum Subspace Methods	PQSE	N/A	Displays improved numerical stability over single-step QSE in the presence of finite sampling noise.	[3]

Quantum Subspace Expansion has established itself as a versatile and powerful algorithm within the quantum computing toolkit for molecular energy calculations. Its core principle—using a quantum computer to inform a small, classically-solvable subspace model—enables a flexible trade-off between quantum and classical resources. The development of advanced protocols incorporating classical shadows for scalability and partitioned approaches for numerical stability is pushing the boundaries of what is possible on today's noisy quantum devices.

As quantum hardware continues to improve, QSE and its generalizations are poised to play a critical role in achieving practical quantum advantage in computational chemistry and materials science, with significant potential implications for drug discovery and the design of novel materials.

The Critical Challenge of Molecular Energy Calculations in Drug Discovery

The process of drug discovery is characterized by significant financial investment, often ranging between $1-3 billion, with a typical timeline of 10 years and a success rate of only 10% [8]. This inefficiency highlights a critical need for innovative approaches to enhance the drug development pipeline. At the heart of this challenge lies the accurate calculation of molecular energies—the precise computational prediction of how potential drug molecules interact with biological targets, their stability, and their reactivity under physiological conditions.

Traditional computing methods struggle to accurately simulate quantum effects in complex molecular systems, particularly for large molecules and complex reactions like covalent bond formation [8]. Quantum subspace expansion (QSE) represents a transformative approach to this problem,

offering a framework for performing electronic structure calculations on quantum computers with theoretical guarantees absent in parameter-optimization-based algorithms [9]. These methods are specially designed to be compatible with near-term quantum hardware constraints while providing pathways to exponential improvements in computational efficiency for specific chemical simulations [9].

This application note details practical protocols for implementing quantum subspace expansion methods to address critical challenges in drug discovery, focusing on real-world applications including prodrug activation and covalent inhibitor design.

Theoretical Framework: Quantum Subspace Expansion Fundamentals

Quantum subspace methods comprise a class of algorithms for learning properties of low-energy quantum states using a quantum computer, with particular relevance for molecular electronic structure calculations [3] [9]. The core principle involves constructing and diagonalizing an effective Hamiltonian within a carefully chosen subspace of the full quantum state space.

The Partitioned Quantum Subspace Expansion (PQSE) algorithm represents an iterative generalization of QSE that uses a Krylov basis [3]. This approach connects a sequence of subspaces via their lowest energy states, offering improved numerical stability over single-step approaches in the presence of finite sampling noise—a critical consideration for implementation on real quantum hardware [3]. The method employs a variance-based criterion for determining optimal iterative sequences and requires only polynomial classical overhead in the subspace dimension.

For molecular systems, quantum subspace methods establish rigorous complexity bounds and convergence guarantees, characterizing the relationship between subspace dimension, basis set selection, and solution accuracy for ground and excited state calculations [9]. The framework incorporates realistic noise models and provides performance predictions for near-term quantum devices, analyzing trade-offs between circuit depth, measurement overhead, and solution quality while identifying optimal operating regimes for different molecular systems.

Application Protocols

Protocol 1: Gibbs Free Energy Calculation for Prodrug Activation

Background: Prodrug activation strategies, particularly those based on carbon-carbon (C-C) bond cleavage, represent innovative approaches to targeted drug delivery [10]. Accurate calculation of the Gibbs free energy profile for these reactions is essential for predicting activation kinetics under physiological conditions.

Experimental Workflow:

- **System Preparation:** Select key molecular structures along the reaction coordinate for C-C bond cleavage. For β-lapachone prodrug analysis, this involves five critical molecules during the cleavage process [10].
- **Active Space Selection:** Employ active space approximation to simplify the quantum chemical region into a manageable two-electron/two-orbital system, reducing the problem to a 2-qubit implementation on superconducting quantum devices [10].
- **Wavefunction Preparation:** Utilize the Variational Quantum Eigensolver (VQE) framework with a hardware-efficient (R_y) ansatz with a single layer as the parameterized quantum circuit [10].
- **Energy Calculation:** Perform single-point energy calculations incorporating solvation effects using the polarizable continuum model (PCM) to simulate physiological conditions [10].
- **Free Energy Profiling:** Compute Gibbs free energy differences between molecular states along the reaction pathway to determine activation energy barriers.

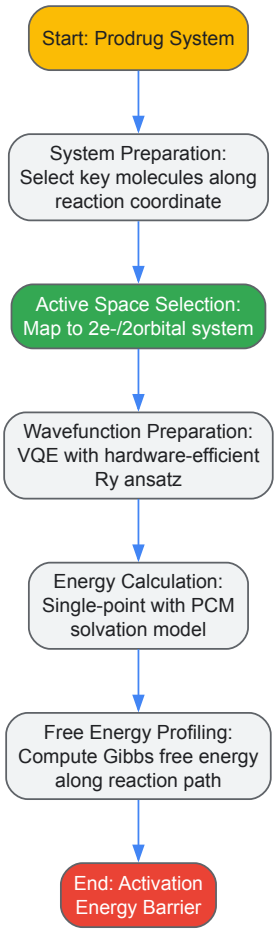


Figure 1: Workflow for prodrug activation energy calculation using a hybrid quantum-classical approach.

Key Considerations: The 6-311G(d,p) basis set is recommended for balanced accuracy and computational efficiency. Classical reference calculations using Hartree-Fock (HF) and Complete Active Space Configuration Interaction (CASCI) methods provide benchmarks for quantum computation results [10].

Protocol 2: Covalent Inhibitor Binding Analysis

Background: Covalent inhibitors, such as Sotorasib targeting the KRAS G12C mutation in cancers, represent a growing class of therapeutics that form covalent bonds with their protein targets [10]. Simulating these interactions requires precise modeling of bond formation energetics.

Experimental Workflow:

- **System Setup:** Prepare the protein-ligand complex structure, focusing on the covalent binding site and surrounding residues.
- **QM/MM Partitioning:** Implement a hybrid quantum mechanics/molecular mechanics (QM/MM) scheme where the covalent bond formation region is treated quantum mechanically while the remainder of the system is handled classically.
- **Force Calculation:** Implement a hybrid quantum computing workflow for molecular forces during QM/MM simulation [10].
- **Binding Energy Calculation:** Compute the interaction energies between the inhibitor and target protein, focusing on the covalent bond formation process.
- **Validation:** Compare results with experimental binding affinity data and classical simulation methods where available.

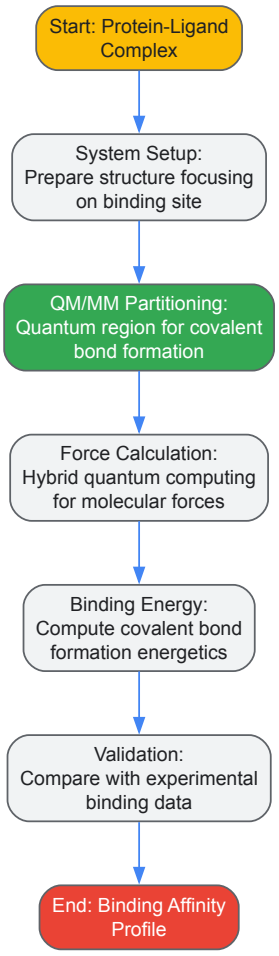


Figure 2: Workflow for covalent inhibitor binding analysis using QM/MM and quantum computing.

Key Considerations: The QM region should include all atoms directly involved in the covalent bond formation and adjacent residues critical to the binding interaction. For KRAS G12C inhibitors, this includes the cysteine residue and surrounding pocket [10].

Research Reagents and Computational Tools

Table 1: Essential Research Reagents and Computational Tools for Quantum Subspace Expansion in Drug Discovery

Category	Item	Function/Application	Implementation Notes
Quantum Algorithms	Variational Quantum Eigensolver (VQE) [10]	Ground state energy calculation	Uses parameterized quantum circuits with classical optimization
	Quantum Subspace Expansion (QSE) [3] [5]	Excited state calculations and error mitigation	Builds Krylov subspace for diagonalization
	Partitioned QSE (PQSE) [3]	Iterative subspace expansion	Improved stability against sampling noise
Classical Computational Methods	Density Functional Theory (DFT) [11] [10]	Reference molecular energy calculations	Multiple functionals (76+) and basis sets available
	Molecular Dynamics (MD) [12]	Conformational sampling and binding free energy	MM-PBSA, LIE, and alchemical methods
	Polarizable Continuum Model (PCM) [10]	Solvation energy calculation	Critical for physiological condition simulation
Datasets & Libraries	MultiXC-QM9 Dataset [11]	Benchmarking and training	Molecular energies with 76 DFT functionals and 3 basis sets
	Quantum Chemistry Packages	Hamiltonian generation and basis set management	TenCirChem [10] and other specialized software

Category	Item	Function/Application	Implementation Notes
Error Mitigation	Readout Error Mitigation [10]	Measurement error correction	Standard technique for noisy quantum devices
	Classical Shadows [5]	Efficient measurement protocol	Reduces required measurement resources

Data Presentation and Analysis

Performance Benchmarks

Table 2: Comparative Analysis of Computational Methods for Molecular Energy Calculations

Method	Computational Scaling	Typical Accuracy	Key Advantages	Implementation Challenges
Classical DFT	$O(N^3)$ to $O(N^4)$ [8]	Chemical accuracy for many systems [10]	Well-established, reliable for medium systems	Struggles with strong correlation, large systems
Quantum Subspace Expansion	Polynomial in subspace dimension [9]	Approaching chemical accuracy [5]	Theoretical convergence guarantees, noise resilience	Measurement overhead, basis selection critical
Partitioned QSE	Polynomial overhead [3]	Improved stability in noise [3]	Parameter-free, enhanced numerical stability	Additional classical processing required
VQE-based Approaches	Circuit depth dependent [10]	Varies with ansatz choice [10]	Suitable for near-term devices, flexibility	Optimization challenges, barren plateaus
MM-PBSA (Classical)	$O(N^2)$ for PB solver [12]	~2-3 kcal/mol for binding [12]	Efficient for large systems, implicit solvent	Limited accuracy for charged systems

Case Study: Quantum Computing Pipeline for Real-World Drug Discovery

Recent work demonstrates a hybrid quantum computing pipeline applied to two critical drug discovery challenges [10]:

- **Prodrug Activation Analysis:** Implementation of quantum computations for C-C bond cleavage in β-lapachone prodrugs, achieving consistent results with wet lab validation. The approach combined active space approximation with the ddCOSMO solvation model, demonstrating the viability of quantum computations for simulating covalent bond cleavage relevant to prodrug activation [10].
- **Covalent Inhibition Study:** Application to KRAS G12C inhibition by Sotorasib, implementing a hybrid quantum computing workflow for molecular forces during QM/MM simulation. This approach enables detailed examination of covalent inhibitor binding mechanisms that are crucial in cancer therapy [10].

The pipeline demonstrated particular advantages for simulating covalent bonding interactions, transitioning quantum computing applications from theoretical models to tangible drug design problems [10].

Technical Validation and Error Mitigation

Accurate technical validation is essential for reliable quantum computations in drug discovery. The following approaches are recommended:

- **Classical Benchmarks:** Compare quantum results with high-accuracy classical methods like CASCI and DFT for small systems where exact solutions are known [10].
- **Statistical Error Characterization:** Implement rigorous statistical error analysis accounting for data covariances, particularly when using informationally complete measurements like classical shadows [5].
- **Regularization Techniques:** Apply singular value decomposition (SVD) with regularization to address ill-conditioning in overlap matrices, discarding smallest singular values to stabilize calculations [5].

For the prodrug activation case study, researchers calculated atomization energy distributions across different basis sets (SZ, DZP, TZP) and observed expected trends where TZP and DZP basis sets produced similar results while SZ basis sets showed relatively different distributions [11]. Error distribution analysis for reaction energies demonstrated that errors typically follow a normal distribution with the majority of reactions exhibiting relatively small error scales due to error cancellation effects [11].

Large-scale implementations of quantum subspace expansion with classical shadows have been successfully demonstrated for systems of up to 80 qubits, accurately recovering ground state energies while effectively mitigating local order parameters [5]. These implementations required substantial measurement resources (32,768 randomized bases per circuit and 4.3×10^{11} to 5.6×10^{13} Pauli traces) but established viable paths to overcoming measurement overhead through rigorous error characterization [5].

How QSE Complements and Enhances Variational Quantum Eigensolver (VQE)

Calculating molecular energies is a fundamental challenge in chemistry with profound implications for drug discovery and materials science. The Variational Quantum Eigensolver (VQE) has emerged as a leading algorithm for near-term quantum computers to tackle this problem, operating by variationally optimizing a parameterized quantum circuit (ansatz) to find the ground state energy of a molecular Hamiltonian [13] [14]. Despite its promise, VQE faces significant practical limitations: its accuracy is constrained by the expressive power of the chosen ansatz, and its trainability is hampered by hardware noise, statistical errors, and the barren plateau phenomenon [15] [16]. These challenges become increasingly severe with molecular size, limiting VQE's practical application.

Quantum Subspace Expansion (QSE) provides a powerful post-processing technique that complements and enhances VQE by systematically improving its results. This hybrid approach allows researchers to balance quantum and classical computational resources effectively [15]. By building upon a VQE-prepared reference state, QSE can generate more accurate ground and excited state energies without increasing quantum circuit depth, thus mitigating some core limitations of standalone VQE implementations. This application note details the methodology, performance, and implementation protocols for integrating QSE with VQE in molecular energy calculations.

Theoretical Foundation: VQE and QSE Synergy

Variational Quantum Eigensolver Fundamentals

The VQE algorithm is a hybrid quantum-classical approach that leverages the variational principle to estimate the ground state energy of a molecular Hamiltonian. The algorithm follows these key steps:

- **Hamiltonian Formulation**: The molecular electronic Hamiltonian is transformed into a second-quantized form and then mapped to a qubit representation using techniques such as the Jordan-Wigner or parity transformation [13] [17].
- **Ansatz Preparation**: A parameterized quantum circuit (ansatz), such as the Unitary Coupled Cluster (UCC) or hardware-efficient ansatz, prepares a trial state ($|\Psi(\vec{\theta})\rangle$) from an initial reference state (typically Hartree-Fock) [13] [18].
- **Measurement and Optimization**: The expectation value ($\langle \Psi(\vec{\theta})|H|\Psi(\vec{\theta})\rangle$) is measured on quantum hardware, and a classical optimizer adjusts parameters ($\vec{\theta}$) to minimize this energy [14].

The accuracy of VQE is fundamentally limited by the ansatz's ability to represent the true ground state. Deep, expressive ansätze often require circuit depths incompatible with current noisy quantum devices, while shallow ansätze may lack the representational power for strongly correlated systems [19].

Quantum Subspace Expansion Framework

QSE enhances VQE by performing a classical diagonalization of the Hamiltonian in a carefully constructed subspace around the VQE solution. The general QSE procedure, following the VQE optimization, is:

- **Subspace Construction**: Generate a set of basis states ($\{|\Psi_i\rangle\}$) *that span a subspace ($\mathcal{C}_K$). This can be a fine-grained Krylov subspace formed by applying Pauli strings (P_j)* to the VQE state: ($|\Psi_i\rangle = P_j|\Psi_{\text{VQE}}\rangle$) [15].
- **Matrix Element Measurement**: Use the quantum computer to measure the matrix elements of the Hamiltonian ($H_{ij} = \langle\Psi_i|H|\Psi_j\rangle$) *and overlap ($S_{ij} = \langle\Psi_i|\Psi_j\rangle$)* operators within this subspace.
- **Classical Diagonalization**: Solve the generalized eigenvalue problem ($H\vec{c} = ES\vec{c}$) on a classical computer to obtain refined energy estimates and wavefunctions within the subspace [15] [16].

This approach increases the expressivity of the final wavefunction without complicating the quantum circuit training, as the diagonalization is performed classically [15]. The subspace expansion effectively captures additional electron correlation effects missing from the original VQE state.

Performance Comparison and Quantitative Analysis

Multiple studies demonstrate that the VQE-QSE combination delivers superior performance compared to standalone VQE, particularly for more complex molecular systems and in the presence of hardware noise.

Table 1: Comparative Performance of VQE and VQE-QSE for Molecular Ground State Calculations

Molecule	Method	Ansatz/Protocol	Energy Accuracy (Absolute Error)	Key Performance Notes
H(_2)/Small Systems	Standalone VQE	UCCSD	Varies (exact for minimal systems)	Baseline for comparison [13]
	VQE-QSE	UCCSD with FGKS expansion	Improved	Systematically approaches exact solution [15]
H(_{24}) (STO-3G)	Fragment-Based VQE	FMO/VQE with UCCSD	0.053 mHa	Reduced qubit requirement [20]
Stretched H(_6)	ADAPT-VQE	QEB-ADAPT	~1000+ CNOT gates for chemical accuracy	Prone to local minima, over-parameterization [19]
Stretched H(_6)	Overlap-ADAPT-VQE	Overlap-guided compact ansatz	Chemically accurate with fewer operators	Avoids energy plateaus, more compact circuits [19]
General Performance in Noise	Standalone VQE	Various	Degraded by noise and barren plateaus	Optimization becomes difficult with system size [15] [16]
General Performance in Noise	VQE-QSE (PIQAE)	Balanced ansatz depth ((L)) & expansion moment ((K))	Order of magnitude improvement per site demonstrated on IBM QPUs	Robustness against noise; accuracy tunable via (L) and (K) [15]

Table 2: QSE Applications Beyond Ground State Energy Calculation

Application	QSE Variant	System Studied	Key Outcome
Excited State & Band Structure Calculation	Standard QSE	Silicon crystal quasiparticle bands	Enabled calculation of excited states from VQE ground state [16]
Scalable Ground State Calculation	QSE with Quantum-Selected CI (QSCI)	--	Avoids full VQE optimization; uses quantum sampling to select configurations [16]
Noise-Resilient Calculation	Paired IQAE (PIQAE)	1D/2D Ising models on ibmq_quito/guadalupe	Effective noise mitigation via optimal subspace selection (e.g., based on overlap matrix trace) [15]

Experimental Protocols

Core VQE-QSE Workflow Protocol

This protocol outlines the essential steps for implementing the combined VQE-QSE method for molecular ground state energy calculation.

Step 1: Molecular Hamiltonian Preparation

- **Input**: Molecular geometry (atomic species and coordinates), basis set (e.g., STO-3G, 6-31G).
- **Procedure**:
  - Classically compute molecular integrals (one- and two-electron integrals in the chosen basis set) using quantum chemistry packages (e.g., PySCF via OpenFermion) [19].
  - Form the second-quantized electronic Hamiltonian (\hat{H} = \sum*{pq} h*{pq} ap^*dagger aq + \frac{1}{2} \sum*{pqrs} h*{pqrs} ap^*dagger aq^\dagger ar as) [13] [20].
  - Map the fermionic Hamiltonian to a qubit Hamiltonian using a transformation (e.g., Jordan-Wigner, parity) [13] [16]. The result is a Pauli string representation: (H = \sumi wi P_i).

Step 2: VQE Optimization

- **Input**: Qubit Hamiltonian, choice of ansatz (e.g., UCCSD, hardware-efficient), initial parameters.
- **Procedure**:
  - **Prepare Initial State**: Prepare the Hartree-Fock state (|0\rangle) on the quantum computer.
  - **Construct Ansatz Circuit**: Implement the parameterized ansatz circuit (U(\vec{\theta})). For UCCSD, this involves Trotterized exponentials of fermionic excitation operators [13].
  - **Measure Expectation Value**: Estimate (\langle H \rangle = \langle 0| U^\dagger(\vec{\theta}) H U(\vec{\theta}) |0\rangle) by measuring the expectation values of all Pauli terms (Pi). *Use grouping strategies (e.g., qubit-wise commuting) to reduce measurement*

overhead [17].

- ***Classical Optimization**: Use a classical optimizer (e.g., BFGS) to minimize (\langle H \rangle) with respect to parameters (\vec{\theta}). The optimized state is (|\Psi{\text{VQE}}\rangle = U(\vec{\theta}^*)|0\rangle) [18].*

Step 3: Quantum Subspace Expansion

- **Input**: Optimized VQE state (|\Psi_{\text{VQE}}\rangle), subspace definition.
- **Procedure**:
  - **Define Subspace Basis**: Choose a set of basis states for expansion. A common choice is the fine-grained Krylov subspace, where basis vectors are (|\Phi*j\rangle = \hat{P}*j |\Psi{\text{VQE}}\rangle) *and* (\hat{P}j) are Pauli strings from a predefined set (e.g., all Paulis up to a certain weight (K)) [15].
  - **Measure Subspace Matrices**: On the quantum computer, measure the matrix elements of the Hamiltonian (H*{ij} = \langle\Phi*i|H|\Phi*j\rangle) *and the overlap* (S{ij} = \langle\Phi*i|\Phi*j\rangle). This does not require deep controlled circuits but can involve a substantial number of measurements [15].
  - **Classically Diagonalize**: Solve the generalized eigenvalue problem (\mathbf{H} \vec{c} = E \mathbf{S} \vec{c}) on a classical computer. The lowest eigenvalue (E_0) is the QSE-refined ground state energy estimate [15] [16].

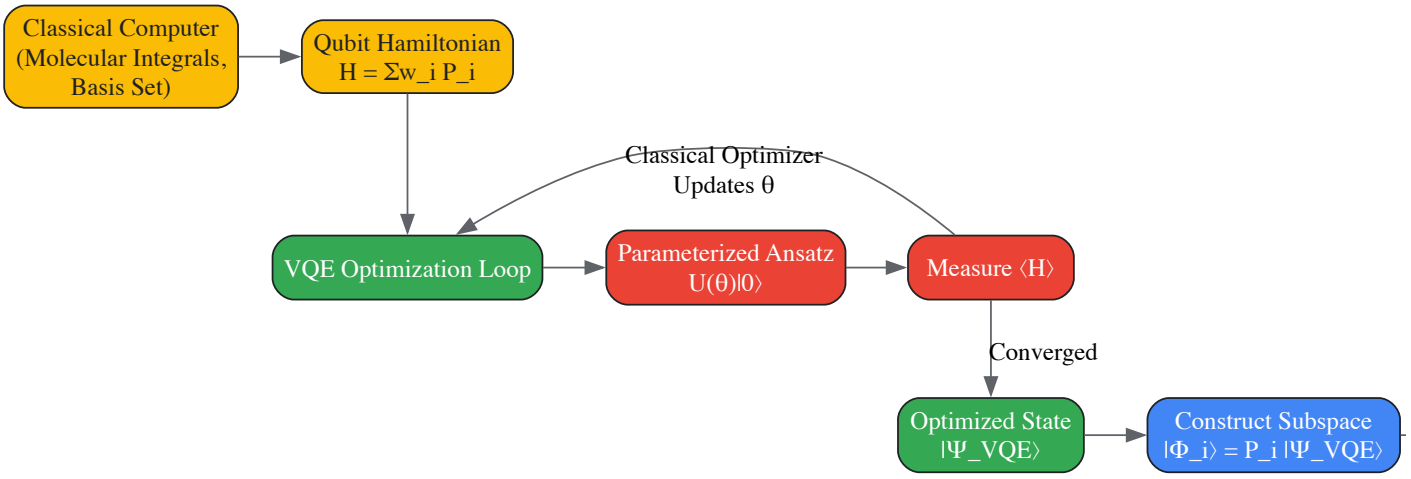


Diagram 1: VQE-QSE Hybrid Algorithm Workflow. The process integrates quantum (red/blue) and classical (yellow/green) computation stages, beginning with Hamiltonian preparation and culminating in a refined energy estimate via subspace diagonalization.

Advanced Protocol: Noise-Aware PIQAE

For calculations on real noisy hardware, the Paired Iterative Quantum-Assisted Eigensolver (PIQAE) protocol provides enhanced robustness [15].

Procedure:

- **Balance Resources**: Choose a VQE ansatz depth (L) compatible with hardware error rates. A shallower, more noise-resilient ansatz is often preferable.
- **Perform VQE**: Run the standard VQE optimization with the chosen ansatz.
- **Iterative Expansion**: Expand the subspace up to a moment (K), limited by the classical and quantum computational budget (e.g., number of measurements).
- **Noise Mitigation and Subspace Selection**:
  - Measure the overlap matrix (\mathbf{S}) under noise.
  - Compute its trace. As noise increases, the trace deviates from its noiseless value (which equals the subspace dimension).
  - Use the trace to select the optimal subspace dimension (\mathcal{M}_0) to truncate the noisy matrices before diagonalization, preventing unphysical energy estimates [15].
- **Optional Error Mitigation**: Apply additional error mitigation techniques (e.g., probabilistic error cancellation) to the measurement results of (H*{ij}) *and* (S{ij}) for further accuracy improvement [15].

Table 3: Key Software Tools and Computational Resources for VQE-QSE Implementation

Tool/Resource	Type	Primary Function	Relevance to VQE-QSE
OpenFermion	Software Library	Manipulation of fermionic Hamiltonians and mapping to qubits [19].	Prepares the qubit Hamiltonian from molecular data. Essential first step.

Tool/Resource	Type	Primary Function	Relevance to VQE-QSE
Qiskit / PennyLane	Quantum Computing SDK	Construction and simulation of quantum circuits, execution on simulators/hardware [13] [14].	Implements the VQE ansatz circuit, manages measurement, and interfaces with classical optimizers.
PySCF	Quantum Chemistry Package	Computes molecular integrals and performs classical electronic structure calculations [19].	Generates the one- and two-electron integrals required to build the second-quantized Hamiltonian.
IBM Quantum / AWS Braket	Cloud QPU Access	Provides access to real noisy intermediate-scale quantum (NISQ) devices [15].	Platform for running the quantum steps (state preparation and measurement) of the VQE-QSE algorithm.
BFGS / SPSA Optimizers	Classical Optimization Algorithm	Finds parameters that minimize the energy cost function [18].	Critical for the VQE parameter optimization loop. Gradient-based optimizers like BFGS are often preferred [18].
Fine-Grained Krylov Subspace	Mathematical Subspace	Defines the basis for the QSE expansion [15].	The specific choice of subspace (e.g., using Pauli strings) directly impacts the accuracy and cost of the QSE refinement.

The integration of Quantum Subspace Expansion with the Variational Quantum Eigensolver represents a significant advancement in quantum computational chemistry. This synergistic approach effectively addresses key limitations of standalone VQE—limited ansatz expressivity and susceptibility to noise—by offloading additional complexity to efficient classical post-processing. The structured protocols and performance data presented herein provide researchers and drug development professionals with a practical framework for implementing these hybrid algorithms. As quantum hardware continues to evolve, the VQE-QSE paradigm is poised to play an increasingly vital role in enabling high-accuracy simulation of complex molecular systems, accelerating the discovery of new therapeutic agents and functional materials.

The Role of Krylov Subspaces in Building an Expansive Computational Basis

Quantum subspace expansion (QSE) represents a powerful class of hybrid quantum-classical algorithms designed to overcome the limitations of variational methods for molecular energy calculations. Unlike variational quantum eigensolver (VQE) approaches that require challenging parameter optimizations, QSE methods formulate the problem as a generalized eigenvalue problem within a carefully constructed subspace of the full Hilbert space [21] [22]. The core mathematical framework involves projecting the molecular Hamiltonian into a smaller subspace spanned by a set of basis states, then solving the eigenvalue problem classically to obtain approximate ground and excited state energies [22]. This approach has demonstrated particular promise for near-term and early fault-tolerant quantum hardware by exchanging quantum circuit depth for additional measurements [3].

Among various subspace constructions, **Krylov subspaces** have emerged as particularly powerful due to their proven convergence properties and foundation in classical numerical analysis [3] [21]. A Krylov subspace of dimension D is typically generated from an initial reference state $|\psi_0\rangle$ through repeated applications of the Hamiltonian H: $\mathcal{K} = \text{span}\{|\psi_0\rangle, H|\psi_0\rangle, H^2|\psi_0\rangle, ..., H^{D-1}|\psi_0\rangle\}$ [22]. The exceptional value of this construction lies in its **exponential convergence** toward the true ground state energy as the subspace dimension increases, a property well-established in classical numerical mathematics that carries over to quantum implementations [21] [22]. For quantum chemistry applications, this enables systematically improvable approximations to molecular energies without exponentially growing quantum resource requirements.

Table: Comparison of Quantum Computational Approaches for Molecular Energy Calculations

Method	Key Principle	Convergence Guarantees	Quantum Resource Requirements
Variational Quantum Eigensolver (VQE)	Parametric circuit optimization	Limited; depends on ansatz and optimization	Moderate circuit depth, many measurements
Quantum Phase Estimation (QPE)	Quantum Fourier transform on phase information	Theoretical guarantees with error correction	Very high circuit depth, fault tolerance needed
Quantum Krylov Subspace Diagonalization	Projected eigenvalue problem in constructed subspace	Exponential with subspace dimension (theoretical)	Low-moderate circuit depth, many measurements
Partitioned Quantum Subspace Expansion (PQSE)	Iterative Krylov subspace construction	Improved numerical stability with exponential convergence	Same quantum resources as QSE, additional classical processing

Krylov Subspace Fundamentals and Theoretical Framework

Mathematical Foundation of Krylov Methods

The theoretical foundation of Krylov subspace methods in quantum computing mirrors their established classical counterparts while addressing quantum-specific challenges. In classical numerical mathematics, the Lanczos algorithm represents the archetypal Krylov method for eigenvalue problems, generating an orthonormal basis for the Krylov subspace through a three-term recurrence relation [23] [24]. This elegant mathematical structure ensures that the Hamiltonian becomes tridiagonal in the Krylov basis, significantly simplifying the eigenvalue solution while maintaining exponential convergence properties [24]. The power of this approach stems from the **global operator** nature of Hamiltonian powers, which efficiently explore the regions of Hilbert space most relevant to the low-energy spectrum.

The transfer of these methods to quantum computers introduces both opportunities and challenges. Quantum Krylov Diagonalization (KQD) leverages the quantum computer's ability to handle exponentially large state spaces while avoiding the memory limitations that constrain classical approaches for large quantum systems [21]. The core mathematical problem solved by KQD is the generalized eigenvalue equation $\tilde{\mathbf{H}}\mathbf{c} = \mathbf{E}\hat{\mathbf{S}}\mathbf{c}$, where $\tilde{H}$ is the Hamiltonian projected into the Krylov subspace with elements $\tilde{H}_{jk} = \langle \psi_j | H | \psi_k \rangle$, and $\hat{S}$ is the overlap matrix with elements $\hat{S}_{jk} = \langle \psi_j | \psi_k \rangle$ [21]. The eigenvectors c provide the coefficients for the optimal linear combination of basis states that approximates the true molecular eigenstates.

For molecular energy calculations, a critical theoretical advantage emerges from the **variational property** of subspace methods: the ground state energy obtained by diagonalizing the projected Hamiltonian is guaranteed to be an upper bound to the true ground state energy [24]. This variational foundation ensures systematic improvability and provides a valuable diagnostic for method accuracy. Furthermore, the accuracy of the ground state approximation can be quantified by the variance of the Hamiltonian in the trial state, with zero variance indicating an exact eigenstate [22].

Krylov Basis Construction Methods

Several technical approaches have been developed for constructing Krylov subspaces on quantum hardware, each with distinct advantages for molecular applications:

- **Unitary Krylov Spaces**: Generated through real-time evolutions $U^j|\psi_0\rangle = e^{-iHjt}|\psi_0\rangle$, these subspaces leverage the natural dynamics of quantum systems and are particularly suitable for near-term devices due to their implementability with relatively shallow circuits [21].
- **Power Krylov Spaces**: Constructed from powers of the Hamiltonian $H^j|\psi_0\rangle$, these subspaces directly mirror classical Krylov methods and can be implemented through qubitization techniques, though they typically require deeper circuits [22].
- **Chebyshev Krylov Spaces**: Using Chebyshev polynomials $T_j(H)|\psi_0\rangle$ as basis states provides numerical advantages and enables efficient implementation through Quantum Signal Processing (QSP), making this approach particularly valuable for fault-tolerant implementations [23].

The choice of construction method involves trade-offs between circuit depth, measurement requirements, and numerical stability. For molecular systems with specific symmetry properties, customized approaches that preserve symmetries can significantly enhance efficiency and accuracy.

Advanced Krylov Techniques and Protocol Implementation

Partitioned Quantum Subspace Expansion (PQSE)

The Partitioned Quantum Subspace Expansion (PQSE) represents a significant advancement in addressing the primary limitation of standard Krylov methods: numerical instability arising from ill-conditioned overlap matrices [3] [22]. As the Krylov dimension increases, higher-order basis states become nearly linearly dependent due to the mathematical properties of power iteration, causing the overlap matrix $\hat{S}$ to become ill-conditioned and amplifying noise in measured matrix elements [22]. PQSE addresses this challenge through an **iterative partitioning** strategy that breaks a large QSE problem into a sequence of smaller, better-conditioned subproblems.

The PQSE protocol operates by connecting a sequence of subspaces through their lowest-energy states, with the output of one small QSE instance serving as input to the next iteration [3] [22]. This approach maintains the same quantum resource requirements as standard QSE while introducing additional classical processing with polynomial overhead in the subspace dimension [3]. The partioning strategy is guided by a **variance-based criterion** for determining effective iterative sequences, with numerical demonstrations showing substantially improved numerical stability in the presence of finite sampling noise compared to both standard QSE and thresholded QSE (TQSE) [22].

Table: Key Research Reagent Solutions for Quantum Krylov Experiments

Research Reagent	Function in Experiment	Implementation Considerations
Krylov Subspace Basis States	Forms the computational basis for diagonalization	Choice between unitary, power, or Chebyshev constructions affects circuit depth and measurement requirements
Overlap Matrix ($\hat{S}$)	Encodes linear dependencies between basis states	Ill-conditioning requires regularization techniques; measured via swap tests or Hadamard tests

Research Reagent	Function in Experiment	Implementation Considerations	
Projected Hamiltonian ($\tilde{H}$)	Reduced Hamiltonian for classical diagonalization	Matrix elements measured through quantum expectation value estimation	
Reference State	$\psi_0\rangle$	Initial state for Krylov basis generation	Typically Hartree-Fock state for molecular systems; affects convergence rate
Time Evolution Operator e^{-iHdt}	Generates unitary Krylov basis states	Implemented through Trotterization or more advanced product formulas; depth depends on accuracy requirements	

Experimental Protocol: Quantum Krylov Subspace Diagonalization

The following protocol details the implementation of Quantum Krylov Subspace Diagonalization for molecular energy calculations, based on established methodologies from recent experimental demonstrations [23] [21].

Step 1: System Preparation and Reference State Initialization

- Prepare the molecular Hamiltonian H in qubit representation using Jordan-Wigner or Bravyi-Kitaev transformation. For the water molecule (H_2O) in a cc-pVDZ basis with 4 electrons in 4 molecular orbitals, this results in a 16-term Hamiltonian [23].
- Initialize the reference state $|\psi_0\rangle$, typically the Hartree-Fock state, which can be prepared through a simple quantum circuit applying X gates to qubits corresponding to occupied orbitals.
- For systems with conserved quantities (e.g., particle number, total spin), exploit these symmetries to simplify circuits and improve convergence [21].

Step 2: Krylov Basis Construction

- Select the Krylov subspace dimension D based on desired accuracy and available quantum resources. For initial experiments, D=4-8 provides reasonable accuracy while managing measurement costs.
- Generate basis states using the chosen construction method:
  - For unitary evolution: Implement $|\psi_j\rangle = e^{-iHjdt}|\psi_0\rangle$ using Trotterized circuits with appropriate time steps dt [21].
  - For Chebyshev approach: Construct $|\psi_j\rangle = T_j(H)|\psi_0\rangle$ using Quantum Signal Processing where feasible [23].
- For near-term hardware, focus on minimizing circuit depth through optimization techniques and hardware-aware compilation.

Step 3: Quantum Measurement of Matrix Elements

- Measure all elements of the overlap matrix $\hat{S}_{jk} = \langle\psi_j|\psi_k\rangle$ and projected Hamiltonian $\tilde{H}_{jk} = \langle\psi_j|H|\psi_k\rangle$ using efficient quantum circuits.
- Employ symmetry relations to reduce measurement requirements: for exact time evolutions, $\tilde{H}_{jk} = \langle\psi_0|HU^{k-j}|\psi_0\rangle$, enabling measurement with a single time evolution [21].
- For each matrix element, perform sufficient measurement shots to achieve desired statistical precision, considering the trade-off between measurement cost and accuracy.

Step 4: Classical Post-Processing and Diagonalization

- Construct the matrices $\tilde{H}$ and $\hat{S}$ from measured matrix elements, applying error mitigation techniques to address measurement noise and hardware errors.
- Solve the generalized eigenvalue problem $\tilde{H}c = E\hat{S}c$ using classical computational resources.
- Apply regularization techniques if $\hat{S}$ is ill-conditioned, such as truncated singular value decomposition with an appropriate threshold [24].
- Extract the ground state energy E_0 as the lowest eigenvalue, and excited states from higher eigenvalues if desired.

Step 5: Validation and Error Analysis

- Compute the energy variance $var(H) = \langle H^2\rangle - \langle H\rangle^2$ for the obtained ground state to assess accuracy, with near-zero variance indicating a good approximation to a true eigenstate [22].
- Perform convergence analysis with respect to Krylov dimension D to ensure sufficient basis size.
- For partitioned approaches (PQSE), iterate the process using the obtained ground state as a new reference until convergence criteria are satisfied [3].

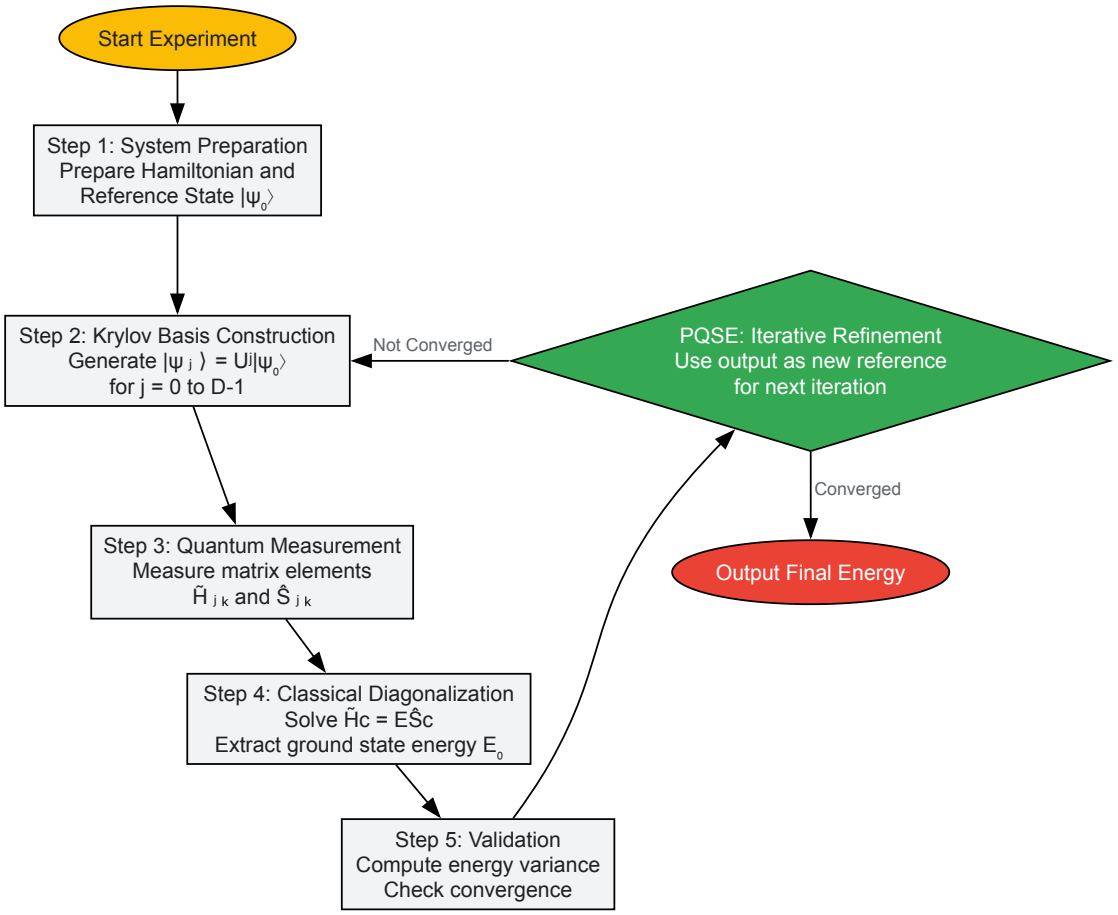


Figure 1: Quantum Krylov Subspace Diagonalization Workflow

Measurement-Efficient Variants and Error Mitigation

Recent advances in quantum Krylov methods have focused on addressing the challenging measurement requirements through algorithmic innovations. The measurement-efficient quantum Krylov subspace diagonalization approach reduces the number of required measurements by expressing the product of power and Gaussian functions of the Hamiltonian as an integral of real-time evolution, evaluable through Monte Carlo sampling on quantum computers [24]. This approach can reduce measurement costs by **orders of magnitude** compared to standard implementations while maintaining accuracy comparable to classical Lanczos algorithms at the same subspace dimension [24].

For error mitigation in near-term implementations, several strategies have proven effective:

- **Symmetry verification**: Exploit conserved quantities (e.g., particle number) to detect and mitigate errors that violate these symmetries [21].
- **Regularization techniques**: Address ill-conditioned overlap matrices through truncated singular value decomposition or Tikhonov regularization, with theoretical guarantees ensuring variational bounds remain valid [24].
- **Noise-aware algorithms**: Design circuits specifically to minimize error accumulation, such as using circuits that preserve the "vacuum state" up to a calculable phase to avoid controlled time evolutions [21].

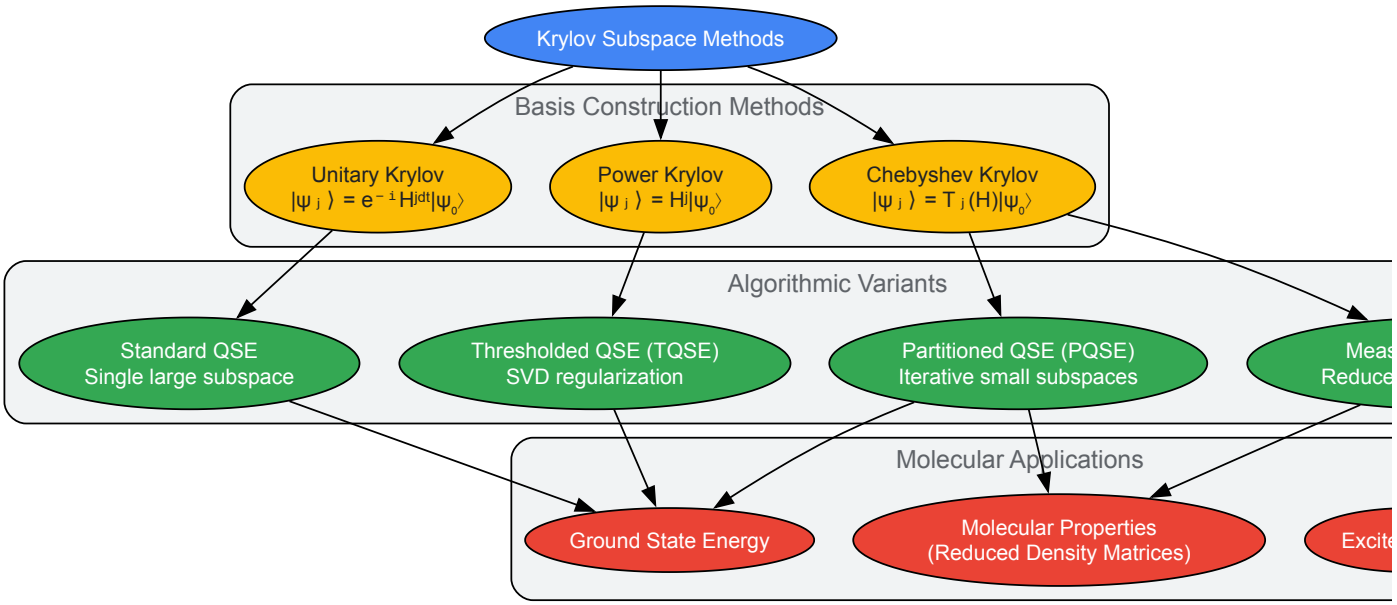


Figure 2: Krylov Method Taxonomy and Applications

Applications in Molecular Energy Calculations

Practical Implementation for Molecular Systems

Quantum Krylov subspace methods have demonstrated significant potential for practical molecular energy calculations, with several experimental implementations showcasing their capabilities. For the water molecule (H_2O) in a cc-pVDZ basis with an active space of 4 electrons in 4 molecular orbitals, the Krylov approach enables accurate ground state energy estimation with relatively modest quantum resources [23]. The protocol involves constructing the Krylov subspace using Chebyshev polynomials $T_j(H)|\psi_0\rangle$ applied to the Hartree-Fock reference state, then solving the generalized eigenvalue problem to obtain the ground state approximation $|\Psi_0\rangle = \sum_k c_k^0 T_k(H)|\psi_0\rangle$ [23].

Beyond ground state energies, Krylov methods facilitate the efficient calculation of molecular properties through reduced density matrices. For the one-particle reduced density matrix with elements $\gamma_{pp'} = \langle\Psi_0|a_p^\dagger a_{p'}|\Psi_0\rangle$, specialized measurement protocols based on Quantum Signal Processing can achieve **constant scaling** with respect to Krylov dimension D, dramatically improving over the quadratic scaling of naive measurement approaches [23]. This capability is essential for computing molecular properties such as dipole moments, bond orders, and other electronic structure indicators directly relevant to drug development.

For drug discovery applications, where accurate molecular energy calculations inform binding affinity predictions and reactivity assessments, the numerical stability and systematic improvability of Krylov methods offer significant advantages over purely variational approaches. The ability to compute both ground and excited states within the same framework further enables studies of photochemical properties and reaction pathways relevant to pharmaceutical development.

Performance Analysis and Resource Requirements

The performance of quantum Krylov methods is characterized by exponential convergence toward the true ground state energy with increasing subspace dimension, providing a systematic path to higher accuracy without increasing quantum circuit depth [21] [22]. This convergence behavior has been experimentally demonstrated on quantum processors for systems of up to 56 qubits, showing that quantum diagonalization algorithms can complement classical counterparts even in the pre-fault-tolerant era [21].

The primary resource requirements for quantum Krylov methods include:

- **Circuit depth**: Scales with the complexity of time evolution operators, typically implemented through Trotterization with depth depending on the molecular Hamiltonian and desired accuracy.
- **Measurement count**: Grows with the square of subspace dimension D^2 for naive implementations, but advanced techniques can substantially reduce this scaling [24].
- **Classical processing**: Involves solving a D×D generalized eigenvalue problem, with polynomial scaling in D that is typically manageable for classically tractable subspace dimensions ($D < 100$).

For the Partitioned Quantum Subspace Expansion (PQSE), the quantum resource requirements remain identical to standard QSE, while classical processing overhead increases polynomially but provides substantially improved numerical stability and ability to reach larger effective Krylov

dimensions [3] [22]. This trade-off favors PQSE for applications where measurement noise rather than quantum circuit performance represents the primary limitation, as is often the case for current quantum hardware.

Table: Performance Comparison of Quantum Krylov Methods for Molecular Systems

Method	Convergence Rate	Measurement Cost	Numerical Stability	Circuit Depth Requirements
Standard QSE	Exponential with D	$O(D^2)$ for naive implementation	Poor at large D due to ill-conditioning	Moderate (Trotterized evolution)
TQSE (Thresholded)	Exponential up to critical D	Similar to standard QSE	Improved via SVD truncation	Moderate (Trotterized evolution)
PQSE (Partitioned)	Exponential with effective dimension	Same as QSE, better stability	Substantially improved via partitioning	Moderate (Trotterized evolution)
Measurement-Efficient QKSD	Comparable to classical Lanczos	Orders of magnitude reduction	Good with regularization	Moderate to high (depends on implementation)

Krylov subspace methods represent a foundational approach for building expansive computational bases in quantum computational chemistry, offering rigorous convergence guarantees and practical implementability on emerging quantum hardware. The iterative structure of Partitioned Quantum Subspace Expansion addresses the critical challenge of numerical instability while maintaining exponential convergence, enabling larger effective Krylov dimensions and higher accuracy for molecular energy calculations [3] [22]. For drug development professionals and researchers, these methods provide a systematically improvable path toward predictive quantum chemistry simulations without requiring fault-tolerant quantum computing.

Future developments in quantum Krylov methods will likely focus on further reducing measurement costs, enhancing error mitigation techniques, and developing problem-specific basis constructions that leverage molecular symmetries and structure. As quantum hardware continues to advance, the integration of Krylov subspace methods with fault-tolerant building blocks such as Quantum Signal Processing will enable increasingly accurate simulations of complex molecular systems relevant to pharmaceutical development and materials design.

Quantum Subspace Expansion (QSE) is a computational technique in quantum chemistry that enables the calculation of molecular excited state energies from a pre-computed ground state. This method is particularly valuable for near-term quantum computers, as it provides a pathway to extract excited state information and additional correlation energy beyond what is achievable with a standalone Variational Quantum Eigensolver (VQE) calculation. The core idea involves constructing a linear subspace of wavefunctions from a reference ground state, followed by the diagonalization of the molecular Hamiltonian within that subspace to obtain a set of energetically low-lying states [25] [4].

The process begins with a correlated ground state wavefunction, $(|\Psi_0\rangle)$, *typically obtained using the VQE algorithm. A subspace is then created by applying a set of excitation operators to this ground state. Commonly, these are single- and double-electron excitation operators from the field of quantum chemistry:* $[\hat{G}_i^a = \hat{a}^\dagger_a \hat{a}_i, \quad \hat{G}_{ij}^{ab} = \hat{a}^\dagger_a \hat{a}^\dagger_b \hat{a}_j \hat{a}_i]$ *where (i, j) denote occupied orbitals and (a, b) denote virtual orbitals in the reference state [26]. The resulting subspace vectors,* $(|\Psi_k\rangle = c_k^\dagger c_j |\Psi_0\rangle)$, are not generally orthogonal, necessitating the solution of a generalized eigenvalue problem to find the optimal energy eigenvalues and eigenvectors within the subspace [25].

Theoretical Foundation: Overlap Matrix and Generalized Eigenvalue Problem

The Overlap Matrix (S)

The overlap matrix, **S**, is a central mathematical object in the QSE formalism. It quantifies the non-orthogonality of the basis vectors that span the constructed subspace. The matrix elements of **S** are defined as the inner products between these basis states [27]: $[S_{jk}^{lm} = \langle \Psi_j^l | \Psi_k^m \rangle = \langle \Psi_0 | c_j^\dagger c_l c_m^\dagger c_k | \Psi_0 \rangle]$ In a quantum computation, this expectation value is measured on a quantum computer or simulator after mapping the fermionic operators to Pauli spin operators via a transformation such as Jordan-Wigner or Bravyi-Kitaev [25]. The overlap matrix is a square, positive definite matrix. If the basis vectors were orthonormal, **S** would be the identity matrix. However, the use of non-orthogonal basis vectors leads to off-diagonal elements, and the matrix's condition number (the ratio of its largest to smallest singular value) becomes a critical factor for the numerical stability of the subsequent eigenvalue solution [26] [27].

The Hamiltonian Matrix (H)

Simultaneously, the Hamiltonian matrix, **H**, is constructed within the same subspace. Its elements are the expectation values of the system's Hamiltonian, $(\hat{H})$, with respect to the subspace basis vectors [4]: $[H_{jk}^{lm} = \langle \Psi_j^l \left| \hat{H} \right| \Psi_k^m \rangle = \langle \Psi_0 | c_j^\dagger c_l \hat{H} c_m^\dagger c_k | \Psi_0 \rangle]$ Like the overlap matrix, these matrix elements are also determined through quantum measurements.

The Generalized Eigenvalue Equation

The optimal energies and wavefunctions within the QSE subspace are found by solving the generalized eigenvalue equation [25] [4]: [\mathbf{H}C = \mathbf{S}CE] Here, (E) is a diagonal matrix containing the eigenvalues (which provide estimates for the ground and excited state energies), and (C) is the matrix of eigenvectors containing the expansion coefficients for the states in the chosen basis. This equation can be solved on a classical computer once the **H** and **S** matrices have been measured on the quantum processor.

Critical Analysis and Numerical Challenges

A significant challenge in practical QSE implementations is the **numerical instability** arising from the generalized eigenvalue problem. This instability is directly linked to the condition number of the overlap matrix, **S** [26].

- **Ill-Conditioning and Sampling Errors**: When the overlap matrix has a high condition number, it is termed "ill-conditioned." In this regime, small errors in the matrix elements of **H** and **S**—which are inevitable due to finite sampling (shot noise) on quantum hardware—can lead to large errors in the computed eigenvalues [26].
- **Singularities and Thresholding**: In severe cases, the overlap matrix can become nearly singular (i.e., its determinant is close to zero), preventing a standard numerical solution. A common technique to mitigate this is **thresholding**, where the smallest eigenvalues of the **S** matrix and their corresponding eigenvectors are discarded before solving the generalized eigenvalue problem in the remaining, more stable subspace. While this can restore solvability, it may also result in the loss of physically meaningful excited states from the calculated spectrum [26].
- **Comparison with Alternative Methods**: The instability of the generalized eigenvalue problem has motivated the development of alternative subspace methods that do not suffer from this issue. The **quantum self-consistent Equation-of-Motion (q-sc-EOM)** method, for example, uses an orthonormal set of excited wavefunctions, leading to an overlap matrix that is the identity matrix. This transforms the problem into a standard eigenvalue equation, which is inherently more robust to statistical sampling errors [26]. Another approach is the **Partitioned Quantum Subspace Expansion (PQSE)**, which iteratively connects a sequence of smaller subspaces. This method has been shown to offer improved numerical stability over single-step QSE in the presence of finite sampling noise [3].

Experimental Protocol: QSE for Molecular Energy Calculation

The following protocol outlines the steps for performing a Quantum Subspace Expansion calculation to compute the ground and excited states of a molecule, using methane (CH_4) in a minimal basis set as an example [25].

System Definition and Qubit Encoding

Step 1: Define the Molecular System

- Specify the molecular geometry, for example, using a Z-matrix.
- Select a basis set (e.g., STO-3G) and run a classical Restricted Hartree-Fock (RHF) calculation to obtain the electronic integrals.
- Freeze a subset of core orbitals to reduce the problem size.

Step 2: Qubit Encoding

- Map the fermionic Hamiltonian and the Hartree-Fock state to the qubit space using an encoding scheme such as Jordan-Wigner or Bravyi-Kitaev.
- Compress the resulting qubit Hamiltonian by discarding terms with negligible weights to reduce the number of required quantum measurements.

Ground State Preparation via VQE

Step 3: Prepare the Ground State Ansatz

- Select a parametrized quantum circuit (ansatz) capable of representing electron correlation. The Chemically Aware Unitary Coupled Cluster Singles and Doubles (UCCSD) ansatz is a common choice.

Step 4: Optimize the Ground State

- Use the VQE algorithm to find the parameters that minimize the energy expectation value of the qubit Hamiltonian. This can be done on a noiseless simulator for initial testing.

Subspace Expansion and Matrix Measurement

Step 5: Define the Expansion Operators

- Choose a set of operators to generate the subspace. For calculating spin-adapted excited states, singlet single excitation operators are a suitable and common starting point.

Step 6: Construct the QSE Matrices Computable

- Define a computational object that will handle the construction of the **H** and **S** matrices.

Step 7: Configure the Quantum Backend and Measurement Protocol

- Select a quantum backend (simulator or hardware). For shot-based simulations, define a measurement protocol that specifies the number of measurements (shots) per circuit.

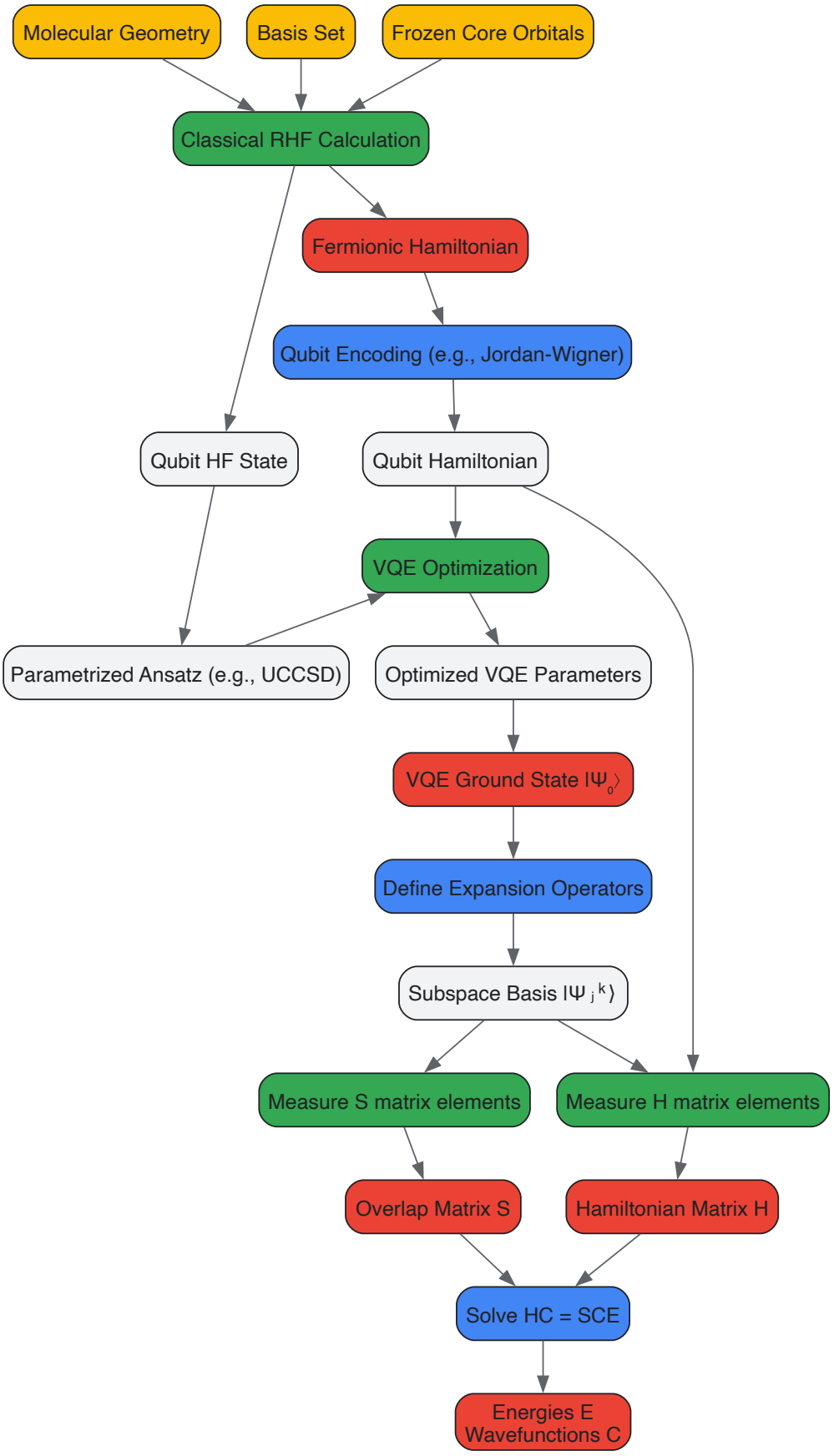
Classical Post-Processing

Step 8: Run the QSE Algorithm and Solve the Generalized Eigenvalue Problem

- Execute the algorithm to measure all required matrix elements. Subsequently, solve the generalized eigenvalue problem (\mathbf{H}C = \mathbf{S}CE) on a classical computer.
- The resulting eigenvalues correspond to the QSE-corrected ground state energy and the energies of the captured excited states. The eigenvectors describe the wavefunctions of these states within the expanded subspace.

Workflow and Data Relationships

The following diagram illustrates the logical workflow and data flow of a typical Quantum Subspace Expansion calculation.



QSE Experimental Workflow

Advanced Methodologies and Error Mitigation

Handling Numerical Instability

To address the inherent instability of the generalized eigenvalue problem in QSE, several advanced methodologies have been developed:

- **Singular Value Decomposition (SVD) and Thresholding**: This is the primary technique for stabilizing the QSE calculation. The overlap matrix **S** is diagonalized via SVD ((\mathbf{S} = U \Sigma V^\dagger)). Singular values in (\Sigma) that fall below a pre-defined threshold are considered to represent numerical noise or linear dependencies and are discarded. The generalized eigenvalue problem is then projected and solved in the truncated subspace spanned by the retained singular vectors, which is numerically well-behaved [26] [5].
- **Partitioned QSE (PQSE)**: This iterative generalization of QSE breaks a single, large Krylov subspace into a sequence of smaller, connected subspaces. By diagonalizing the Hamiltonian in each smaller subspace and using the lowest-energy state to initiate the next, PQSE can substantially alleviate numerical instability in a parameter-free manner, albeit with additional classical processing [3].
- **Reformulation as a Constrained Optimization**: Another approach reformulates the QSE problem as a constrained optimization problem, which allows for rigorous statistical error estimates and can avoid numerical instability associated with direct matrix inversion. This method has been successfully demonstrated in large-scale experiments using classical shadows for measurement [5].

Measurement Techniques

The efficient measurement of the **H** and **S** matrix elements is a significant bottleneck. Advanced measurement strategies can drastically reduce the required quantum resources.

- **Classical Shadows**: This technique uses informationally complete (IC) measurements, randomizing the measurement basis over many circuit executions. It allows for the efficient estimation of many observables (like all the matrix elements of **H** and **S**) from a single set of measurements, making it scalable to large systems (dozens of qubits) [5].
- **Pauli Averaging**: A more conventional shot-based protocol where the expectation values of the Pauli terms that make up the **H** and **S** matrices are measured directly and averaged [4].

The Scientist's Toolkit: Essential Research Reagents

The table below details the key computational "reagents" and their functions required to implement a QSE experiment.

Item Name	Function in QSE Protocol	Specification Notes
Molecular Hamiltonian	Defines the quantum mechanical system and its energy levels; the operator to be diagonalized.	Generated from electronic structure integrals via a classical Hartree-Fock calculation [25].
Qubit Mapping	Transforms the fermionic Hamiltonian and operators into a form executable on a qubit-based quantum processor.	Jordan-Wigner and Bravyi-Kitaev are common mappings [25] [4].
VQE Ansatz	A parametrized quantum circuit that prepares an approximation of the true molecular ground state.	UCCSD is a chemically inspired, common choice. Other ansatzes like hardware-efficient can be used [25].
Expansion Operator Set	A set of operators that generate the subspace when applied to the ground state.	Typically single and double excitations. Singlet singles ensure spin adaptation [26] [4].
Quantum Backend/Simulator	The computational engine that executes the quantum circuits to measure expectation values.	Can be a noiseless simulator (e.g., Qulacs) for algorithm development, or shot-based simulators/hardware for realistic results [25] [4].
Measurement Protocol	The strategy for estimating expectation values from quantum circuit executions.	Protocols include Pauli Averaging (direct measurement) or more advanced techniques like Classical Shadows [4] [5].
Eigensolver	A classical numerical routine that solves the generalized eigenvalue problem (\mathbf{H}C = \mathbf{S}CE).	Should include regularization (e.g., SVD thresholding) to handle ill-conditioned overlap matrices [26].

Implementing QSE: Methods and Real-World Applications in Molecular Modeling

Step-by-Step Workflow of a Quantum Subspace Expansion Calculation

Quantum Subspace Expansion (QSE) is a **post-processing variational algorithm** designed to find accurate ground and excited state energies of molecular systems on quantum computers. It operates by constructing a subspace of wavefunctions from a single, efficiently prepared reference state, often found using the Variational Quantum Eigensolver (VQE). A generalized eigenvalue problem is then solved entirely classically within this subspace to yield refined energy estimates [25] [3].

This protocol details the application of QSE for molecular energy calculations, a critical task in fields like **drug discovery and materials science**. The method is particularly suited for near-term quantum hardware, as it can mitigate errors and improve accuracy without a significant increase in

quantum circuit depth, exchanging this depth for additional measurements [3] [1].

Principle of the Method

The foundational principle of QSE is to expand a computed ground state into a subspace to obtain a better approximation of the true ground state and low-lying excited states.

Consider a molecular Hamiltonian (H) and an approximate ground state (|\Psi0\rangle) *obtained from a VQE calculation. The QSE method constructs a subspace of state vectors (*|\Psij^k\rangle) formed by applying excitation operators to the ground state wavefunction: [|\Psi{j}^{k}\rangle = ck^{\dagger}c{j}|\Psi0\rangle] where (ck^{\dagger}) and (c{j}) are the fermionic creation and annihilation operators, respectively [25].

Within this subspace, one solves the generalized eigenvalue problem: [H C = S C E] where:

- (H) is the subspace-projected Hamiltonian with matrix elements (H{jk}^{lm} = \langle \Psi0 | c{j}^\dagger c{l} H c{m}^{\dagger}c{k} | \Psi_0 \rangle).
- (S) is the overlap matrix with matrix elements (S{jk}^{lm} = \langle \Psi0 | c{j}^\dagger c{l} c{m}^{\dagger}c{k} | \Psi_0 \rangle).
- (C) is the matrix of eigenvectors.
- (E) is the vector of eigenvalues, which provides an estimate of the excited state energies and a refined ground state energy [25] [1].

The solution of this generalized eigenvalue equation is performed on a classical computer after the matrix elements of (H) and (S) have been measured on a quantum computer.

Experimental Protocols

This section provides a detailed, step-by-step workflow for performing a QSE calculation, using the methane molecule (CH_4) as a representative example [25].

Step 1: Define the Molecular System and Hamiltonian

The first step involves defining the molecular geometry and obtaining the corresponding electronic structure Hamiltonian.

- **Procedure**:
  - Define the molecular geometry, for example, using a Z-matrix.
  - Choose a basis set (e.g., `STO-3G`) and run a restricted Hartree-Fock (RHF) calculation using a classical computational chemistry package to obtain the molecular orbitals and electronic integrals.
  - Freeze core orbitals and apply point group symmetry to reduce the computational cost.
  - Generate the fermionic Hamiltonian in second quantization.
  - Map the fermionic Hamiltonian to a qubit Hamiltonian using an encoding scheme such as Jordan-Wigner or Bravyi-Kitaev.
  - Compress the qubit Hamiltonian by discarding terms with coefficients below a specified threshold (e.g., `abs_tol=1e-6`) to reduce the number of measurements required [25].
- **Example Parameters for Methane (CH_4)**:
  - **Z-matrix**: Defined with a C-H bond length of 1.083 Å and H-C-H angles of 109.471°.
  - **Basis Set**: `STO-3G`.
  - **Charge**: `0`.
  - **Frozen Orbitals**: `[0, 1, 2, 3, 7, 8]`.
  - **Qubit Encoding**: Jordan-Wigner.
  - **Hamiltonian Compression Tolerance**: `1e-6`. After compression, the Hamiltonian for this methane example contained 34 terms [25].

Step 2: Prepare the Approximate Ground State

Prepare a reference state (|\Psi_0\rangle) that has a non-zero overlap with the true ground state. This is typically done using the VQE algorithm.

- **Procedure**:
  - **Select an Ansatz**: Choose a parameterized quantum circuit (ansatz) that is expressive yet efficient. The Chemically Aware Unitary Coupled Cluster Singlet and Doubles (UCCSD) ansatz is a common choice for molecular systems.
  - **Optimize Parameters**: Use a classical optimizer to variationally minimize the expectation value of the Hamiltonian with respect to the ansatz parameters, (\langle \Psi0(\vec{\theta}) | H | \Psi0(\vec{\theta}) \rangle).

- The final optimized parameters ($\vec{\theta}_\text{opt}$) define the ground state preparation circuit [25].

- **Example Parameters for Methane**:
  - **Ansatz**: `FermionSpaceAnsatzChemicallyAwareUCCSD`.
  - **VQE Backend**: A noiseless state-vector simulator (e.g., `QulacsBackend`) can be used for initial testing and validation [25].

Step 3: Configure the Quantum Backend and Error Mitigation

Configure the settings for the quantum device or simulator that will execute the circuits.

- **Procedure**:
  - Select a quantum backend (e.g., a Quantinuum emulator, IBM's `ibm_cleveland`, or others accessible via cloud services).
  - Configure error mitigation techniques to improve result quality. For example, the Pauli Measurement Error Reduction by Symmetry Verification (PMSV) technique can be applied [25]. Other advanced error mitigation methods like dynamical decoupling and gate twirling are also used in practice [7] [6].

Step 4: Define and Measure QSE Matrix Elements

This is the most critical quantum step, where the matrix elements for the Hamiltonian (H) and overlap (S) matrices are measured.

- **Procedure**:
  - **Define Expansion Operators**: Choose a set of L operators ($\{O_i\}$) to define the subspace. Common choices include:
    - **Single Fermionic Excitations**: ($c_k^\dagger c_j$) for a selected set of spin orbitals (j) and (k) [25].
    - **Pauli String Operators**: Low-weight Pauli operators generated from the system's symmetry or the reference state [1].
    - **Krylov Vectors**: Powers of the Hamiltonian, (H^p), applied to the reference state [3] [1].
  - **Construct Measurement Circuits**: For each matrix element ($\mathcal{O}_{ij} = \langle \Psi_0 | O_i^\dagger H O_j | \Psi_0 \rangle$) and ($\mathcal{S}_{ij} = \langle \Psi_0 | O_i^\dagger O_j | \Psi_0 \rangle$), transform the expression into a linear combination of Pauli observables.
  - **Measure Expectation Values**: Execute the prepared quantum circuits on the backend and measure the expectation values of the required Pauli terms. This can be a significant bottleneck, as the number of measurements scales with (L^2).
  - **Advanced Measurement Techniques**: To overcome the measurement bottleneck, Informationally Complete (IC) measurements, such as **Classical Shadows (CS)**, can be employed. CS uses randomized measurements to reconstruct the state and estimate many observables simultaneously, significantly reducing the measurement overhead [1].

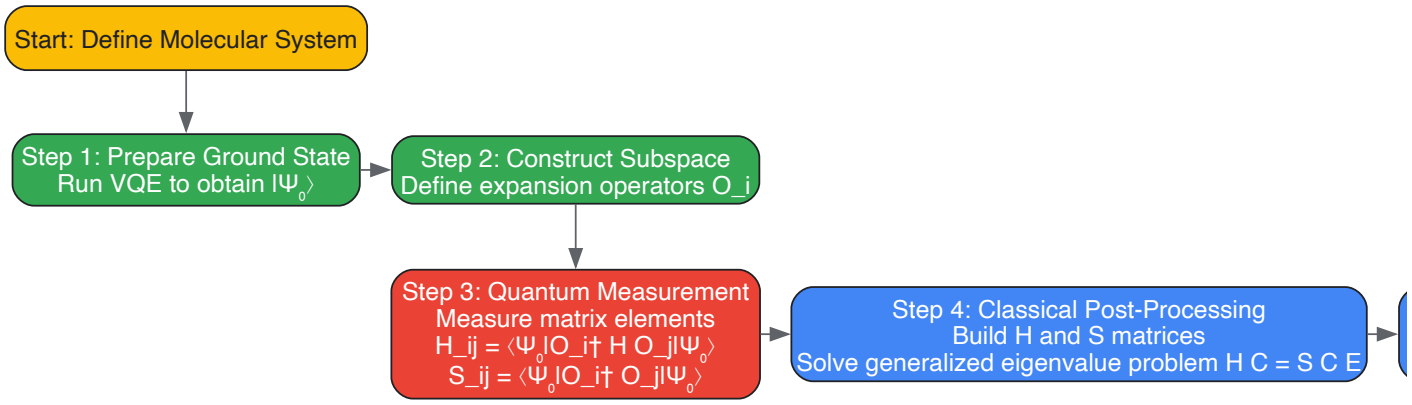
- **Example from Large-Scale Implementation**:
  - A recent 80-qubit QSE experiment used over (3×10^4) measurement basis randomizations per circuit, evaluating ($\mathcal{O}(10^{14})$) Pauli traces, showcasing the scale required for large systems [1].

Step 5: Classically Solve the Generalized Eigenvalue Problem

The final step is entirely classical and involves solving for the energies of the system within the constructed subspace.

- **Procedure**:
  - Assemble the estimated (H) and (S) matrices from the measured data.
  - Solve the generalized eigenvalue problem ($H C = S C E$) for the eigenvalues (E) and eigenvectors (C).
  - **Address Numerical Instability**: The overlap matrix (S) can be ill-conditioned due to noise or non-orthogonal expansion operators. Mitigation strategies include:
    - **Truncation**: Discarding eigenvectors corresponding to singular values of (S) below a certain threshold [1].
    - **Partitioned QSE (PQSE)**: An iterative algorithm that breaks down a single Krylov basis and recombines it to ensure stability against statistical noise [3].
    - **Constrained Optimization**: Reformulating the problem as a constrained optimization to avoid direct inversion of (S) [1].
  - The lowest eigenvalue is the refined ground-state energy, and the higher eigenvalues correspond to excited states.

The following diagram illustrates the complete QSE workflow, integrating both quantum and classical processing stages.



The Scientist's Toolkit

This section lists key resources, software, and hardware required to implement the QSE protocol.

- **Software and Libraries**
  - **InQuanto**: A computational chemistry library for quantum computers that provides high-level tools for implementing algorithms like VQE and QSE [25].
  - **PySCF**: A classical computational chemistry package used for initial molecular Hartree-Fock calculations and integral generation [25] [6].
  - **Qiskit / Tangelo**: Quantum computing software development kits (SDKs) that provide tools for building and executing quantum circuits, as well as implementations of algorithms like SQD [7] [6].
  - **Open Source Pipelines**: Frameworks like `FreeQuantum` provide modular, open-source pipelines for binding energy calculations, designed to integrate quantum computing subroutines [28].
- **Quantum Backends**
  - **Quantinuum Emulators**: High-performance emulators accessible via cloud, often used for algorithm development and testing [25].
  - **IBM Quantum Systems**: Superconducting quantum computers (e.g., `ibm_cleveland`, `ibm_kyiv`) that are available for cloud-based research and have been used for demonstrations of hybrid quantum-classical algorithms like DMET-SQD on molecules such as cyclohexane [7] [6].
- **Key Theoretical Components**
  - **Reference State ($|\Psi_0\rangle$)**: The initial state, typically the VQE-optimized UCCSD or pUCCD ansatz wavefunction [25] [29].
  - **Expansion Operators ($\{O_i\}$)**: A set of operators used to create the subspace, such as fermionic excitation operators or low-weight Pauli strings [25] [1].
  - **Classical Shadows**: An informationally complete measurement technique that dramatically reduces the number of measurements needed for QSE on large systems [1].

Anticipated Results

When applied to a molecular system like methane (CH_4) in a minimal basis (STO-3G), the QSE protocol is expected to produce a set of energies corresponding to the ground and excited states.

- **Output Data**:
  - A refined ground-state energy lower than the initial VQE result.
  - A set of excited-state energies.
  - The energy difference between the ground state and the first excited state is a key spectroscopic observable.
- **Benchmarking and Validation**:
  - The results should be benchmarked against high-accuracy classical methods like CCSD(T) or Full Configuration Interaction (FCI) where computationally feasible.
  - For the methane dimer, a 36-qubit QSE simulation achieved deviations within **1.000 kcal/mol** from leading classical methods like CCSD(T), demonstrating chemical accuracy for non-covalent interactions [6].
  - Another study on cyclohexane conformers using a hybrid DMET-SQD method produced energy differences within **1 kcal/mol** of classical benchmarks [7].
- **Troubleshooting**:

- **Ill-conditioned Overlap Matrix (S)**: This is a common issue. Use the regularization techniques described in Section 3.5. The Partitioned QSE (PQSE) algorithm is specifically designed to alleviate this numerical instability [3].
- **Large Statistical Errors**: Increase the number of measurement shots or employ advanced techniques like Classical Shadows to reduce variance [1].
- **Inaccurate Initial State**: Ensure the VQE has converged to a state with high fidelity relative to the true ground state. A poor initial state will lead to a poor QSE result.

Method Notes

Limitations

The standard QSE algorithm has two primary limitations:

- **Measurement Overhead**: The number of measurements required to build the (H) and (S) matrices scales quadratically with the number of expansion operators (L), which can be prohibitive for large subspaces [1].
- **Numerical Instability**: The overlap matrix (S) is often singular or ill-conditioned, making the solution of the generalized eigenvalue problem sensitive to noise and statistical errors [3] [1].

Method Modifications

Several modifications have been developed to address these limitations:

- **Partitioned QSE (PQSE)**: An iterative generalization that breaks a large Krylov subspace into a sequence of smaller, interconnected subspaces. This improves numerical stability in the presence of finite sampling noise without requiring parameter tuning [3].
- **QSE with Classical Shadows (CS)**: Uses randomized measurements to efficiently estimate the required matrices, enabling the application of QSE to very large systems (e.g., 80 qubits) [1].
- **Generalized QSE (GQSE)**: Extends the framework to a wider class of expansion operators and can be used for error mitigation [3].

Applications in Drug Development

Accurate molecular energy calculations are the foundation of understanding molecular interactions in drug discovery. QSE and related hybrid quantum-classical methods are being developed to tackle challenges beyond the reach of classical computers.

- **Binding Energy Calculations**: Predicting how tightly a drug candidate binds to its protein target is crucial. A computational pipeline like `FreeQuantum` is a blueprint for integrating quantum computing to achieve quantum advantage in these calculations. For a ruthenium-based anticancer drug, such a pipeline predicted a binding free energy that differed significantly from classical force field predictions, highlighting the impact of quantum-level accuracy [28].
- **Simulating Biologically Relevant Molecules**: Hybrid methods like DMET-SQD have been used to simulate the conformers of cyclohexane, a common benchmark, and show potential for simulating protein-drug interactions. These methods leverage current quantum hardware (using 27-32 qubits) to handle electronically complex fragments of larger molecules, achieving chemical accuracy [7].
- **Non-Covalent Interactions**: Accurately modeling weak interactions (e.g., hydrogen bonds, dispersion forces) is essential for drug binding. Quantum-centric simulations using SQD have successfully simulated the potential energy surfaces of the water and methane dimers, registering deviations within 1.000 kcal/mol from classical gold-standard methods [6].

Leveraging Classical Shadows for Large-Scale, Efficient Implementation

Classical shadows constitute a powerful protocol for estimating properties of an unknown quantum state from a minimal number of measurements, making them particularly valuable for near-term quantum hardware [30]. Unlike full quantum state tomography, this quantum-classical protocol enables many predictions with a high success probability using comparatively few quantum measurements [30]. The technique involves performing randomized measurements on copies of a quantum state, processing the results classically to build a "shadow" representation that can predict many properties with rigorous performance guarantees.

Quantum Subspace Expansion (QSE) represents a promising pathway for performing spectral calculations on quantum processors, particularly for molecular energy calculations [1]. This post-VQE methodology projects the quantum Hamiltonian into a subspace built upon a prepared "root state," which is then diagonalized classically to extract ground and excited state energies. However, traditional QSE implementations face a substantial measurement overhead, which has hindered scalability beyond small proof-of-principle demonstrations [1].

The integration of classical shadows with QSE creates a powerful synergy for large-scale quantum computational chemistry. Informationally complete (IC) measurements, such as classical shadows, can estimate all necessary observables from a single set of measurement samples, dramatically reducing the resource requirements for QSE [1]. This combination has recently enabled the first large-scale implementation of QSE, with demonstrations on systems of up to 80 qubits [1].

Theoretical Foundations

Classical Shadow Formalism

The classical shadow protocol begins with a quantum state ρ. Rather than performing full state tomography, the protocol collects measurement outcomes from randomized measurements. For each copy of ρ, a random unitary U is selected from a fixed ensemble (typically random Clifford circuits or random Pauli measurements), transforming the state to $U\rho U^\dagger$. The system is then measured in the computational basis, yielding a bitstring $|\hat{b}\rangle$. The classical snapshot of the state is reconstructed as $U^\dagger|\hat{b}\rangle\langle\hat{b}|U$ [30] [31].

After collecting many such snapshots, the classical shadow representation is built as the average of these snapshots. This ensemble of classical snapshots can then be used to estimate expectation values of various observables $O_1, O_2, \ldots, O_m$ through classical post-processing. The remarkable efficiency of this protocol stems from its ability to predict many properties from a single set of measurements, with proven bounds on sample complexity [30].

Recent advancements have extended the classical shadows formalism to noisy quantum scenarios. Modified estimators have been developed that remain unbiased even in the presence of various known noise channels, including depolarizing noise and amplitude damping [30]. This robustness is particularly valuable for implementations on current noisy intermediate-scale quantum (NISQ) devices.

Quantum Subspace Expansion Framework

Quantum Subspace Expansion provides a systematic approach to improve upon an initial quantum state preparation. Consider a scenario where a root state ρ_0—ideally close to the true ground state $|\psi_{gs}\rangle$—can be prepared on a quantum computer. QSE expands this state into a subspace spanned by Hermitian expansion operators $\{\sigma_1, \sigma_2, \ldots, \sigma_l\}$ according to:

$$\rho_{SE}(\vec{c}) = W^\dagger\rho_0 W/\mathrm{Tr}[W^\dagger\rho_0 W], \text{ with } W = \Sigma_i c_i\sigma_i$$

The expectation value of an observable O within this expanded subspace becomes:

$$\mathrm{Tr}[O\rho_{SE}(\vec{c})] = \Sigma_{i,j}\, c_i^* c_j\, \mathcal{O}_{ij} \,/\, \Sigma_{i,j}\, c_i^* c_j\, \mathcal{S}_{ij}$$

where $\mathcal{O}_{ij} = \mathrm{Tr}[\sigma_i^\dagger\rho_0\sigma_j O]$ and $\mathcal{S}_{ij} = \mathrm{Tr}[\sigma_i^\dagger\rho_0\sigma_j]$ [1].

For ground state energy calculations, the optimal coefficients $\vec{c}$ are found by solving the generalized eigenvalue problem $\mathcal{H}\vec{c} = \lambda\mathcal{S}\vec{c}$, where $\mathcal{H}$ is the subspace-projected Hamiltonian and $\mathcal{S}$ is the overlap matrix. The smallest eigenvalue λ provides the best approximation to the ground state energy within the subspace [1].

Table 1: Key Mathematical Components in Quantum Subspace Expansion

Component	Mathematical Expression	Physical Significance
Subspace State	$\rho_{SE}(\vec{c}) = W^\dagger\rho_0 W/\mathrm{Tr}[W^\dagger\rho_0 W]$	Expanded state beyond initial approximation
Expansion Operator	$W = \Sigma_i c_i\sigma_i$	Linear combination of basis operations
Projected Hamiltonian	$\mathcal{H}_{ij} = \mathrm{Tr}[\sigma_i^\dagger\rho_0\sigma_j H]$	Hamiltonian restricted to subspace
Overlap Matrix	$\mathcal{S}_{ij} = \mathrm{Tr}[\sigma_i^\dagger\rho_0\sigma_j]$	Metric tensor for non-orthogonal basis

Resource-Efficient Implementation Strategies

Advanced Grouping and Measurement Allocation

Significant improvements in measurement efficiency can be achieved through sophisticated grouping strategies. The Resource-Optimized Grouping Shadow (ROGS) algorithm employs a three-stage process to minimize both the number of measurements and the number of unique quantum circuits required [32]:

- **Max-Min Grouping of Pauli Observables**: This stage begins with applying a Minimum Clique Cover (MCC) algorithm to the qubit-wise commuting (QWC) graph of Pauli observables in the Hamiltonian. The groups are then expanded to include the maximal number of additional observables while maintaining the QWC property, effectively minimizing the number of groups while maximizing the size of each group [32].
- **Measurement Resource Allocation via Convex Optimization**: This innovative approach formulates shot allocation as a convex optimization problem. The objective function, based on confidence bounds from mean estimation techniques like Hoeffding's inequality, minimizes the estimation error for the Hamiltonian energy given a fixed total measurement budget [32].
- **Measurement and Estimation**: After determining the optimal allocation, the corresponding Pauli-basis measurements are performed on the quantum state, with results used to estimate expectation values of the target Hamiltonian [32].

This approach is particularly valuable because it addresses both the number of measurements and the number of distinct quantum circuits—the latter being a critical but often overlooked cost factor in quantum computing due to compilation and loading overheads [32].

Constrained Optimization for Numerical Stability

A major challenge in practical QSE implementations is the numerical instability that arises from inverting the overlap matrix $\mathcal{S}$ under shot noise and experimental imperfections. Traditional regularization approaches, such as discarding dimensions corresponding to small singular values, can be effective but lack rigorous statistical treatment [1].

Recent work has introduced a constrained optimization formulation that avoids direct matrix inversion altogether. This approach defines an estimator $\hat{H}(\vec{c})$ for the energy $\mathrm{Tr}[H\rho SE(\vec{c})]$ along with its statistical error $\varepsilon\hat{H}(\vec{c})$. The problem then becomes finding the lowest possible subspace energy subject to a maximum tolerated statistical error, effectively trading potential bias for controllable variance [1].

This methodology naturally handles the covariances between matrix entries that arise when using informationally complete measurements, providing more reliable error bars for the resulting energy estimates [1].

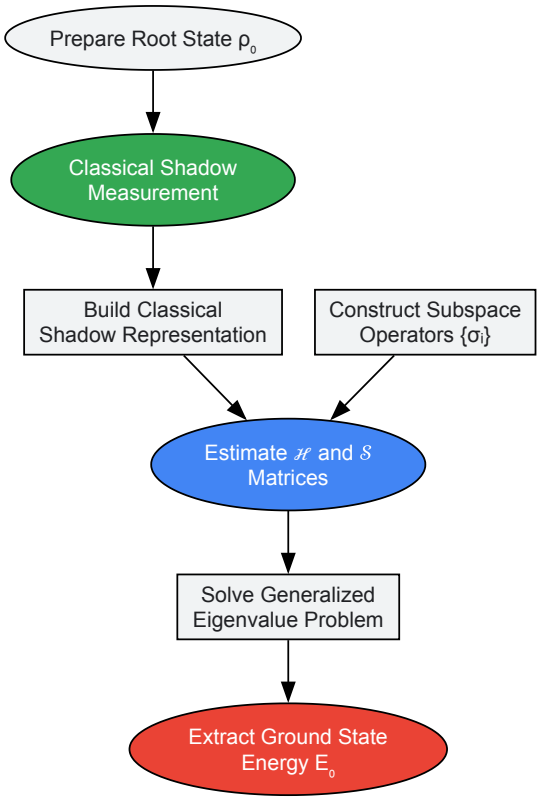
Table 2: Resource Optimization Techniques for Classical Shadows

Technique	Key Innovation	Advantage
ROGS Algorithm [32]	Overlapped grouping with convex optimization	Minimizes both measurement count and circuit count
Constrained Optimization QSE [1]	Formulates QSE as constrained optimization problem	Avoids numerical ill-conditioning, provides reliable error estimates
Noise-Robust Shadows [30]	Modified estimators for known noise channels	Maintains unbiased estimation under realistic noise conditions
Matrix Product Operator Learning [31]	Sequential tensor optimization via DMRG-inspired approach	Provably efficient for short-range correlated states

Experimental Protocols and Workflows

Large-Scale QSE with Classical Shadows Protocol

The following workflow diagram illustrates the complete protocol for implementing quantum subspace expansion with classical shadows:



Protocol Steps:

- **Root State Preparation**: Prepare the initial quantum state ρ_0 on the quantum processor. This state should have non-zero overlap with the true ground state and can be generated using a variational quantum circuit or other state preparation method.
- **Classical Shadow Measurement**: Perform randomized measurements on the root state. For large-scale implementations, this typically involves over 3×10^4 measurement basis randomizations per circuit [1]. The specific measurement bases are determined by the grouping and allocation strategy (e.g., ROGS) to maximize efficiency.
- **Classical Shadow Construction**: Process the measurement outcomes to build the classical shadow representation. This involves reconstructing classical snapshots for each measurement and storing them efficiently for subsequent processing.
- **Subspace Operator Selection**: Choose appropriate expansion operators $\{\sigma_i\}$ that define the subspace. Common choices include powers of the Hamiltonian H^p or low-weight Pauli operators. The selection should balance expressiveness with practical computational constraints.
- **Matrix Estimation**: Use the classical shadow to estimate all matrix elements $\mathcal{H}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j H]$ and $\mathcal{S}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j]$ simultaneously. For large systems, this may involve evaluating $\mathcal{O}(10^{14})$ Pauli traces [1].
- **Constrained Optimization**: Solve the generalized eigenvalue problem using the constrained optimization approach to avoid numerical instability while accounting for statistical errors in the matrix estimates.
- **Energy Extraction**: Extract the ground state energy estimate from the smallest generalized eigenvalue, along with rigorous statistical error bars derived from the constrained optimization framework.

Molecular Dynamics Integration

Classical shadows have also been successfully integrated with quantum molecular dynamics simulations. In Quantum Car-Parrinello Molecular Dynamics (QCPMD), classical shadows enable simultaneous estimation of forces on all nuclei, significantly improving resource efficiency as system size increases [33]. The workflow involves:

- Preparing a parameterized quantum state $|\Psi(\theta)\rangle = U(\theta)|\Psi_0\rangle$
- Using classical shadows to estimate the energy $L(R,\theta) = \mathrm{Tr}[H(R)|\Psi(\theta)\rangle\langle\Psi(\theta)|]$
- Computing forces $F_{I,a}(R,\theta) = -\partial L(R,\theta)/\partial R_{I,a}$ for all nuclei simultaneously
- Updating nuclear positions and circuit parameters in parallel

This approach has been demonstrated for H_2 molecules, showing promising results for efficient AIMD simulation on near-term quantum devices [33].

The Scientist's Toolkit

Table 3: Essential Research Reagents for QSE with Classical Shadows

Resource/Reagent	Function/Role	Implementation Notes
Classical Shadows Framework	Informationally complete measurement protocol	Core engine for efficient observable estimation [30] [1]
Qubit-Wise Commuting Grouping	Measurement optimization	Groups simultaneously measurable observables [32]
Constrained Optimization Solver	Numerical stability	Avoids ill-conditioning in matrix inversion [1]
Matrix Product Operator (MPO) Representation	State approximation for large systems	Efficient representation for short-range correlated states [31]
Resource-Optimized Grouping Shadow (ROGS)	Measurement allocation	Optimizes shot distribution across groups [32]
Quantum Subspace Expansion Library	Subspace construction and diagonalization	Implements generalized eigenvalue solver with error handling

Application to Molecular Energy Calculations

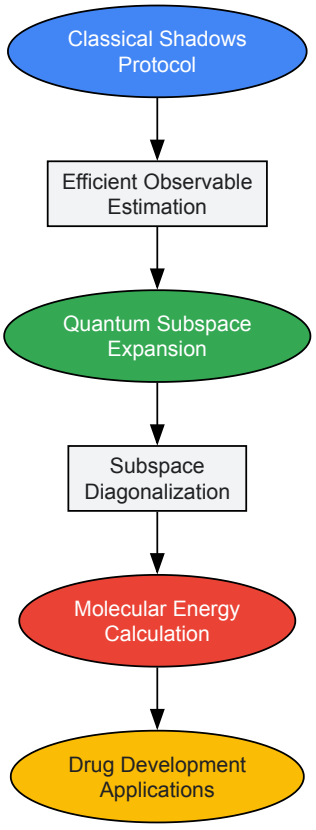
The integration of classical shadows with quantum subspace expansion has demonstrated remarkable success in molecular energy calculations. Recent experimental implementations have achieved accurate ground state energy recovery across system sizes of up to 80 qubits, representing one of the most significant experimental realizations of classical shadows to date [1].

In the context of molecular systems, this approach enables the efficient calculation of potential energy surfaces—a fundamental requirement for molecular dynamics simulations and drug development applications. By significantly reducing the measurement overhead required for accurate

energy estimation, the classical shadows approach makes practical quantum computational chemistry more accessible on current hardware.

For drug development professionals, this methodology offers a pathway to more efficient in silico screening of drug candidates by providing accurate molecular energy calculations at scales previously impractical with classical computational methods. The resource efficiency gains are particularly valuable for simulating the large molecular systems relevant to pharmaceutical applications.

The workflow below illustrates the logical relationship between classical shadows, quantum subspace expansion, and their application in molecular energy calculations:



As quantum hardware continues to advance, the combination of classical shadows and quantum subspace expansion represents a promising framework for tackling increasingly complex molecular systems, potentially revolutionizing computational approaches to drug discovery and materials design.

Quantum subspace expansion (QSE) has emerged as a powerful post-processing technique to enhance the accuracy of ground state energy calculations on quantum computers. By constructing a linear subspace around a pre-prepared quantum state, QSE enables the recovery of accurate energies and other molecular properties, effectively mitigating errors from noise or approximate circuit preparations [1]. This method is particularly valuable for near-term quantum devices, where circuit depth and hardware noise remain significant constraints. Framed within the broader thesis of advancing quantum computational chemistry, these application notes detail the experimental protocols and quantitative performance of QSE methods, providing researchers and drug development professionals with practical guidelines for their implementation in molecular electronic structure problems.

Theoretical Framework and Key Concepts

Quantum subspace expansion operates on the principle of constructing a classically tractable linear subspace from a quantumly prepared root state, denoted as ρ_0 [1]. A set of Hermitian expansion operators $\{\sigma_i\}$ is used to span this subspace, creating states of the form $\rho_{SE}(\vec{c}) = W^\dagger \rho_0 W / \mathrm{Tr}[W^\dagger \rho_0 W]$, where $W = \Sigma_i c_i \sigma_i$ [1]. The optimal energy within this subspace is found by solving the generalized eigenvalue problem $\square \vec{c} = \lambda \mathcal{S} \vec{c}$, where $\square$ is the subspace-projected Hamiltonian and $\square$ is the overlap matrix [1]. The lowest eigenvalue λ provides the refined ground state energy estimate.

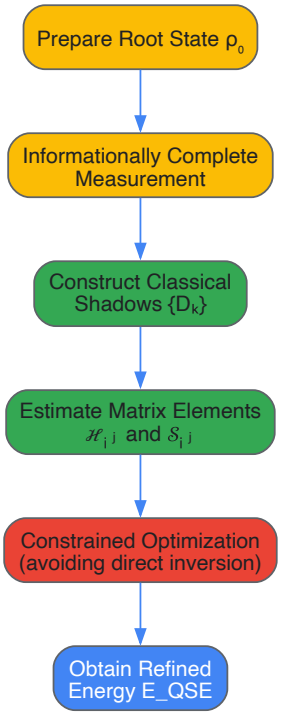
A particularly effective choice for expansion operators are powers of the Hamiltonian, H^p, which generates a Krylov subspace. The overlap of the states $H^p \rho_0 H^{p\dagger}$ with the true ground state increases exponentially with p, provided ρ_0 has non-zero initial overlap [1]. This approach forms the basis for rigorous complexity bounds and convergence guarantees in molecular electronic structure calculations [9].

Experimental Protocols and Methodologies

Quantum Subspace Expansion with Classical Shadows

The integration of informationally complete (IC) measurements, specifically classical shadows, with QSE significantly reduces the measurement overhead traditionally associated with estimating matrix elements [1].

- **Protocol Workflow**:
  - **State Preparation**: Prepare the root state ρ_0 on the quantum processor using an efficient quantum circuit (e.g., pUCCD or LUCJ ansatz) [29] [6].
  - **Informationally Complete Measurement**: Perform randomized measurements on multiple copies of ρ_0. The POVM operators $\{M_k\}$ satisfy $\Sigma_k M_k = \mathbb{1}$, enabling the reconstruction of expectation values for any observable [1].
  - **Classical Shadow Reconstruction**: For each measurement outcome $k^{(s)}$, construct the dual operator D_k. The expectation value of an observable O is then estimated as the mean of $Tr[OD_k]$ over all shots [1].
  - **Constrained Optimization**: Reformulate the QSE generalized eigenvalue problem as a constrained optimization to avoid numerical instabilities from direct matrix inversion. This approach provides rigorous statistical error estimates and allows trading potential bias for variance [1].



Sample-Based Quantum Diagonalization (SQD)

SQD is a quantum-centric workflow that leverages quantum sampling and classical high-performance computing (HPC) for electronic structure calculations [6].

- **Protocol Workflow**:
  - **Ansatz Preparation and Sampling**: Prepare a wavefunction using an efficient ansatz (e.g., Local Unitary Coupled Cluster - LUCJ) on a quantum computer. Sample electronic configurations from this state [6].
  - **Configuration Recovery and Post-processing**: Use classical HPC resources to post-process quantum measurements against known symmetries to recover configurations that may have been corrupted by device noise [6].
  - **Classical Diagonalization**: Construct the Hamiltonian matrix within the subspace spanned by the recovered configurations. Diagonalize this matrix classically to obtain refined energies and states [6].
  - **Energy Extrapolation (Optional)**: Employ total energy extrapolation using Hamiltonian variance to produce accurate energy estimates, particularly useful for mapping potential energy surfaces [6].

Table 1: Key Research Reagents and Computational Resources for QSE Experiments

Resource / Reagent	Type / Specification	Function in Protocol
Classical Shadows [1]	Informationally Complete POVM	Enables efficient estimation of multiple observables from a single set of quantum measurements, drastically reducing measurement overhead.
pUCCD Ansatz [29]	Paired Unitary Coupled Cluster Doubles	Provides an efficient, low-depth parameterized quantum circuit for preparing initial root states in the seniority-zero subspace.

Resource / Reagent	Type / Specification	Function in Protocol
LUCJ Ansatz [6]	Local Unitary Coupled Cluster	Approximates UCCSD with reduced circuit depth, enabling feasible execution on NISQ devices for configuration sampling in SQD.
S-CORE Procedure [6]	Self-Consistent Configuration Recovery	Classical post-processing algorithm that mitigates quantum hardware noise by recovering corrupted electronic configurations.
Constrained Optimizer [1]	Classical Algorithm	Solves the QSE problem while avoiding numerical instabilities from ill-conditioned overlap matrices and provides statistical error bars.

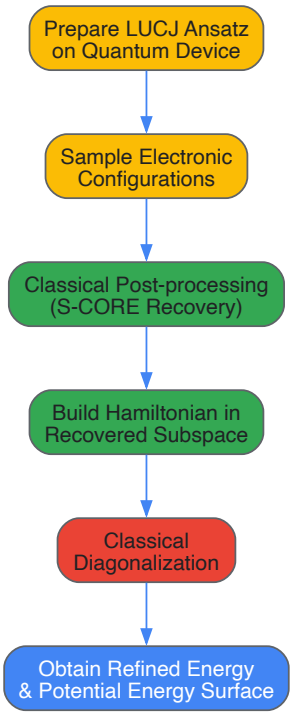
Performance Data and Benchmarking

Large-scale experimental implementations demonstrate the scalability and accuracy of QSE methods. A key experiment successfully applied QSE with classical shadows to study the quantum phase transition of a spin model, achieving accurate ground state energy recovery for systems of up to **80 qubits** [1]. This implementation required over **30,000 measurement basis randomizations** per circuit and the evaluation of approximately **10^{14} Pauli traces**, representing one of the most significant realizations of classical shadows to date [1].

In applications closer to chemistry, the SQD method has been used to simulate non-covalent interactions in molecular dimers. For the methane dimer (a 36-qubit system), SQD energies agreed nearly exactly with CASCI results and deviated from the gold-standard CCSD(T) by less than **1.0 kcal/mol** in the equilibrium region, achieving chemical accuracy [6]. The method has also demonstrated scalability, successfully diagonalizing subspaces of up to **2.49×10^8 configurations** [6].

Table 2: Quantitative Performance of Quantum Subspace Methods

Method / System	System Size (Qubits)	Key Metric	Reported Performance
QSE + Classical Shadows [1]	Up to 80	Energy Accuracy	Accurate ground state energy recovery observed.
QSE + Classical Shadows [1]	Up to 80	Measurement Scale	~30,000 randomizations; evaluation of ~10^{14} Pauli traces.
SQD (Methane Dimer) [6]	36	Accuracy vs. CCSD(T)	Deviation within 1.0 kcal/mol (chemical accuracy).
SQD (Methane Dimer) [6]	54	Subspace Dimension	Diagonalization of 2.49×10^8 configurations.
Hybrid pUNN [29]	N/A	Accuracy vs. CCSD(T)	Near-chemical accuracy for N_2, CH_4; validated on superconducting hardware.



Application in Drug Development and Materials Design

The accurate simulation of non-covalent interactions is critical in drug discovery and materials science, influencing processes like protein-ligand binding, protein folding, and molecular assembly [6]. Quantum subspace methods, by achieving chemical accuracy for interaction energies, offer a path toward more reliable *in silico* drug screening and materials design.

Theoretical analyses have established that for specific challenges, such as transition-state mapping in chemical reactions, adaptive quantum subspace selection can achieve an exponential reduction in required measurements compared to uniform sampling [9]. This efficiency gain is particularly valuable for modeling complex reactions in battery electrolyte development or catalytic processes. The QCSC paradigm, which combines quantum processors with classical HPC, enables the study of these problems at a scale heretofore out of reach for standalone quantum computers [6].

Quantum subspace expansion represents a rigorous and powerful framework for molecular electronic structure calculations on near-term quantum processors. Protocols such as QSE with classical shadows and sample-based quantum diagonalization have demonstrated scalability to dozens of qubits and the ability to achieve chemical accuracy in benchmark systems. By providing mitigation for hardware noise and algorithmic approximations, these methods offer a promising route to practical quantum advantage in computational chemistry. Their successful application to problems like non-covalent interaction energy calculation lays a solid foundation for future applications in drug development and materials design, where high accuracy and reliability are paramount.

This document provides detailed application notes and protocols for employing **Quantum Subspace Expansion (QSE)** in two critical research domains: simulating complex electrochemical reactions in lithium-ion battery electrolytes and predicting drug-target interactions (DTI) for pharmaceutical discovery. The content is framed within a broader thesis on advancing molecular energy calculations, demonstrating how QSE can transcend its traditional use in quantum chemistry to address pressing challenges in materials science and bioinformatics. By leveraging quantum processors to create a subspace of excited states around a computed ground state, QSE enables the calculation of key electronic properties with enhanced accuracy, facilitating the prediction of reaction pathways in electrolytes and the binding affinities between drugs and their protein targets. [25] [1]

The integration of these seemingly disparate fields is rooted in a shared computational foundation. Accurate spectral calculations and the mitigation of errors from noisy quantum computations are universal needs. This note outlines how QSE methodologies, particularly when combined with classical shadow techniques for scalability, can be adapted to model the electronic transitions in battery electrolyte molecules and the interaction landscapes between small drug molecules and large protein targets, thereby providing a unified tool for next-generation simulation-driven research and development. [1]

Quantum Subspace Expansion (QSE) Fundamentals

Quantum Subspace Expansion (QSE) is a post-processing technique that enhances the results from a variational quantum eigensolver (VQE). The core principle involves taking a pre-prepared ground state wavefunction, $|\Psi_0\rangle$, and constructing a subspace of state vectors by applying a set of excitation operators. [25] For a molecule, these are typically fermionic excitation operators:

[\left|\Psi_{j}^{k}\right\rangle = c_k^{\dagger}c_{j}\left|\Psi_0\right\rangle]

where (*ck^{\dagger}) and (c*{j}) are the fermionic creation and annihilation operators, respectively. [25] A generalized eigenvalue problem is then solved within this subspace:

[HC=SCE]

where H is the matrix of the Hamiltonian in the subspace, S is the overlap matrix, C is the matrix of eigenvectors, and E is the vector of eigenvalues providing estimates for the ground and excited state energies. [25] This approach provides a powerful framework for noise-agnostic error mitigation and accessing excited states from an approximate ground state. [1]

Relevance to Battery and Drug Discovery Simulations

The applicability of QSE to both battery and drug discovery simulations stems from its ability to provide precise electronic structure information.

- **Battery Electrolyte Reactions:** The stability and reactivity of electrolyte molecules at the electrode interface are governed by their redox properties and the energy landscapes of possible decomposition pathways. QSE can be used to calculate the excited states and ionization potentials of these molecules with higher accuracy than ground-state methods alone, enabling the *in silico* prediction of reaction products and degradation mechanisms. [34]
- **Drug-Target Interactions (DTI):** The binding affinity between a drug and its protein target is a critical determinant of efficacy. From a quantum chemistry perspective, this interaction involves complex electronic rearrangements. While classical machine learning models like SaeGraphDTI use sequence and graph data for prediction, a QSE-based approach can provide a more fundamental understanding by calculating interaction energies and electronic transition states for drug-target complexes, potentially informing and validating classical models. [35] [36]

Table 1: Performance Metrics of QSE and Comparative DTI Models

Model / Method	Key Metric	Performance / Value	Dataset / System	Notes
QSE with Classical Shadows [1]	System Size Scalability	Up to 80 qubits	Spin model with three-body interactions	Enables large-scale simulation; over $3x10^4$ measurement basis randomizations.
SaeGraphDTI (DTI Prediction) [36]	AUC	>0.98 (Davis)	Davis, E, GPCR, IC	State-of-the-art classical deep learning model for DTI.
	AUPR	>0.71 (Davis)		
QSE (General Framework) [25]	Energy Refinement	Provides refined ground state and excited state energies	Molecular systems (e.g., Methane)	Mitigates noise and approximations from initial VQE calculation.

Detailed Experimental Protocols

Protocol 1: QSE for Molecular Energy Calculations

This protocol outlines the steps for performing a Quantum Subspace Expansion calculation to obtain ground and excited state energies for a molecule, such as a battery electrolyte component or a small drug molecule.

1. System Definition and Hamiltonian Preparation

- **Objective:** Generate the fermionic Hamiltonian of the target system.
- **Procedure:**
  - Define the molecular geometry (e.g., using a Z-matrix or Cartesian coordinates).
  - Perform a classical restricted Hartree-Fock (RHF) calculation using a quantum chemistry driver (e.g., via PySCF).
  - Extract the fermionic Hamiltonian, Fock space, and Hartree-Fock state.
  - Encode the fermionic Hamiltonian into a qubit Hamiltonian using a mapping such as Jordan-Wigner or Bravyi-Kitaev.
  - Compress the qubit Hamiltonian by removing terms with coefficients below a specified tolerance (e.g., `abs_tol=1e-6`) to reduce quantum resource requirements. [25]

2. Ground State Preparation via VQE

- **Objective:** Prepare an approximate ground state wavefunction, $|\Psi_0\rangle$.
- **Procedure:**
  - Select a parameterized ansatz, such as the Chemically Aware Unitary Coupled Cluster Singles and Doubles (UCCSD).
  - Choose a quantum backend (simulator or hardware).
  - Run the VQE algorithm to minimize the expectation value of the qubit Hamiltonian with respect to the ansatz parameters.

- Record the optimized parameters that define the ground state wavefunction. [25]

3. Quantum Subspace Expansion

- **Objective:** Construct and solve the generalized eigenvalue problem in the QSE subspace.
- **Procedure:**
  - **Define Expansion Operators:** Select a set of operators $\{\sigma_i\}$ to define the subspace. Common choices include single-electron excitation operators ($c_k^\dagger c_j$) or powers of the Hamiltonian (H^p) for Krylov subspace methods. [25] [1]
  - **Construct Matrices:** Calculate the matrix elements of the Hamiltonian (H) and overlap (S) matrices in the subspace.
    - $H_{jk}^{lm} = \langle \Psi_{ij}^l | \hat{H} | \Psi_{k}^m \rangle = \langle \Psi_0 | c_j^\dagger c_l \hat{H} c_m^\dagger c_k | \Psi_0 \rangle$
    - $S_{jk}^{lm} = \langle \Psi_{ij}^l | \Psi_{k}^m \rangle = \langle \Psi_0 | c_j^\dagger c_l c_m^\dagger c_k | \Psi_0 \rangle$
  - **Measurement:** For each matrix element, transform the corresponding fermionic operators into Pauli gates via the chosen mapping. Prepare the ground state circuit with the optimized VQE parameters, apply the gate sequence for the specific matrix element, and perform measurements to obtain the expectation values. On hardware, employ error mitigation techniques like probabilistic error cancellation (PMSV). For large systems, use Informationally Complete (IC) measurements like classical shadows to reduce measurement overhead. [25] [1]
  - **Classical Solution:** On a classical computer, solve the generalized eigenvalue problem ($HC=SCE$). The eigenvalues E provide the QSE-refined ground state energy and excited state energies. [25]

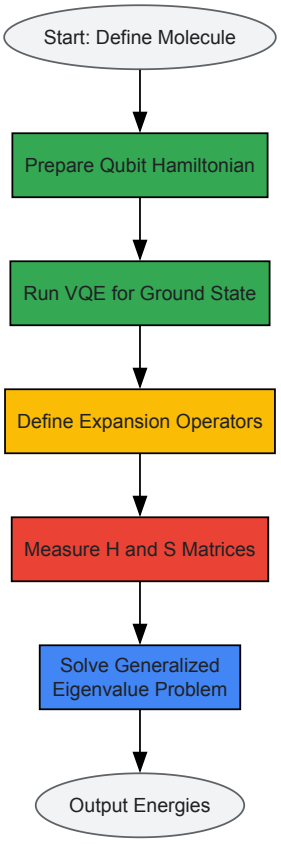


Diagram 1: QSE Computational Workflow

Protocol 2: Integrating QSE with DTI Prediction Pipelines

This protocol describes how insights from QSE calculations can be integrated with classical deep learning models for enhanced drug-target interaction prediction.

1. Target and Ligand Selection

- **Objective:** Identify a protein target and a set of small molecule ligands (drug candidates) for study.
- **Procedure:** Curate datasets from public databases such as Davis, BindingDB, or the Protein Data Bank (PDB). Pre-process the data, including standardizing SMILES strings for drugs and amino acid sequences for targets. [36]

2. Classical DTI Model Feature Extraction

- **Objective:** Generate initial feature representations for the drugs and targets.

- **Procedure:**
  - **Drug Feature Extraction:** Use a sequence attribute extractor on the SMILES string. This involves encoding each character, padding/trimming to a fixed length, and passing through an embedding layer and convolutional layers to produce a fixed-dimensional feature vector. [36]
  - **Target Feature Extraction:** Similarly, process the amino acid sequence of the target protein through an embedding layer and convolutional layers to extract a fixed-dimensional feature vector. [36]

3. QSE-Based Energy Calculation for Key Complexes

- **Objective:** Obtain quantum-mechanically refined interaction energies for a subset of drug-target complexes to serve as high-fidelity labels or validation.
- **Procedure:**
  - For a select number of complexes (e.g., those with uncertain predictions from the classical model), generate a 3D structure of the binding pose.
  - Define the quantum chemical system for the binding site, including key residues and the bound ligand.
  - Perform a QSE calculation (as per Protocol 1) to compute the interaction energy with high accuracy, going beyond classical force fields or DFT.

4. Model Integration and Prediction

- **Objective:** Leverage both classical and quantum features for final DTI prediction.
- **Procedure:**
  - **Graph Construction:** Build a heterogeneous network incorporating drugs, targets, their similarity relationships, and known interactions. [36]
  - **Feature Integration:** Input the classically extracted features and the QSE-derived energies (where available) into a Graph Neural Network (GNN) encoder. The GNN updates the node features by propagating information through the network topology. [36]
  - **Interaction Prediction:** Use a graph decoder to compute the probability of an interaction edge existing between a drug and target node, producing the final prediction. [36]

The Scientist's Toolkit: Research Reagent Solutions

Table 2: Essential Computational Tools and Resources

Item Name	Function / Purpose	Specific Example / Note
Quantum Computing SDKs	Provides libraries for constructing and running quantum circuits.	InQuanto (used in [25]), pytket, Qiskit, Cirq.
Classical Shadows	An informationally complete measurement technique to drastically reduce the number of measurements needed for QSE on large systems. [1]	Critical for scaling to 80+ qubit systems.
Molecular Graph Encoder	A neural network component that updates molecular features based on the topological relationships in a larger bioinformatics network.	Part of the SaeGraphDTI model; incorporates network information beyond the molecule itself. [36]
Sequence Attribute Extractor	Transforms variable-length drug (SMILES) and target (amino acid) sequences into fixed-length, aligned feature vectors.	Uses 1D convolution; key for deep learning-based DTI prediction. [36]
3D Molecular Visualization	Software for visualizing molecular structures, binding poses, and simulation results.	ChimeraX, PyMOL, VMD (listed in [37]).
Public DTI Datasets	Curated datasets for training and benchmarking DTI prediction models.	Davis (kinase inhibitors), E (Enzymes), GPCR, IC (Ion Channels). [36]

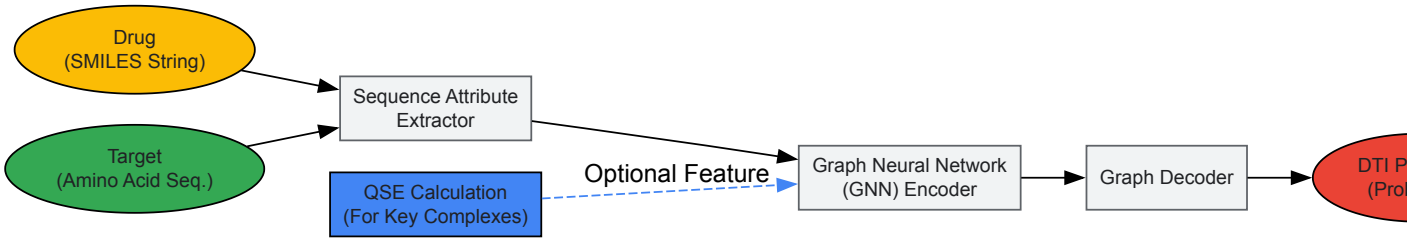


Diagram 2: Hybrid DTI Prediction Pipeline

Hybrid Quantum-Classical Workflows for Practical Drug Discovery Pipelines

Application Note: Advancing Molecular Energy Calculations with Quantum Subspace Expansion

The accurate calculation of molecular energies represents a fundamental challenge in computational drug discovery. Classical computational methods often face significant limitations in modeling complex quantum mechanical phenomena, particularly for large molecular systems or states exhibiting strong electron correlation. **Quantum subspace expansion (QSE)** has emerged as a powerful post-processing technique that enhances the accuracy of quantum computations by classically diagonalizing a Hamiltonian in a carefully constructed subspace of quantum states [1]. This application note details the integration of QSE methodologies into practical, hybrid quantum-classical workflows for pharmaceutical research, enabling researchers to achieve **near-chemical accuracy** in molecular simulations while mitigating hardware noise and algorithmic constraints [29].

The **theoretical foundation** of QSE rests on expanding a computationally prepared quantum state, often approximated due to hardware limitations or algorithmic approximations, into a subspace spanned by a set of expansion operators. When applied to molecular energy calculations, this approach allows for the correction and refinement of energy estimates obtained from variational quantum eigensolvers (VQE) or other variational algorithms, providing a pathway to more accurate determination of ground and excited state energies essential for predicting molecular reactivity and binding affinities [1].

Recent experimental implementations have demonstrated the scalability and effectiveness of QSE approaches. A large-scale implementation of QSE with classically shadowed measurements successfully probed quantum phase transitions in spin models for systems of up to 80 qubits, achieving accurate ground state energy recovery with rigorous statistical error estimates [1]. This represents one of the most significant experimental realizations of classical shadows to date, with over 30,000 measurement basis randomizations per circuit and the evaluation of approximately 10^{14} Pauli traces [1].

The table below summarizes key performance indicators for hybrid quantum-classical methods in molecular energy calculations:

Table 1: Performance Metrics of Hybrid Quantum-Classical Methods for Molecular Energy Calculations

Method	System Size	Accuracy Achieved	Key Innovation	Experimental Validation
QSE with Classical Shadows	Up to 80 qubits	Accurate ground state energy recovery	Constrained optimization with statistical error estimates	Spin model phase transitions [1]
Hybrid Quantum-Neural Wavefunction (pUNN)	Diatomic and polyatomic molecules (N_2, CH_4)	Near-chemical accuracy, comparable to CCSD(T)	Neural network correction of seniority-zero subspace	Cyclobutadiene isomerization on superconducting quantum processor [29]
ADAPT-GQE	Pharmaceutical molecules (e.g., imipramine)	234x speed-up in training data generation	Transformer-based generative quantum AI for circuit synthesis	Ground state preparation on Quantinuum Helios quantum computer [38]

Quantum Subspace Expansion Protocol for Molecular Energy Calculations

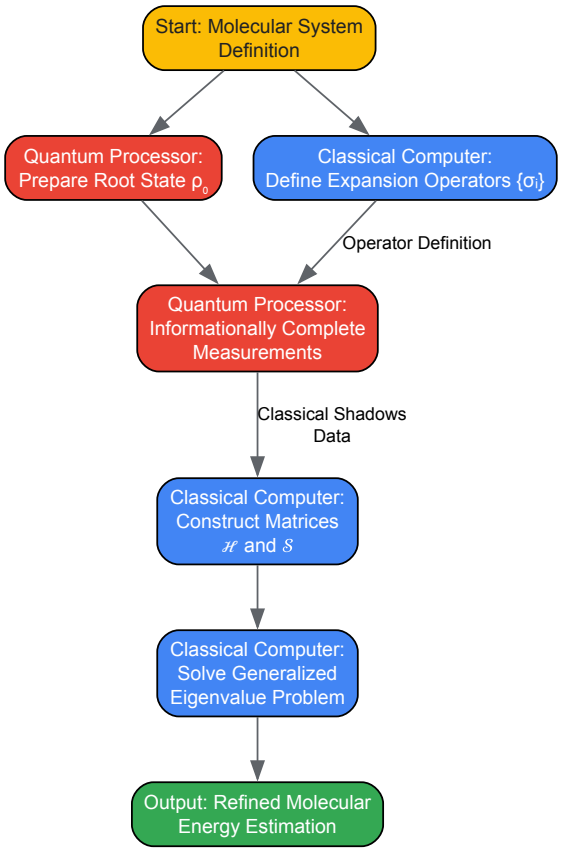
Background and Principles

Quantum subspace expansion operates on the principle that a quantum state prepared on a quantum processor, while potentially inaccurate due to noise or algorithmic approximations, can serve as a valuable starting point for subsequent classical post-processing. The **mathematical foundation** of QSE involves constructing a subspace around a prepared root state ρ_0 using a set of expansion operators $\{\sigma_i\}$. The resulting subspace states take the form $\rho_SE(\vec{c}) = W^\dagger\rho_0 W/\mathrm{Tr}[W^\dagger\rho_0 W]$, where $W = \Sigma_i c_i\sigma_i$ is parameterized by coefficient vector $\vec{c}$ [1]. The optimal energy within this subspace is obtained by solving a generalized eigenvalue problem of the form $\square\vec{c} = \lambda\mathcal{S}\vec{c}$, where $\square$ is the subspace-projected Hamiltonian and $\square$ is the overlap matrix [1].

For molecular energy calculations, the **Krylov subspace approach**—using powers of the Hamiltonian H^p as expansion operators—has proven particularly effective. This method exponentially improves the overlap with the true ground state as the expansion order increases, provided the initial state has non-zero overlap with the target ground state [1].

Experimental Workflow

The following diagram illustrates the complete hybrid workflow for quantum subspace expansion in molecular energy calculations:



Step-by-Step Protocol

- **Step 1: Quantum State Preparation**
  - Initialize the quantum processor to prepare an approximate ground state ρ_0 of the target molecular Hamiltonian H. This can be achieved through variational quantum eigensolver (VQE) circuits or other state preparation methods.
  - For molecular systems, employ Jordan-Wigner or Bravyi-Kitaev transformations to map the electronic Hamiltonian to qubit representations [39].
- **Step 2: Expansion Operator Selection**
  - Select a set of L Hermitian expansion operators $\{\sigma_i\}$ that define the subspace. For molecular energy calculations, consider:
    - Powers of the Hamiltonian H^p (Krylov subspace method)
    - Low-weight Pauli operators [1]
    - Excitation operators relevant to the molecular system
  - The choice of operators balances expressivity against measurement overhead.
- **Step 3: Informationally Complete Measurement**
  - Perform classical shadow measurements on the prepared state ρ_0 by applying random unitary rotations before computational basis measurements [1].
  - For each measurement setting, collect a sufficient number of shots S to achieve desired statistical precision. Recent large-scale implementations used over 30,000 measurement basis randomizations per circuit [1].
  - Store the measurement outcomes $\{k^{(1)}, ..., k^{(S)}\}$ for classical post-processing.
- **Step 4: Classical Post-Processing**
  - Construct estimators for the matrix elements $\mathcal{H}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j H]$ and $\mathcal{S}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j]$ using the classical shadow data [1].
  - To address numerical instability from ill-conditioned overlap matrices, employ regularization techniques such as truncated singular value decomposition or reformulate as a constrained optimization problem [1].
- **Step 5: Generalized Eigenvalue Solution**
  - Solve the generalized eigenvalue problem $\mathcal{H}\vec{c} = \lambda \mathcal{S}\vec{c}$ for the smallest eigenvalue λ_{min}, which provides the refined estimate of the ground state energy [1].
  - Compute statistical error bars through error propagation or resampling methods to quantify uncertainty in the energy estimate.

Integration with Drug Discovery Pipelines

The integration of QSE-based energy calculations into practical drug discovery workflows enables more accurate prediction of key pharmaceutical properties. **Binding affinity predictions** benefit from precise calculations of interaction energies between drug candidates and their protein targets. Additionally, **reaction barrier calculations** for metabolic pathways can be refined through accurate determination of transition state energies [40].

In recent applications, researchers have successfully employed hybrid quantum-classical approaches to study the isomerization of cyclobutadiene, a challenging multi-reference system, demonstrating both high accuracy and significant resilience to noise when implemented on superconducting quantum processors [29]. This capability is particularly valuable for investigating photochemical reactions and excited state dynamics relevant to drug stability and reactivity.

Protocol: Hybrid Quantum-Neural Wavefunction for Molecular Energy Calculations

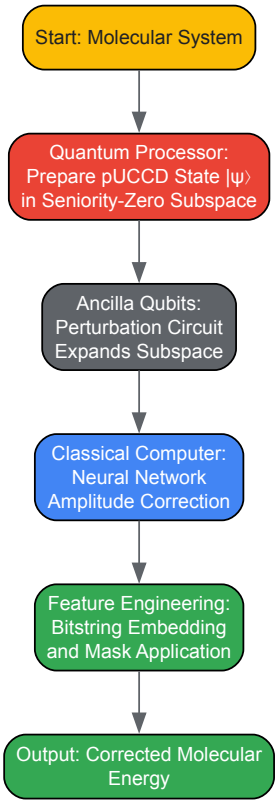
Background

The hybrid quantum-neural wavefunction approach, exemplified by the pUNN (paired Unitary Coupled-Cluster with Neural Networks) algorithm, addresses fundamental limitations in quantum computational chemistry by combining the complementary strengths of quantum circuits and neural networks [29]. This protocol details the implementation of pUNN for molecular energy calculations, achieving **near-chemical accuracy** while maintaining practical computational costs and demonstrating notable **noise resilience** on current quantum hardware.

The **theoretical innovation** of pUNN lies in its decomposition of the molecular wavefunction representation tasks: the quantum circuit captures the complex phase structure and entanglement patterns that are challenging for classical neural networks, while the neural network component corrects for approximations inherent in the quantum ansatz, particularly for configurations outside the seniority-zero subspace [29].

Experimental Workflow

The following diagram illustrates the hybrid quantum-neural wavefunction methodology:



Step-by-Step Protocol

- **Step 1: Quantum Circuit Preparation**
  - Implement the paired Unitary Coupled-Cluster with double excitations (pUCCD) ansatz on the quantum processor to prepare a reference state |ψ⟩ in the seniority-zero subspace [29].
  - The pUCCD circuit captures dominant correlation effects while maintaining computational tractability with linear circuit depth relative to system size.

- **Step 2: Hilbert Space Expansion**
  - Add N ancilla qubits to the original N-qubit system, expanding the Hilbert space from N to 2N qubits.
  - Apply an entanglement circuit $\hat{E}$ consisting of N parallel CNOT gates to create correlations between original and ancilla qubits: $|\Phi\rangle = \hat{E}(|\psi\rangle \otimes |0\rangle)$ [29].
- **Step 3: Ancilla Perturbation**
  - Apply a low-depth perturbation circuit to the ancilla qubits to slightly rotate them from $|0\rangle$ state, introducing components outside the seniority-zero subspace.
  - Use single-qubit rotation gates (e.g., R_y) with small angles (typically 0.2 radians) to ensure the perturbation remains manageable [29].
- **Step 4: Neural Network Architecture**
  - Design a feedforward neural network with the following specifications:
    - Input: Binary representation of the combined bitstring $|k\rangle \otimes |j\rangle$ (size 2N vector with elements ±1)
    - Hidden layers: L = N-3 dense layers with ReLU activation functions
    - Width: 2KN neurons per hidden layer (K is a tunable integer, typically K=2)
    - Output: Single value b_{kj} before mask application [29]
  - The number of parameters in the neural network scales as K^2N^3, providing sufficient expressivity while maintaining computational efficiency.
- **Step 5: Particle Number Conservation**
  - Apply a conservation mask m(k,j) that eliminates configurations $|k\rangle \otimes |j\rangle$ not preserving the correct number of spin-up and spin-down electrons [29].
  - The mask function equals 1 for valid configurations and 0 otherwise, ensuring physical meaningfulness of the wavefunction.
- **Step 6: Energy Evaluation**
  - Compute the energy expectation value using the ratio $E = \langle\Psi|\hat{H}|\Psi\rangle/\langle\Psi|\Psi\rangle$, where $|\Psi\rangle$ is the complete hybrid quantum-neural wavefunction.
  - Employ an efficient measurement protocol that avoids quantum state tomography, leveraging the specific structure of the pUNN ansatz to minimize measurement overhead [29].

Performance Validation

The pUNN method has been validated through numerical simulations of diatomic and polyatomic molecules including N_2 and CH_4, achieving accuracy comparable to high-level classical methods such as CCSD(T) while maintaining the computational efficiency of shallow quantum circuits [29]. Experimental implementation on superconducting quantum processors for the isomerization reaction of cyclobutadiene demonstrated both high accuracy and significant resilience to hardware noise, highlighting the practical utility of this approach for realistic chemical problems on current quantum devices [29].

Protocol: Variational Quantum Linear Solver (VQLS) for Digital Twin and CFD Simulations

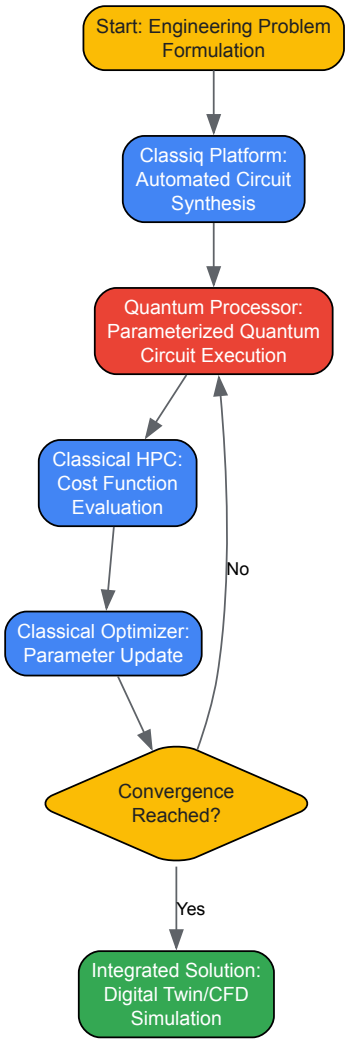
Background

The Variational Quantum Linear Solver (VQLS) algorithm has found significant application in engineering domains relevant to pharmaceutical development, particularly in digital twin simulations and computational fluid dynamics (CFD) for drug delivery system optimization [41]. This protocol outlines the implementation of an optimized VQLS workflow developed through collaboration between Classiq, BQP, and NVIDIA, demonstrating practical **hybrid quantum-classical workflows** integrated directly into existing high-performance computing (HPC) environments [41].

The **key innovation** in this implementation lies in the automated circuit synthesis that reduces qubit counts, circuit size, and the number of trainable parameters compared to traditional quantum linear solver formulations, thereby improving the scaling behavior of matrix-based problems commonly encountered in pharmaceutical simulation workflows [41].

Experimental Workflow

The following diagram illustrates the VQLS workflow for engineering simulations:



Step-by-Step Protocol

- **Step 1: Problem Formulation**
  - Transform the engineering problem (e.g., fluid dynamics for drug delivery systems, molecular transport phenomena) into a linear system of equations $A|x\rangle = |b\rangle$.
  - Encode the matrix A and vector $|b\rangle$ into quantum-mechanical representations suitable for quantum processing.
- **Step 2: Automated Circuit Synthesis**
  - Utilize the Classiq platform to automatically generate optimized quantum circuits for the VQLS algorithm [41].
  - The synthesis process specifically minimizes resource requirements by:
    - Reducing qubit counts through efficient encoding schemes
    - Minimizing circuit depth via gate compilation optimizations
    - Decreasing the number of trainable parameters to improve convergence behavior [41]
- **Step 3: Hybrid Quantum-Classical Optimization**
  - Execute the parameterized quantum circuit on the quantum processor (utilizing NVIDIA CUDA-Q platform for hybrid computation) [41].
  - Compute the cost function $C(\theta) = \langle\psi(\theta)|A^\dagger(I - |b\rangle\langle b|)A|\psi(\theta)\rangle$ on classical HPC resources to evaluate solution quality.
  - Employ classical optimizers (e.g., gradient-based methods) to update circuit parameters θ to minimize the cost function.
- **Step 4: Convergence Checking**
  - Monitor the cost function value across iterations to determine convergence to the solution $|x\rangle$.
  - Implement termination criteria based on either reaching a threshold cost value or exceeding a maximum number of iterations.
- **Step 5: Solution Integration**

- Extract the solution vector from the quantum state and integrate it into the larger engineering simulation framework (digital twin or CFD workflow) [41].
- Utilize the solution for predictive modeling and optimization of pharmaceutical systems, such as drug delivery device design or biomolecular transport simulations.

Implementation Considerations

This optimized VQLS implementation has been incorporated into BQP's BQPhy platform offerings available to clients today, enabling enterprises to explore and apply hybrid quantum-classical techniques while maintaining the structure and reliability of their existing HPC systems [41]. The workflow demonstrates how hybrid quantum-classical approaches can currently support demanding engineering workloads in pharmaceutical development without requiring complete overhaul of established simulation environments.

Table 2: Essential Research Reagents and Computational Resources for Hybrid Quantum-Classical Drug Discovery

Resource Category	Specific Tool/Platform	Function in Workflow	Key Features
Quantum Hardware Platforms	Quantinuum Helios [38]	Trapped-ion quantum computer for high-accuracy molecular simulations	Industry-leading low error rates; integrated with NVIDIA Grace Blackwell platform
	IonQ Forte [42]	Quantum processing for hybrid workflows	99.99% two-qubit gate fidelity; cloud-accessible
	Quantum Circuits Aqumen Seeker [43]	8-qubit processor with dual-rail qubits for molecular simulation	Built-in error detection; "correct first, then scale" architecture
Quantum Software Platforms	NVIDIA CUDA-Q [41] [38]	Hybrid quantum-classical computing platform	Integration with HPC environments; support for quantum machine learning
	Classiq Platform [41]	Automated quantum circuit synthesis	Optimizes circuit design to reduce qubit counts and circuit depth
	InQuanto [38]	Computational chemistry platform	Specialized for molecular simulations on quantum hardware
Classical HPC Integration	NVIDIA Grace Blackwell [38]	Accelerated computing for hybrid workflows	Paired with quantum systems for real-time decoding and error correction
	Exascale Supercomputers [40]	Generation of quantum-accurate training data	Creates synthetic quantum chemistry datasets for AI model training
Algorithmic Frameworks	Quantum Subspace Expansion (QSE) [1]	Post-processing technique for energy refinement	Enhances accuracy of quantum computations; enables error mitigation
	Hybrid Quantum-Neural Networks [29]	Combined quantum-classical wavefunction representation	Achieves chemical accuracy with noise resilience
	Variational Quantum Linear Solver (VQLS) [41]	Hybrid algorithm for linear systems	Applied to digital twin and CFD simulations in pharmaceutical engineering

The integration of hybrid quantum-classical workflows into drug discovery pipelines represents a transformative approach to addressing long-standing computational challenges in pharmaceutical research. Quantum subspace expansion methods, combined with advanced hybrid algorithms like quantum-neural wavefunctions and variational quantum linear solvers, provide **practical pathways** to more accurate molecular energy calculations and complex simulations relevant to drug development.

As quantum hardware continues to advance—with companies like Quantinuum, IonQ, and Quantum Circuits demonstrating increasingly capable systems—and software platforms such as NVIDIA CUDA-Q and Classiq mature, these hybrid approaches are positioned to move from experimental demonstrations to production-scale implementations in pharmaceutical R&D. The protocols and application notes detailed herein provide researchers with practical methodologies for leveraging these emerging technologies to accelerate and enhance the drug discovery process.

Organizations that strategically invest in building quantum capabilities, forming technology partnerships, and developing integrated hybrid workflows will be best positioned to leverage these advances for creating more effective therapeutics with reduced development timelines and costs.

Overcoming Practical Challenges: Noise Mitigation and Optimization of QSE

Addressing Hardware Noise and Ill-Conditioning in NISQ Devices

The pursuit of quantum utility on contemporary Noisy Intermediate-Scale Quantum (NISQ) hardware is fundamentally constrained by inherent physical noise and the mathematical ill-conditioning of computational tasks. These processors, while increasingly sophisticated, accumulate errors from decoherence, imperfect gate operations, and noisy measurements that systematically bias computational outcomes [44] [45] [46]. For research in quantum subspace expansion (QSE) applied to molecular energy calculations, this noise sensitivity presents a critical bottleneck. The extraction of excited-state properties and dynamical correlations relies on manipulating small eigenvalue separations in the quantum subspace—a process severely compromised by even minor perturbations that are amplified in ill-conditioned systems [47]. This Application Note provides a structured framework of protocols and analytical tools designed to characterize, mitigate, and model these effects, thereby enhancing the reliability of quantum simulations for chemical and pharmaceutical applications.

Quantitative Error Characterization and Modeling

Effective error mitigation begins with accurate noise characterization. The following metrics and models are essential for diagnosing hardware performance and informing mitigation strategies.

Table 1: Key Metrics for Quantum Hardware Error Characterization

Metric	Description	Typical Values in NISQ Era	Impact on QSE
Qubit Error Probability (QEP) [44]	Probability a qubit suffers an error during a cycle.	Varies by hardware and qubit.	Directly impacts fidelity of overlap and Hamiltonian matrices.
Two-Qubit Gate Error Rate [46]	Infidelity of two-qubit gate operations.	~0.1% on best superconducting devices.	Limits maximum feasible circuit depth for VQE preparation.
Measurement Error Rate [45]	Probability of misidentifying a qubit state during readout.	<1% on advanced devices.	Corrupts measurement statistics for expectation values.
Coherence Times [44]	Duration qubits maintain quantum state (T_1, T_2).	Hundreds of microseconds.	Constraints total algorithm runtime and circuit depth.

Advanced noise modeling moves beyond simplistic approximations to capture the correlated nature of errors across space and time. Recent breakthroughs leverage mathematical symmetries and root space decomposition to classify noise into distinct categories, enabling the application of targeted mitigation techniques [48]. Furthermore, scalable noise models calibrated to real superconducting hardware can predict the performance of 20-qubit quantum computers, providing a crucial tool for pre-runtime algorithm selection and benchmarking [49].

Error Mitigation Techniques for Subspace Methods

A multi-layered error mitigation strategy is recommended to protect quantum subspace expansion workflows from initialization through to measurement.

Zero Error Probability Extrapolation (ZEPE)

Zero-Noise Extrapolation (ZNE) systematically amplifies noise to extrapolate results back to a zero-noise limit. The novel **Zero Error Probability Extrapolation (ZEPE)** method enhances this by using the more accurate Qubit Error Probability (QEP) as the scaling metric, rather than simplistic circuit copy factors. This provides a superior extrapolation, especially for mid-depth circuits common in variational quantum eigensolver (VQE) state preparation [44].

Protocol:

- **Calibration:** Determine the mean QEP for your target hardware and circuit.
- **Noise Amplification:** Run the circuit at several intentionally amplified noise levels (e.g., 1x, 1.5x, 2x the base QEP). This can be achieved via pulse stretching or gate duplication.
- **Extrapolation:** Measure the observable of interest (e.g., energy expectation) at each noise level. Fit the data to a linear or exponential model and extrapolate to the QEP=0 point.

Measurement Error Mitigation

This technique corrects for readout errors, which is vital for accurate calculation of the overlap (**S**) and Hamiltonian (**H**) matrices in QSE [50].

Protocol:

- **Characterization:** Prepare and measure a complete set of computational basis states (e.g., $|00...0\rangle$, $|00...1\rangle$, ..., $|11...1\rangle$).

- **Construct Calibration Matrix:** Build a matrix **A** where each element $A_{i\,j}$ is the probability of measuring bitstring *i* when preparing bitstring *j*.
- **Inversion:** For a subsequent experiment yielding raw measured probabilities ***praw****, compute the corrected probabilities via* **pcorrected = A⁻¹ p_raw**.

Symmetry Verification and Subspace Methods

Many molecular systems possess inherent symmetries (e.g., particle number conservation, spin symmetry). Noise often drives the quantum state into symmetry-violating subspaces. By performing additional measurements to check these symmetries, researchers can post-select or re-weight results to discard or correct corrupted runs, significantly improving the conditioning of the QSE problem [50] [47].

Experimental Protocol for Robust Fermionic Subspace Expansion

The following detailed protocol outlines the process for conducting a noise-resilient QSE calculation for molecular excited states, from initial ground state preparation to final mitigated result.

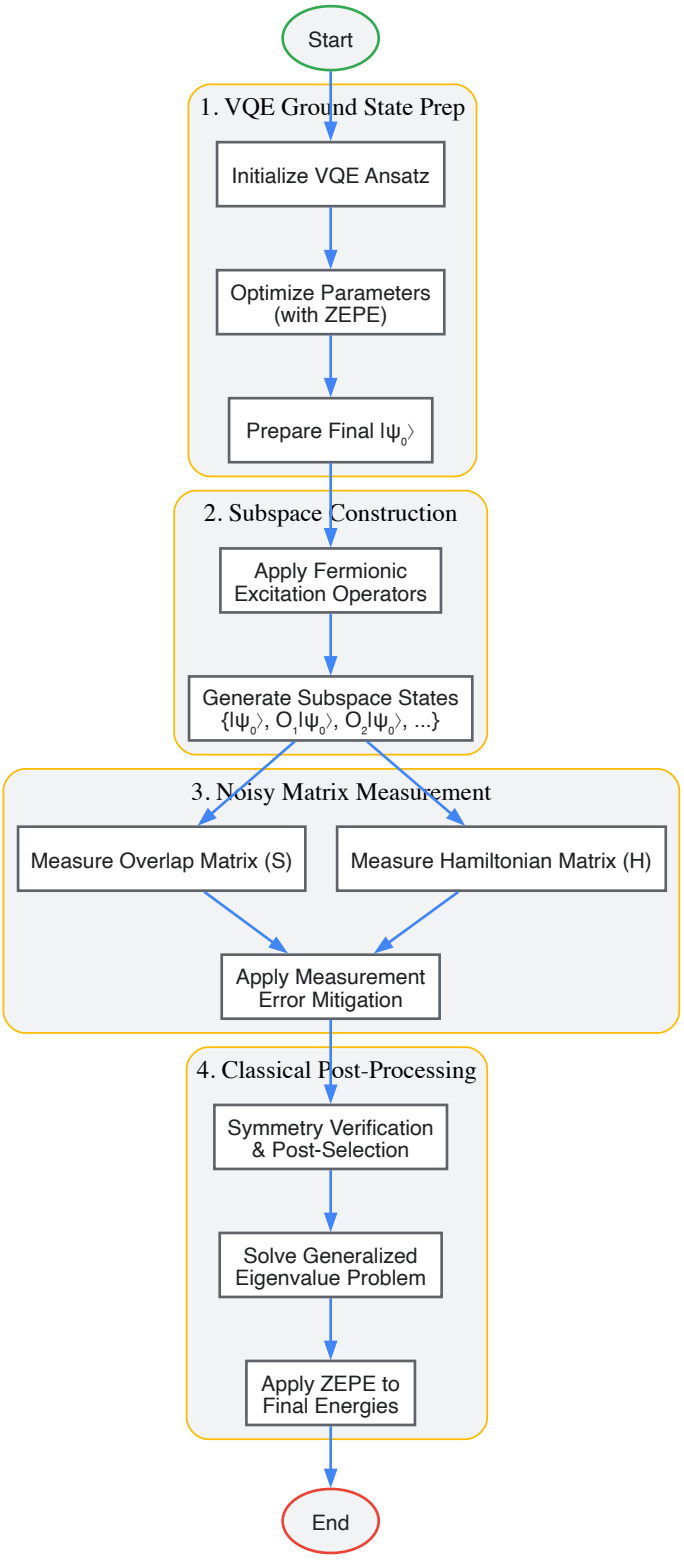


Figure 1: Experimental workflow for noise-resilient Quantum Subspace Expansion.

Step 1: VQE Ground State Preparation

- **Objective:** Prepare a high-fidelity approximation of the molecular ground state, $|\psi_0\rangle$.
- **Procedure:**
  - **Ansatz Selection:** Choose a hardware-efficient or chemically inspired ansatz (e.g., UCCSD) compatible with the target molecular Hamiltonian mapped to qubits [47].
  - **Noise-Aware Optimization:** Execute the VQE optimization loop, employing ZEPE on the energy expectation value at each function evaluation to guide the optimizer with noise-mitigated gradients.
- **Validation:** Verify the prepared state satisfies expected molecular symmetries (e.g., particle number) as an initial check.

Step 2: Subspace Construction

- **Objective:** Generate a basis for the excited subspace.
- **Procedure:**
  - **Operator Selection:** Define a set of fermionic excitation operators $\{O_i\}$ (e.g., single and double excitations) relevant to the target molecular system [47].
  - **State Generation:** Apply these operators to the prepared ground state to generate the subspace basis: $\{ |\psi_0\rangle, O_1|\psi_0\rangle, O_2|\psi_0\rangle, \dots \}$.

Step 3: Noisy Matrix Measurement

- **Objective:** Accurately measure the overlap (**S**) and Hamiltonian (**H**) matrices within the subspace, where $S_{ij} = \langle \psi_i|\psi_j \rangle$ and $H_{ij} = \langle \psi_i|H|\psi_j \rangle$.
- **Procedure:**
  - **Circuit Execution:** Run a series of quantum circuits to compute the matrix elements between all basis states in the subspace.
  - **Measurement Mitigation:** Apply the Measurement Error Mitigation protocol (Section 3.2) to the raw results for each matrix element.

Step 4: Classical Post-Processing

- **Objective:** Solve for the excited states and energies from the noisy matrices.
- **Procedure:**
  - **Symmetry Verification:** Check the symmetry of the measured states. Discard (post-select) or down-weight results that violate known symmetries.
  - **Solve Generalized Eigenvalue Problem (GEVP):** Classically solve $H c = E S c$ for the energies E and coefficients c. If the mitigated **S** matrix remains ill-conditioned, use regularization techniques (e.g., Tikhonov regularization).
  - **Final Extrapolation:** Apply the ZEPE protocol (Section 3.1) to the resulting energies from multiple noise levels to obtain the final, mitigated energy spectrum.

Table 2: Essential Research Reagents and Computational Tools

Item / Resource	Function / Purpose	Example Application in Protocol
Tool for Error Description (TED-qc) [44]	Pre-processing tool to calculate error probabilities in a quantum circuit given hardware specs.	Estimating QEP for ZEPE protocol before running on hardware.
Density Matrix Simulator (e.g., DM1) [45]	Simulator that represents quantum states as density matrices, enabling realistic noise simulation.	Testing and validating mitigation protocols with simulated noise before hardware deployment.
Mitiq Library [50]	Open-source Python toolkit for error mitigation (implements ZNE, PEC, etc.).	Can be integrated to perform ZNE/ZEPE extrapolation in the post-processing step.
Fermionic Excitation Operators [47]	Set of operators (e.g., $a_p^\dagger a_q$, $a_p^\dagger a_q^\dagger a_r a_s$) that generate excited states from a reference.	Constructing the subspace basis in Step 2 of the experimental protocol.
Calibration Matrix (A) [50]	Matrix characterizing readout error probabilities between classical bitstrings.	Correcting measurement outcomes for the **S** and **H** matrices in Step 3.

The path toward reliable molecular energy calculations on NISQ devices necessitates a co-design of algorithms and error resilience strategies. The protocols and application notes detailed herein—centered on precise error quantification via QEP, robust mitigation via ZEPE and symmetry verification, and rigorous experimental workflows—provide a tangible roadmap for researchers in quantum chemistry and drug development. While these techniques extend the utility of current hardware, the long-term trajectory points toward Fault-Tolerant Application-Scale Quantum (FASQ) systems, where quantum error correction will ultimately manage noise [46]. Until that era dawns, the systematic application of these error-aware methodologies will be paramount in extracting chemically accurate and scientifically valuable results from noisy quantum processors.

Quantum Subspace Expansion (QSE) is a class of hybrid quantum-classical algorithms for computing molecular energies and properties by diagonalizing a Hamiltonian within a subspace constructed from quantum computer measurements [26] [4]. While promising for near-term quantum devices, standard QSE implementations solve a **generalized eigenvalue equation** (\mathbf{H}C = \mathbf{S}CE), where (\mathbf{H}) is the Hamiltonian matrix and (\mathbf{S}) is the overlap matrix between non-orthogonal basis states [26] [51]. This formulation introduces severe **numerical instability** when the overlap matrix becomes ill-conditioned—a frequent occurrence due to statistical sampling errors inherent in quantum measurements and near-linear dependencies in the subspace basis [26] [51].

The core instability arises from the **condition number** of the overlap matrix (\mathbf{S}). As the condition number increases, errors in the estimated eigenvalues grow drastically, eventually rendering the generalized eigenvalue equation unsolvable without intervention [26]. These instabilities have been reported in practical chemical calculations, limiting QSE's application to realistic molecular systems [26]. This application note

details two advanced techniques—Thresholded QSE (TQSE) and Partitioned QSE (PQSE)—developed to mitigate these stability issues while maintaining accuracy for molecular energy calculations.

Technical Foundations and Comparative Analysis

Fundamental Mechanisms of TQSE and PQSE

Thresholded QSE (TQSE) addresses numerical instability by projecting the generalized eigenvalue problem onto a subspace dominated by the largest singular values of the overlap matrix [51]. The technique performs a **singular value decomposition** of the overlap matrix ($\mathbf{S}$) and eliminates eigenvalues below a predetermined threshold, along with their associated eigenvectors [26]. This effectively removes the near-linear dependencies from the basis before solving the generalized eigenvalue problem, preventing the solution from being dominated by noise [26] [51].

Partitioned QSE (PQSE) employs an alternative strategy by breaking a large, ill-conditioned Krylov subspace into smaller, well-conditioned subspaces connected through an iterative sequence [51] [3]. Rather than solving one large QSE problem, PQSE solves a sequence of smaller problems where the lowest-energy state from one subspace serves as the reference state for the next subspace [51]. This iterative recombination maintains numerical stability while enabling the use of larger effective basis sizes than possible with standard QSE or TQSE [51] [3].

Quantitative Performance Comparison

The table below summarizes key characteristics and performance metrics of standard QSE, TQSE, and PQSE based on theoretical analyses and numerical experiments:

Table 1: Comparative Analysis of Standard and Advanced QSE Techniques

Method	Core Approach	Matrix Type Solved	Stability to Sampling Noise	Basis Size Limit	Classical Processing
Standard QSE	Single generalized eigenvalue problem	Generalized ($\mathbf{H}C = \mathbf{S}CE$)	Low - errors increase with condition number [26]	Limited by onset of ill-conditioning [26]	Standard diagonalization
TQSE	Projection to dominant singular values	Generalized with thresholded ($\mathbf{S}$) [51]	Moderate - stable up to critical basis size [51]	Limited by threshold choice [51]	SVD plus thresholding
PQSE	Iterative sequence of small subspaces	Multiple generalized eigenvalue problems [51]	High - improved stability beyond TQSE [51] [3]	Can extend beyond TQSE limit [51]	Polynomial overhead in subspace dimension

PQSE introduces a **variance-based criterion** for determining effective iterative sequences, with numerical experiments demonstrating substantially improved accuracy and stability for both molecular Hamiltonians and spin systems [51]. The method maintains the same quantum resource requirements as standard QSE while adding polynomial classical processing overhead [51] [3].

Methodological Protocols

Protocol for Thresholded QSE (TQSE) Implementation

Objective: Solve for ground and excited state energies of a molecular Hamiltonian using TQSE to overcome numerical instability from ill-conditioned overlap matrices.

Pre-requisites: Quantum computer access, optimized ground state wavefunction ($|\Psi_{\mathrm{VQE}}\rangle$) *from VQE, set of excitation operators* ($\{O_i\}$) for subspace construction.

Procedure:

- **Subspace Construction**:
  - Prepare the ground state ($|\Psi_{\mathrm{VQE}}\rangle$) on quantum hardware.
  - Construct subspace basis states ($\{|\phi_i\rangle\} = \{O_i |\Psi_{\mathrm{VQE}}\rangle\}$) using excitation operators [4].
- **Quantum Measurement**:
  - Measure all matrix elements ($H_{ij} = \langle\phi_i|H|\phi_j\rangle$) and ($S_{ij} = \langle\phi_i|\phi_j\rangle$) on quantum hardware [4].
  - Use shot-based measurement protocols (e.g., PauliAveraging) with sufficient shots per circuit to control statistical errors [4].
- **Thresholding Procedure**:

- Perform singular value decomposition (SVD) of the overlap matrix: (\mathbf{S} = U\Sigma V^\dagger) [51].
- Identify singular values (\sigma_i) below threshold (\epsilon) (typically (10^{-8}) to (10^{-6}), system-dependent).
- Construct projection operator (P) retaining only singular values above (\epsilon).
- Project Hamiltonian and overlap matrices: (\mathbf{H}*{\mathrm{proj}} = P^\dagger \mathbf{H} P), (\mathbf{S}*{\mathrm{proj}} = P^\dagger \mathbf{S} P) [51].

- **Classical Diagonalization**:
  - Solve the projected generalized eigenvalue problem: (\mathbf{H}*{\mathrm{proj}} C = \mathbf{S}*{\mathrm{proj}} C E) [26] [51].
  - Extract eigenvalues (E) (energies) and eigenvectors (C) for ground and excited states.

Troubleshooting: If solution lacks desired number of states, gradually decrease threshold (\epsilon) until acceptable balance between stability and state recovery is achieved.

Protocol for Partitioned QSE (PQSE) Implementation

Objective: Compute molecular energies with enhanced numerical stability and extended effective basis size using partitioned subspace approach.

Pre-requisites: Quantum computer access, initial reference state (e.g., Hartree-Fock or VQE ground state), defined Krylov basis order for each partition.

Procedure:

- **Initial Partition Setup**:
  - Define initial Krylov subspace (K0 = {|\phi0\rangle, H|\phi0\rangle, H^2|\phi0\rangle, \dots, H^{d-1}|\phi_0\rangle}) with moderate dimension (d) to ensure well-conditioning [51].
  - Construct and measure (\mathbf{H}) and (\mathbf{S}) matrices for this initial subspace.

- **Iterative Subspace Sequence**:
  - Solve the generalized eigenvalue problem for the current partition to obtain the lowest-energy state (|\psi^{(i)}\rangle) [51].
  - Use this optimized state as the new reference state for the next partition: (|\phi_{i+1}\rangle = |\psi^{(i)}\rangle).
  - Construct subsequent Krylov subspace around this new reference state.
  - Repeat until variance criterion is satisfied: (\mathrm{var}(H, |\psi\rangle) = \frac{\langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2}{\langle\psi|\psi\rangle} < \delta) for desired accuracy threshold (\delta) [51].

- **Variance-Based Sequence Optimization**:
  - For each partition, monitor the energy variance of the solution.
  - Identify "good sequences" that maintain monotonic energy convergence and avoid variance spikes [51].
  - Adjust partition sizes dynamically based on variance trends.

- **Final Energy Extraction**:
  - The final energy in the sequence provides the improved ground state estimate.
  - Excited states can be extracted from the eigenstates of the final partition or by tracking states across sequences.

Validation: Compare energy convergence with increasing partitions against theoretical benchmarks where available. Monitor condition numbers of overlap matrices in each partition to verify maintained stability.

Workflow Visualization

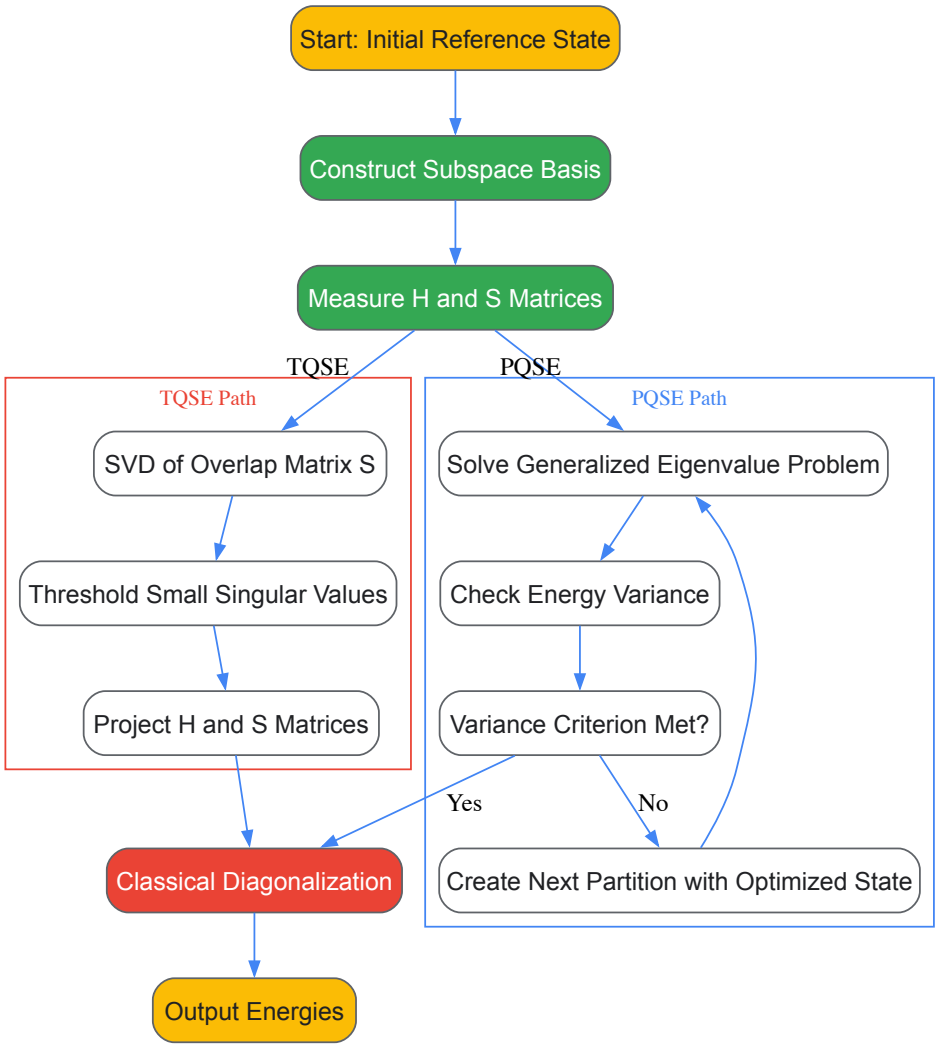


Figure 1: Unified Workflow for Advanced QSE Techniques

Research Reagent Solutions

Table 2: Essential Computational Tools for Advanced QSE Implementation

Tool/Resource	Function	Example Implementation
Quantum Hardware Interface	Executes quantum circuits for state preparation and measurement	Amazon Braket, IBM Quantum Experience, Azure Quantum [52]
Shot-Based Measurement Protocol	Estimates expectation values with finite sampling	PauliAveraging with configurable shots per circuit [4]
Classical Diagonalizer	Solves generalized eigenvalue problems	Scientific computing libraries (SciPy, LAPACK) [26]
Singular Value Decomposition	Decomposes overlap matrix for thresholding	Standard numerical linear algebra packages [51]
Variance Calculator	Monitors convergence in PQSE iterations	Custom implementation based on (\langle H^2 \rangle) measurement [51]
Domain-Specific Platform	Streamlines computational chemistry workflows	Kvantify Qrunch for chemistry-specific quantum algorithms [52]

Thresholded QSE and Partitioned QSE represent significant advances in stabilizing quantum subspace methods against the statistical noise inherent in current quantum hardware. While TQSE provides a direct approach to handling ill-conditioned overlap matrices through singular value thresholding, PQSE offers a more fundamental reorganization of the subspace approach that exchanges circuit depth for additional measurements and classical processing [51] [3]. For researchers targeting molecular energy calculations, these techniques extend the practical applicability of quantum subspace methods to larger, more chemically relevant systems while maintaining robustness against the sampling errors that have

previously limited their predictive power [26] [51]. Implementation requires careful attention to threshold selection for TQSE and sequence construction for PQSE, but both methods provide pathways to more reliable quantum computational chemistry on emerging quantum hardware.

Regularization Strategies and Managing Numerical Instability

In the pursuit of accurate molecular energy calculations, quantum subspace expansion (QSE) has emerged as a powerful algorithmic framework for near-term quantum hardware. A significant obstacle to its practical implementation, however, is a pronounced sensitivity to statistical noise and numerical instability, which can severely limit the accuracy of the computed energies [3]. Regularization strategies provide a mathematical foundation to manage these instabilities, ensuring that the solutions obtained are both physically meaningful and computationally robust. This document details the application of these strategies within the context of QSE, providing structured protocols and data to guide researchers in drug development and related fields.

Regularization Methods in Electronic Structure Theory

The core challenge of numerical instability is not unique to quantum subspace methods but is a recurring theme in electronic structure theory, particularly in methods built on perturbation theory.

Regularized Second-Order Perturbation Theories

Traditional second-order Møller-Plesset (MP2) perturbation theory, a workhorse of quantum chemistry, is known to diverge when the energy gap between occupied and virtual orbitals closes, such as in metallic or strongly correlated systems [53]. Furthermore, it can overestimate interaction energies in large, polarizable systems. Regularized MP2 expressions have been developed to circumvent these issues and, in doing so, also alleviate numerical pathologies.

The table below summarizes key properties of several prominent regularized second-order energy expressions, which inform the development of regularization for quantum subspace methods.

Table 1: Properties of Regularized Second-Order Energy Expressions [53]

Method	Size Extensivity	Size Consistency	Invariance	Iterative	Performance in Strong Correlation
MP2	Yes	Yes	Yes	No	No
DCPT2	No	Yes	Yes	No	Partial
QDPT2	No	Yes	No	No	Partial
BGE2	No	Yes	No	Yes	Partial
κ-MP2	**Yes**	**Yes**	**Yes**	**No**	**No**
iQPMP2	**Yes**	**Yes**	**Yes**	**Yes**	**Partial**

Among these, **κ-MP2** introduces an empirical multiplicative damping term dependent on the parameter κ and the MP2 denominator [53] [54]. Its key advantage is the preservation of size extensivity and consistency while being non-iterative. Conversely, the iterative quasi-particle MP2 (**iQPMP2**) method incorporates a regularized denominator derived from a renormalized first-order pair correlation energy, making it an iterative procedure that partially addresses strong correlation [53]. The choice of method involves a trade-off between desirable physical properties and computational cost.

Regularization Protocols for Quantum Subspace Expansion

The Quantum Subspace Expansion (QSE) algorithm generates a set of basis states by applying excitation operators to a variational quantum eigensolver (VQE) ground state. A generalized eigenvalue problem is then solved to find optimal energy states within this subspace [4]: [H\mathbf{c} = S\mathbf{c}E] where (H) is the Hamiltonian matrix, (S) is the overlap matrix, (E) is a diagonal matrix of eigenvalues, and (\mathbf{c}) is the matrix of eigenvectors. Numerical instability arises when the overlap matrix (S) is ill-conditioned or near-singular, often due to finite sampling noise on quantum hardware or linear dependencies in the basis.

Protocol 1: Partitioned Quantum Subspace Expansion (PQSE)

The Partitioned Quantum Subspace Expansion (PQSE) is a generalization of QSE designed to enhance numerical stability in the presence of finite sampling noise without relying on empirically chosen parameters [55] [3].

Detailed Methodology:

- **Krylov Basis Generation:** Generate a Krylov basis ({ |\Psi0\rangle, H|\Psi0\rangle, H^2|\Psi0\rangle, ..., H^{K-1}|\Psi0\rangle }) from an initial state (|\Psi_0\rangle), which is typically the VQE ground state. The Hamiltonian matrices (H) and (S) are constructed in this subspace.

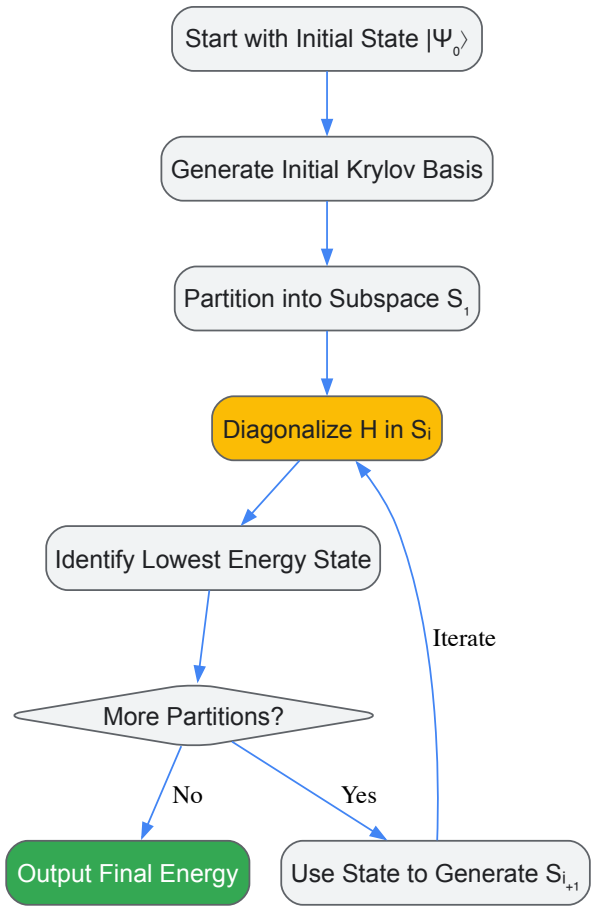
- **Iterative Partitioning:** Instead of diagonalizing the Hamiltonian in the entire Krylov subspace of dimension (K) in a single step, the PQSE algorithm connects a sequence of smaller subspaces.

  - The first subspace, (\mathcal{S}_1), is spanned by the first (m) vectors of the Krylov basis (where (m < K)).
  - The Hamiltonian is diagonalized in (\mathcal{S}_1), and its lowest energy eigenstate is identified.
  - This lowest energy state is used to generate a new, subsequent subspace (\mathcal{S}_2), effectively "restarting" the Krylov process from a refined initial state.
  - This procedure is repeated sequentially, building a chain of subspaces (\mathcal{S}1 \rightarrow \mathcal{S}2 \rightarrow ... \rightarrow \mathcal{S}_L).

- **Variance-Based Criterion:** A key component of PQSE is the use of a variance-based criterion to determine high-quality iterative sequences. This criterion helps select sequences that are less susceptible to noise, thereby improving the stability of the final energy estimate.

- **Classical Processing:** The diagonalizations of the smaller subspace matrices are performed on a classical computer. This process incurs only a polynomial overhead in the subspace dimension while substantially alleviating numerical instability.

The following workflow diagram illustrates the PQSE protocol:



Protocol 2: Generalized Quantum Subspace Expansion (GQSE)

The Generalized Quantum Subspace Expansion (GQSE) method extends the subspace to handle stochastic, coherent, and algorithmic errors agnostically, without prior knowledge of the noise model [56].

Detailed Methodology:

- **Subspace Expansion:** Extend the quantum subspace beyond the traditional Krylov or excitation bases. Two highly practical setups are:

  - **Powers of the Noisy State:** Span the subspace using powers of the noisy density matrix, (\rho^m), where (\rho) is the state prepared by the quantum computer.

- **Error-Boosted States:** Construct the subspace from a set of states that have been intentionally evolved to amplify and subsequently characterize errors.

- **Matrix Construction and Diagonalization:** Construct the Hamiltonian (H) and overlap (S) matrices within this substantially expanded subspace. The generalized eigenvalue problem is then solved classically.

- **Error Mitigation:** By fully exploiting the expanded subspace, which contains a more diverse set of states, the protocol can efficiently mitigate the noise present in the spectra of the Hamiltonian. The optimal solution within this larger subspace effectively projects out contributions from erroneous states.

The Scientist's Toolkit: Essential Research Reagents

The following table lists key computational tools and their functions, as featured in the discussed experiments and the broader field.

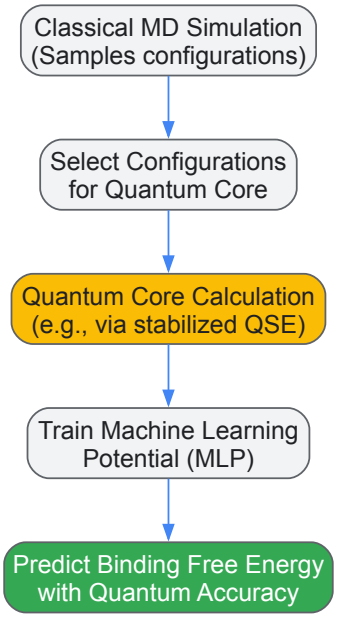
Table 2: Key Research Reagent Solutions for QSE and Regularization

Item Name	Function / Application
Krylov Basis	A subspace generated by repeated action of the Hamiltonian on an initial state; used in QSE and PQSE for its known convergence properties [3].
Generalized Eigenvalue Solver	A classical algorithm that solves $H\mathbf{c} = S\mathbf{c}E$; finds the optimal energies and states in a non-orthogonal basis [4].
Regularization Parameter (κ)	An empirical parameter in methods like κ-MP2 that damps the contribution from small energy denominators, preventing divergence [53] [54].
Variance Criterion	A metric used in PQSE to identify iterative sequences that are stable against finite sampling noise, ensuring numerical robustness [55] [3].
Deep Potential (DP) Framework	A machine learning interatomic potential used to generate high-quality, DFT-level data for training and validation of quantum chemistry models [57].
Open Molecules 2025 (OMol25) Dataset	A massive dataset of over 100 million molecular snapshots with DFT-calculated properties; used for training and benchmarking ML models and electronic structure methods [58].

Integrated Workflow for Molecular Energy Calculations

Combining quantum subspace methods with classical machine learning potentials offers a powerful pipeline for achieving quantum advantage in critical tasks like binding energy calculations, which are foundational to drug discovery.

The FreeQuantum pipeline is a blueprint for such an integrated approach [28]. It embeds high-accuracy quantum calculations (which can be performed using stabilized QSE) into larger classical molecular simulations. The following diagram outlines this modular workflow:



Protocol Application: This workflow was tested on a ruthenium-based anticancer drug (NKP-1339) binding to its protein target, GRP78 [28]. For the transition metal complex, classical force fields failed to capture key quantum interactions. The "quantum core" step, which could be executed with a high-accuracy method like a regularized QSE, provided the precise energy data needed to train a machine-learning potential. The final pipeline predicted a binding free energy of **-11.3 ± 2.9 kJ/mol**, a significant deviation from the classical force field prediction of -19.1 kJ/mol, underscoring the critical importance of quantum-level accuracy in drug design [28].

Balancing Quantum Circuit Depth with Measurement Overhead

In the pursuit of simulating molecular systems on quantum hardware, researchers are confronted with a fundamental trade-off: **deepening quantum circuits** to improve accuracy versus **managing the exponential measurement overhead** required to read out the results. This balance is critically important within the framework of **quantum subspace expansion (QSE)**, a leading methodology for calculating molecular energies and excited states on noisy intermediate-scale quantum (NISQ) devices.

QSE algorithms work by constructing a subspace of quantum states, often generated from a single, efficiently prepared reference state. The accuracy of the final energy calculation is directly tied to the **expressiveness of this subspace**, which typically requires deeper quantum circuits or a larger set of expansion operators. However, each additional operator significantly increases the number of measurements needed to characterize the subspace, creating a direct tension between computational accuracy and experimental feasibility. This application note details practical strategies and recent methodological advances for navigating this trade-off in research aimed at achieving chemical accuracy in molecular energy calculations.

Core Concepts and Quantitative Trade-offs

The Quantum Subspace Expansion (QSE) Framework

Quantum Subspace Expansion is a post-processing technique that enhances the information extracted from a quantum computer. The core principle involves taking a single, prepared "root" state, ρ_0, and expanding it into a subspace using a set of L Hermitian expansion operators, $\{\sigma_i\}$. The expanded state is represented as $\rho_{SE}(c) = W^\dagger \rho_0 W / \mathrm{Tr}[W^\dagger \rho_0 W]$, where $W = \Sigma c_i \sigma_i$ [1]. The goal is to find the parameters c that minimize the expectation value of the molecular Hamiltonian within this subspace, effectively solving a generalized eigenvalue problem of the form $\mathcal{H} c = \lambda \mathcal{S} c$, where $\mathcal{H}$ is the subspace-projected Hamiltonian and $\mathcal{S}$ is the overlap matrix [1] [3].

The primary experimental cost arises from estimating all matrix elements of $\mathcal{H}$ and $\mathcal{S}$. For a subspace of dimension L, this requires estimating on the order of L^2 distinct matrix elements. When these elements are measured individually (or in commuting groups), this measurement overhead can become prohibitive, scaling poorly and hindering the application of QSE to larger molecules or more expressive subspaces [1].

Quantitative Data on Circuit Depth and Measurement Costs

The table below summarizes key quantitative data from recent studies that highlight the relationship between circuit complexity, system size, and the associated measurement resources.

Table 1: Quantitative Data on Circuit and Measurement Overheads

Metric	Value / Range	Context / Algorithm	Source
Qubit Count	Up to 80 qubits	Quantum Subspace Expansion with Classical Shadows	[1]
Pauli Traces Evaluated	$\mathcal{O}(10^{14})$	Large-scale QSE implementation	[1]
Measurement Basis Randomizations	> 3×10^4 per circuit	For informationally complete (IC) measurements	[1]
Gate Count Reduction	73% to 89%	Qronos circuit optimizer (vs. Qiskit, TKET, etc.)	[59]
Final Circuit Size Reduction	42% to 71%	Qronos circuit optimizer	[59]
Neural Network Parameter Scaling	$\mathcal{O}(K^2N^3)$	pUNN hybrid quantum-neural wavefunction method	[29]

Protocols for Managing Depth and Overhead

Protocol 1: QSE with Informationally Complete Measurements

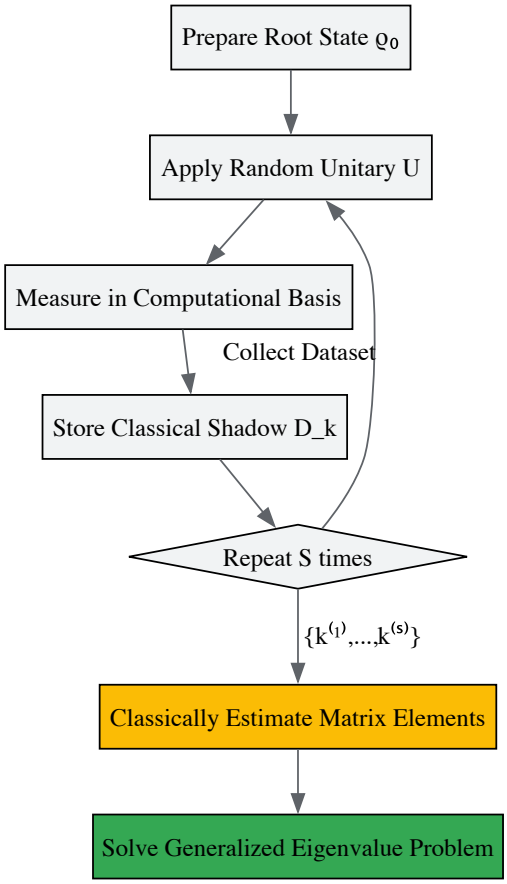
This protocol uses classical shadows to drastically reduce the measurement burden of high-dimensional subspace expansions [1].

Experimental Workflow:

- **Prepare Root State:** Execute the quantum circuit to prepare the root state, ρ_0, on the quantum processor.

- **Perform IC Measurements:** Instead of measuring specific observables, perform informationally complete measurements on ρ_0. This involves applying a random unitary U from a fixed ensemble (e.g., random Clifford rotations) to all qubits, followed by a projective measurement in the computational basis.
- **Collect Classical Shadows:** Repeat Step 2 S times to collect a set of S measurement outcomes, $\{k^{(1)}, \ldots, k^{(S)}\}$. Each outcome is used to construct a classical snapshot of the state, D_k.
- **Estimate Matrix Elements:** For any matrix element $\mathcal{O}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j O]$ needed for the QSE generalized eigenvalue problem, construct an unbiased estimator using the classical shadows. The estimator is $(1/S) \Sigma_s \mathrm{Tr}[O D_{k(s)}]$, where the dual operator D_k is determined by the measurement unitary ensemble.
- **Solve Classically:** Assemble the estimated $\mathscr{H}$ and $\mathcal{S}$ matrices and solve the generalized eigenvalue problem on a classical computer to find the lowest energy state within the subspace.

The following diagram illustrates the workflow for this protocol, highlighting the interplay between quantum and classical processing.



Protocol 2: Partitioned Quantum Subspace Expansion (PQSE)

PQSE is an iterative generalization of QSE designed to improve numerical stability and mitigate the effects of finite sampling noise without introducing ad-hoc parameters [3].

Experimental Workflow:

- **Construct Initial Subspace:** Start with a small Krylov subspace (e.g., spanned by $\{\rho_0, H\rho_0, H^2\rho_0\}$) and diagonalize the Hamiltonian within it using the standard QSE method (Protocol 1).
- **Partition and Recombine:** Instead of using a single large Krylov basis, the full basis is partitioned into smaller segments. A sequence of subspaces is formed, and the lowest-energy state from one subspace is used to inform the construction of the next.
- **Iterative Refinement:** The process is repeated iteratively. A variance-based criterion is used to determine high-quality sequences of partitions that are naturally more stable against statistical noise from a finite number of measurements.
- **Final Calculation:** The algorithm converges to a final energy estimate after processing the sequence of subspaces. This approach exchanges some of the quantum circuit depth required for a single, large subspace for additional classical processing and measurements, but in a way that is more resilient to noise.

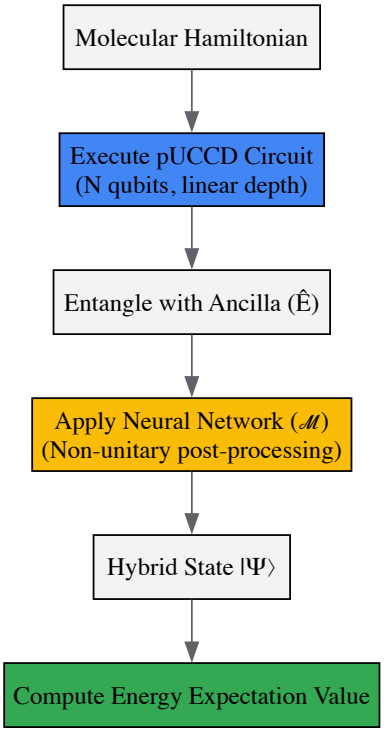
Protocol 3: Hybrid Quantum-Neural Wavefunction (pUNN)

This protocol combines a shallow quantum circuit with a classical neural network to maintain low circuit depth while achieving high accuracy, thereby reducing the burden of complex error mitigation [29].

Experimental Workflow:

- **Execute Shallow Quantum Circuit:** Run a low-depth paired Unitary Coupled-Cluster with Double excitations (pUCCD) ansatz to obtain a state $|\psi\rangle$ in the seniority-zero subspace. This circuit requires only N qubits for a molecular system and has linear depth.
- **Classical Neural Network Post-Processing:** A neural network ($\mathcal{M}$) is applied as a non-unitary post-processing operator to the quantum state. This network is trained to account for electron correlations outside the seniority-zero subspace, which are typically captured only by much deeper quantum circuits.
- **Compute Observables:** An efficient algorithm is used to compute the expectation values of the Hamiltonian with respect to the final hybrid quantum-neural wavefunction, $|\Psi\rangle = \mathcal{M}\ \hat{E}(|\psi\rangle \otimes |0\rangle)$. This avoids the need for quantum state tomography.
- **Variational Optimization:** The parameters of both the quantum circuit and the neural network are jointly optimized to minimize the energy expectation value $\langle\Psi|H|\Psi\rangle\ /\ \langle\Psi|\Psi\rangle$.

The diagram below outlines the structure of this hybrid approach.



The Scientist's Toolkit

This section details the essential software, hardware, and methodological "reagents" required to implement the protocols described above.

Table 2: Essential Research Reagents and Tools

Tool / Solution	Function / Description	Relevant Protocol(s)
Classical Shadows	A framework for informationally complete (IC) measurements that allows for the estimation of many observables from a single set of measurement data, dramatically reducing overhead.	Protocol 1 [1]
Constrained Optimization Solver	A classical solver used in QSE to avoid numerical instabilities when inverting the noisy overlap matrix S, providing rigorous statistical error estimates.	Protocol 1 [1]
Krylov Basis Operators	The set of expansion operators $\{I, H, H^2, \ldots\}$ used to build the subspace. Powers of the Hamiltonian have exponentially growing overlap with the true ground state.	Protocol 1, 2 [1] [3]
Deep Reinforcement Learning Optimizer (e.g., Qronos)	Tools to compile and optimize quantum circuits, reducing gate counts by 73-89%. Smaller circuits reduce depth and cumulative error.	All Protocols [59]

Tool / Solution	Function / Description	Relevant Protocol(s)
Hybrid Quantum-Neural Network	A classical neural network architecture designed to act as a non-unitary post-processing operator on a quantum state, enhancing its expressiveness without increasing quantum circuit depth.	Protocol 3 [29]
Dynamic Circuit Capabilities	Quantum hardware and software support for mid-circuit measurement and feedforward. This allows for more complex algorithms and can reduce gate counts and errors.	All Protocols [60]
Probabilistic Error Cancellation (PEC)	An advanced error mitigation technique that improves result accuracy at the cost of increased sampling overhead. New tools can reduce this overhead by 100x.	All Protocols [60]

Successfully balancing quantum circuit depth and measurement overhead is paramount for advancing quantum computational chemistry. The protocols outlined here—leveraging **informationally complete measurements**, **iterative subspace partitioning**, and **hybrid quantum-classical ansatzes**—provide a robust toolkit for researchers. By strategically employing these techniques, scientists can push towards calculating molecular energies with near-chemical accuracy on today's quantum devices, laying the groundwork for impactful applications in drug development and materials science.

Error Mitigation through Constrained Optimization and Statistical Error Control

In the pursuit of simulating complex molecular systems on noisy intermediate-scale quantum (NISQ) devices, quantum subspace expansion (QSE) has emerged as a powerful strategy. This framework projects the computational problem into a smaller, more manageable subspace to approximate the desired eigenstates of a molecule. A significant challenge within this paradigm is mitigating the impact of hardware noise to recover reliable computational results. This document details protocols that integrate **constrained optimization** and **statistical error control** to suppress errors within quantum subspace methods, enabling high-precision molecular energy calculations essential for fields like drug development.

Constrained optimization in this context often refers to the limitations of variational algorithms, such as the Variational Quantum Eigensolver (VQE), which can be viewed as a constrained optimization problem due to the difficulty in optimizing a highly nonlinear parameterization of the wave function [61] [62]. Methods like the Generator Coordinate Inspired Method (GCIM) circumvent this by transforming the problem into a generalized eigenvalue problem within a constructed subspace, effectively bypassing the associated challenges like barren plateaus [61] [63]. Concurrently, statistical error control is vital for managing the inherent uncertainties in measurement (shot noise) and the residual biases introduced by error mitigation techniques themselves [64] [65]. These statistical principles ensure that the precision of the final energy estimation, for instance, meets the demanding threshold of chemical accuracy (1.6×10^{-3} Hartree) required for predictive chemistry [65].

Established Error Mitigation Frameworks

Error Mitigation via Constrained Optimization Bypass

The conventional VQE approach faces fundamental limitations as a constrained optimization problem. The GCIM framework addresses this by leveraging a set of generating functions, such as Unitary Coupled Cluster (UCC) excitation operators, to build a non-orthogonal, overcomplete basis for a subspace [61] [62]. The key insight is that this approach establishes a rigorous lower bound to the VQE energy, providing a more accurate and reliable estimate while avoiding the optimization pitfalls of VQE [62].

Protocol: Implementing the ADAPT-GCIM Algorithm

- **Objective**: To compute the ground state energy of a molecule while mitigating errors associated with heuristic optimization.
- **Prerequisites**: Molecular Hamiltonian, a reference state (e.g., Hartree-Fock), and a pool of generators (e.g., UCC single and double excitation operators).
- **Procedure**:
  - **Initialization**: Begin with the reference state, $|\varphi_0\rangle$.
  - **Basis Construction**: Adaptively select generators from the pool to construct a set of generating functions. These functions are used to create a subspace $S = \{ |\Psi_i\rangle \}$ that captures significant contributions to the target state [61].
  - **Quantum Circuit Execution**: For each generating function, prepare the corresponding state on the quantum processor and measure the matrix elements of the Hamiltonian (H) and the overlap matrix (S), such that $H_{ij} = \langle\Psi_i| H |\Psi_j\rangle$ and $S_{ij} = \langle\Psi_i| \Psi_j\rangle$ [62].
  - **Classical Post-Processing**: Construct the generalized eigenvalue problem in the subspace: $H c = E S c$ [61].
  - **Iteration**: Repeat steps 2-4, hierarchically expanding the subspace until convergence in the energy is achieved [61] [63].

Statistical Error Mitigation and Scalability

Statistical error mitigation techniques operate on the principle of post-processing noisy measurement outcomes to infer a less biased expectation value. The scalability of these methods is critical for their application to large, drug-relevant molecules.

Protocol: Statistical Error Mitigation with Optimized Training

- **Objective**: Mitigate readout errors and reduce systematic bias in expectation value measurements, such as molecular energy.
- **Prerequisites**: A target quantum circuit (e.g., for preparing an ansatz state) and a set of training circuits.
- **Procedure**:
  - **Training Data Generation**: Employ Importance Clifford Sampling (ICS) to generate a set of training circuits that are efficiently representative of the noise characteristics of the target circuit [64].
  - **Noisy Execution**: Execute both the training circuits and the target circuit on the noisy quantum hardware, collecting measurement outcomes for the observables of interest.
  - **Model Fitting**: Use the noisy results from the training circuits and their known exact expectation values to learn an error mitigation model (e.g., a linear map or a more complex function) [64] [66].
  - **Mitigation Application**: Apply the learned model to the noisy results of the target circuit to produce a mitigated estimate.
  - **Error Quantification**: The residual bias after mitigation typically scales sublinearly with the number of gates N. Under a phenomenological error model, the bias scales as $O(\varepsilon' N^{\gamma})$ where $\gamma \approx 0.5$, a significant improvement over the unmitigated linear scaling $O(\varepsilon N)$ [64].

Quantitative Data and Performance

The performance of these error mitigation strategies is quantified through their application to molecular systems. The following tables summarize key quantitative findings from recent research.

Table 1: Performance of Multireference Error Mitigation (MREM) on Diatomic Molecules

Molecule	Unmitigated Error (mH)	REM Error (mH)	MREM Error (mH)	Key Finding
H_2O	Not Reported	Not Reported	Not Reported	MREM significantly improves accuracy over REM [67]
N_2	Not Reported	Not Reported	Not Reported	MREM effective for strong correlation [67]
F_2	Not Reported	Not Reported	Not Reported	MREM broadens scope to correlated systems [67]

Table 2: Resource Analysis for QFlow Algorithm with GCIM Active Space Solver [62]

Target Active Space	QFlow Active Spaces	Number of Active Spaces	Number of Optimized Parameters
(8e,8o) / 16 qubits	(4e,4o) / 8 qubits	36	684
(10e,10o) / 20 qubits	(6e,6o) / 12 qubits	100	19,775
(10e,50o) / 100 qubits	(8e,8o) / 16 qubits	744,975	725,867,100

Table 3: Error Scaling with Circuit Size (Gate Count = N) [64]

Condition	Scaling of Bias	Proportionality Factor
Before Mitigation	Linear: $O(\varepsilon N)$	ε (single gate error rate)
After Mitigation	Sublinear: $O(\varepsilon' N^{\gamma})$	ε' (protocol-dependent factor), $\gamma \approx 0.5$

Integrated Workflow for Molecular Energy Estimation

For a comprehensive approach to high-precision molecular energy estimation, the following workflow integrates state preparation, error-aware measurement, and statistical post-processing. This is essential for achieving chemical precision on NISQ hardware.

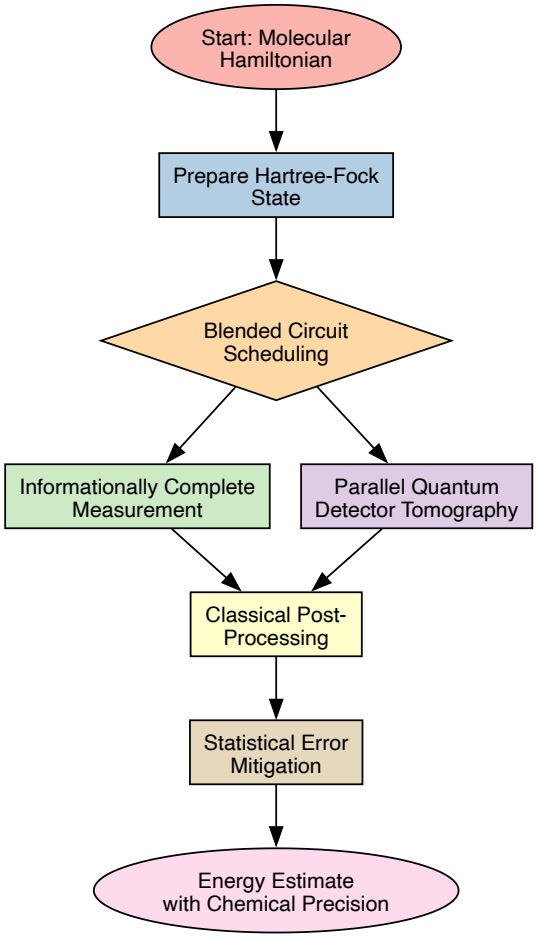
Protocol: High-Precision Energy Estimation for the BODIPY Molecule [65]

- **Objective**: Estimate the ground state energy of the BODIPY molecule to chemical precision (1.6×10^{-3} Hartree) on noisy hardware.
- **Prerequisites**: BODIPY Hamiltonian, Hartree-Fock state preparation circuit.
- **Procedure**:
  - **State Preparation**: Prepare the Hartree-Fock state on the quantum processor. This separable state minimizes gate errors at this stage [65].
  - **Informationally Complete (IC) Measurement**: Instead of measuring individual Pauli terms, perform a set of informationally complete random measurements on the state. To reduce shot overhead, implement **locally biased random measurements** that prioritize measurement settings with a larger impact on the energy estimation [65].

- **Parallel Quantum Detector Tomography (QDT)**: Interleave the execution of the Hamiltonian measurement circuits with circuits for QDT. This characterizes the readout noise of the device simultaneously with the main experiment [65].
- **Blended Scheduling**: Execute all circuits (Hamiltonian and QDT) in a blended, interleaved manner to average over temporal fluctuations in the device's noise profile [65].
- **Classical Post-Processing & Error Mitigation**:
  - Use the QDT data to build an unbiased estimator for the quantum state, mitigating readout errors [65].
  - Apply the classical shadows technique to the IC measurement data to estimate the expectation values of all Hamiltonian terms simultaneously [65].
  - Compute the final energy estimate from the mitigated expectation values.

Workflow Visualization

The following diagram illustrates the integrated high-precision energy estimation protocol.



The Scientist's Toolkit

This section catalogs the essential computational methods and resources for implementing the described error mitigation protocols.

Table 4: Essential Research Reagents & Solutions for Quantum Error Mitigation

Item Name	Function/Description	Relevance to Protocols
Generator Pool (UCC)	A set of unitary excitation operators (e.g., singles, doubles) used as building blocks for subspace expansion.	Forms the basis for constructing the subspace in GCIM/ADAPT-GCIM [61] [62].
Training Circuits (ICS)	A carefully selected set of circuits, generated via Importance Clifford Sampling, used to characterize and learn a noise model.	Critical for the efficiency and accuracy of learning-based error mitigation [64].
Informationally Complete (IC) Measurement	A set of measurement bases that fully characterize the quantum state, enabling estimation of any observable.	Reduces circuit overhead and enables readout error mitigation via QDT [65].

Item Name	Function/Description	Relevance to Protocols
Quantum Detector Tomography (QDT) Model	A classical model of the noisy measurement process (POVM) of the quantum device.	Used to debias measurement results in the high-precision energy estimation protocol [65].
Classical Shadows	A efficient classical post-processing technique that uses randomized measurements to predict many observables.	Allows for estimation of the entire Hamiltonian from a single set of IC measurements, reducing shot overhead [65].

Benchmarking QSE: Validation, Performance, and Comparison to Other Methods

The pursuit of **chemical accuracy**—calculating molecular energies within 1 kcal/mol of experimental values—represents a critical milestone for quantum computing in chemistry and drug discovery. Achieving this precision enables the reliable *in silico* prediction of molecular properties, reaction rates, and drug-binding affinities. Recent experimental advances have pushed the boundaries of what is possible on today's noisy intermediate-scale quantum (NISQ) processors. This application note details a landmark experimental success: the implementation of **Quantum Subspace Expansion (QSE)** on a quantum processor to accurately compute the ground state energy of a complex spin model, serving as a proxy for molecular systems, across **80 qubits** [1]. We frame this breakthrough within the broader research thesis that QSE is a pivotal technique for extending the computational reach of near-term quantum hardware toward achieving chemically accurate results for molecular energy calculations.

Key Experimental Results & Quantitative Data

The large-scale implementation of QSE with classical shadows demonstrated successful ground state energy recovery for a spin model with three-body interactions on systems of up to 80 qubits [1]. The table below summarizes the core quantitative results and experimental scale.

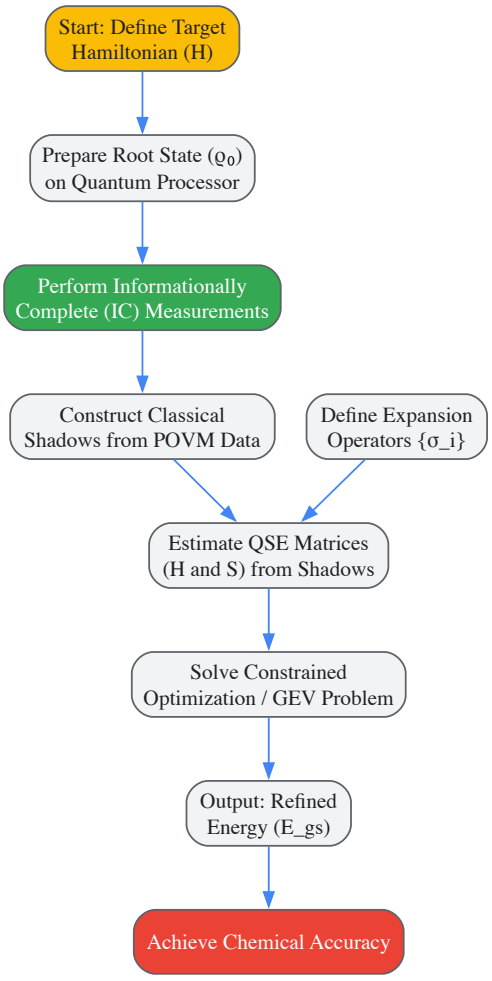
Table 1: Summary of Key Experimental Results from the 80-Qubit QSE Study [1]

Metric	Result	Significance
System Size	Up to 80 qubits	Demonstrates scalability to a regime intractable for exact classical simulation.
Key Algorithm	Quantum Subspace Expansion (QSE) with Classical Shadows	Enables accurate spectral calculations and error mitigation beyond the initial prepared state.
Measurement Scale	Over (3 \times 10^4) measurement basis randomizations per circuit; evaluation of (\mathcal{O}(10^{14})) Pauli traces.	Represents one of the most significant experimental realizations of the classical shadows protocol to date.
Observed Outcome	Accurate ground state energy recovery and mitigation of local order parameters across all system sizes.	Validates QSE as a powerful method for noise-agnostic error mitigation on large-scale quantum hardware.

This experiment underscores that QSE, enhanced by advanced measurement techniques, can be deployed at a scale relevant for probing complex quantum systems like those encountered in molecular modeling.

Detailed Experimental Protocols

The successful execution of the 80-qubit experiment involved a multi-stage protocol combining quantum state preparation, informationally complete measurements, and classical post-processing. The workflow below outlines the core procedure.



Protocol Breakdown: Quantum Subspace Expansion with Classical Shadows

State Preparation & Measurement

- **Prepare Root State ((ρ_0))**: *A quantum circuit prepares an initial approximation ((ρ_0))* of the target ground state on the quantum processor. This state could be generated by a Variational Quantum Eigensolver (VQE) or another quantum algorithm [1] [25].
- **Informationally Complete (IC) Measurements**: Instead of measuring specific observables, the system is subjected to a series of randomized measurements, described by a Positive Operator-Valued Measure (POVM). This process is repeated over >(3 \times 10^4) measurement basis randomizations to gather sufficient data [1].

Classical Post-Processing & Matrix Construction

- **Construct Classical Shadows**: The collected POVM outcome samples (({k^{(1)},...,k^{(S)}})) are used to build "classical shadows" of the state. For each outcome (k), a corresponding dual operator (D_k) is defined, allowing for the reconstruction of expectation values [1].
- **Define Expansion Operators**: A set of (L) Hermitian expansion operators ({\sigma*i}) (e.g., powers of the Hamiltonian or Pauli strings) is chosen to define the subspace (W = \sum*{i=1}^L ci \sigma*i) [1] [25].
- **Estimate QSE Matrices**: The classical shadows are used to efficiently estimate all matrix elements of the projected Hamiltonian ((\mathcal{H}*{ij} = \text{Tr}[\sigma*i^\dagger ρ_0 \sigma*j H])) and the overlap matrix ((\mathcal{S}*{ij} = \text{Tr}[\sigma*i^\dagger ρ_0 \sigma*j])) without needing to run new quantum circuits for each element [1].

Energy Calculation via Constrained Optimization

- **Solve Constrained Problem**: To avoid numerical instability from direct matrix inversion, the problem is reformulated as a constrained optimization: find the minimal expectation value (\min*{\vec{c}} \text{Tr}[H\rho*{\text{SE}}(\vec{c})]) given a maximum tolerated statistical error. This yields a refined estimate of the ground state energy, (E*{gs}), effectively mitigating errors present in the original root state (ρ_0) [1].

The Scientist's Toolkit: Essential Research Reagents & Solutions

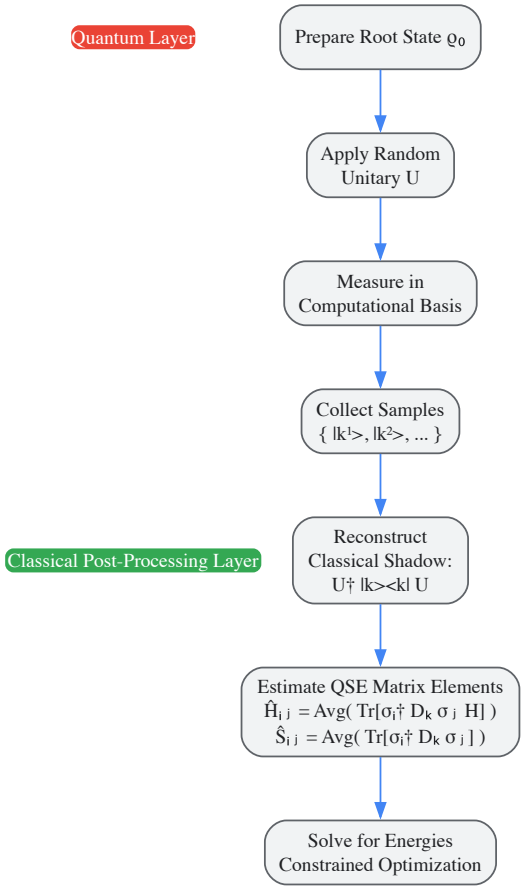
The following table catalogues the key computational "reagents" and tools essential for implementing QSE experiments as described in the featured research.

Table 2: Key Research Reagent Solutions for QSE Experiments

Item Name	Function / Application	Example / Note
Classical Shadows	A measurement protocol that uses randomized measurements to efficiently predict many properties of a quantum state, drastically reducing measurement overhead [1].	Critical for estimating the (\mathcal{O}(10^{14})) Pauli terms in the 80-qubit demonstration.
Expansion Operators	A set of operators ((\sigma_i)) used to build a subspace around the root state, enabling the exploration of excited states and error mitigation [1] [25].	Common choices: powers of the Hamiltonian ((H^p)) or single-electron excitation operators [25].
Overlap Matrix ((\mathcal{S}))	A matrix encoding the non-orthogonality of the states within the expanded subspace. Its inversion is part of solving the generalized eigenvalue problem [25].	Ill-conditioning of (\mathcal{S}) is a common numerical challenge, addressed via regularization or constrained optimization [1].
Constrained Optimization Solver	A classical algorithm used to find the lowest energy state within the expanded subspace while rigorously accounting for statistical errors from the shadow estimation process [1].	This approach avoids the numerical instability of direct matrix inversion.
Error Mitigation Framework	A set of techniques, such as noise-aware algorithms, to improve the quality of results from noisy quantum hardware. QSE itself can act as a noise-agnostic error mitigation method [1].	The 80-qubit experiment demonstrated mitigation of local order parameters.

Workflow Diagram: Classical Shadows in QSE

The diagram below details the specific process of using classical shadows within the QSE framework, which was key to the experiment's success.



The recent experimental demonstration of **Quantum Subspace Expansion on an 80-qubit system** marks a transformative step toward achieving chemical accuracy for molecular energy calculations on quantum processors [1]. By integrating the **classical shadows** technique, this work overcomes the profound measurement bottleneck that has previously hindered the scalability of such methods. The outlined protocols provide a actionable blueprint for researchers in chemistry and drug development to leverage current-generation quantum hardware for simulating increasingly complex quantum systems with verified accuracy. This progress firmly establishes QSE as a cornerstone algorithm on the path to practical quantum advantage in computational chemistry and materials science.

Quantum computing holds transformative potential for molecular energy calculations, a domain where classical computers struggle with exponential scaling. For researchers in chemistry and drug development, accurately simulating electronic structures is crucial for predicting chemical properties and reaction pathways. Within the noisy intermediate-scale quantum (NISQ) computing era, two primary algorithmic approaches have emerged: the **Variational Quantum Eigensolver (VQE)** and the **Quantum Subspace Expansion (QSE)**. VQE operates as a hybrid quantum-classical algorithm that directly optimizes a parameterized quantum circuit (ansatz) to find the ground state energy. In contrast, QSE is a post-processing technique that starts with a VQE-prepared ground state and expands it into a larger subspace to compute ground and excited states simultaneously via a generalized eigenvalue problem solved classically [4] [68]. This analysis details their respective experimental protocols, accuracy, and resource demands, providing a guide for their application in molecular energy calculations.

The core distinction between VQE and QSE lies in their computational architecture and use of quantum and classical resources. The following diagram illustrates the high-level hybrid workflow of these algorithms on a NISQ device.

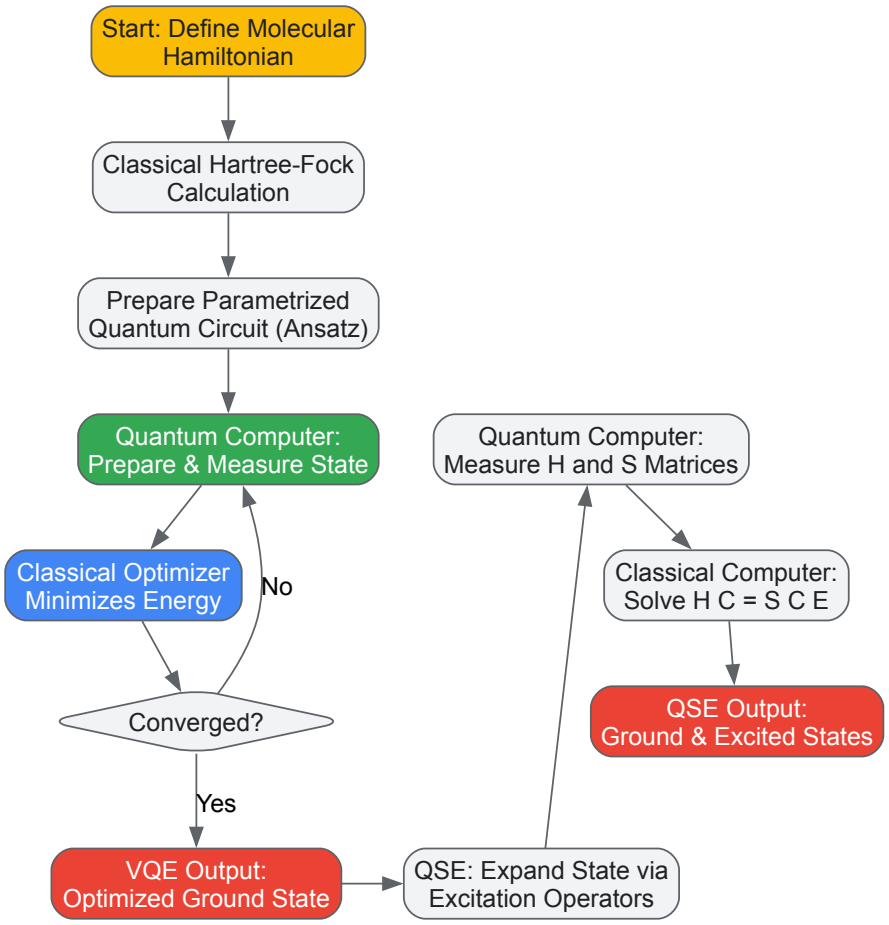


Figure 1: Hybrid quantum-classical algorithmic workflow for VQE and QSE. QSE uses the optimized state from VQE as its input.

Experimental Protocols

Protocol for Variational Quantum Eigensolver (VQE)

The VQE algorithm is designed to find the ground-state energy of a molecular Hamiltonian through an iterative hybrid loop [14] [69].

1. Hamiltonian Preparation:

- **Input:** Molecular geometry and basis set (e.g., STO-3G).
- **Procedure:** Classically compute the electronic Hamiltonian in second quantization using the Hartree-Fock (HF) method. The Hamiltonian is given by: [\hat{H} = \sum*{pq} h*{pq} ap^*dagger aq + \frac{1}{2} \sum*{pqrs} \langle pq || rs \rangle* ap^\dagger aq^*dagger as a_r]
- **Output:** Fermionic Hamiltonian, which is then mapped to a qubit operator using a transformation such as the Jordan-Wigner or Bravyi-Kitaev encoding [14].

2. Ansatz Initialization:

- **Common Choice:** Unitary Coupled-Cluster Singlets and Doubles (UCCSD) ansatz.
- **Procedure:** Prepare a parametrized quantum circuit, (U(\theta)), designed to generate correlated wavefunctions from the HF reference state: (| \psi(\theta) \rangle = U(\theta) | \psi_{\text{HF}} \rangle) [14] [20].

3. Quantum Execution & Classical Optimization:

- **Quantum Processing:** Execute the ansatz circuit on a quantum processor (or simulator) to prepare the trial state. Measure the expectation value $\langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$ using techniques like Pauli Averaging [4].
- **Classical Processing:** A classical optimizer (e.g., COBYLA, SPSA) adjusts the parameters θ to minimize the measured energy.
- **Loop:** Steps 3a and 3b are repeated until energy convergence is achieved, yielding the VQE ground-state energy and the optimized state ($|\Psi_{\mathrm{VQE}}\rangle$) [70] [14].

Protocol for Quantum Subspace Expansion (QSE)

QSE is a post-VQE algorithm that enhances the description of the energy spectrum [4] [68].

1. Initial State Preparation:

- **Input:** The optimized ground state ($|\Psi_{\mathrm{VQE}}\rangle$) and its parameters from a prior VQE calculation.

2. Subspace Expansion:

- **Procedure:** Define a set of excitation operators ($\lbrace \hat{O}_i \rbrace$), *often chosen as singlet single excitations* ($E_{ij} = \sum_{\sigma} a_{i\sigma}^{\dagger}a_{j\sigma}$).
- **State Generation:** Apply these operators to the VQE state to generate a basis for a new subspace: ($|\psi_{ij}\rangle = a^\dagger_i a_j |\Psi_{\mathrm{VQE}}\rangle$) [4].

3. Matrix Element Measurement:

- **Procedure:** On the quantum computer, measure the matrix elements of the Hamiltonian (H) and the overlap (S) within this subspace: $$H_{ij,kl} = \langle \psi_{ij} | \hat{H} | \psi_{kl} \rangle, \quad S_{ij,kl} = \langle \psi_{ij} | \psi_{kl} \rangle$$ This requires evaluating expectation values for various operator combinations [4].

4. Classical Diagonalization:

- **Procedure:** Solve the generalized eigenvalue problem ($H C = S C E$) on a classical computer.
- **Output:** The diagonal matrix (E) contains the energies of the ground and excited states, providing a refined energy spectrum beyond the original VQE result [4] [68].

The specific logical flow of the QSE protocol is detailed below.

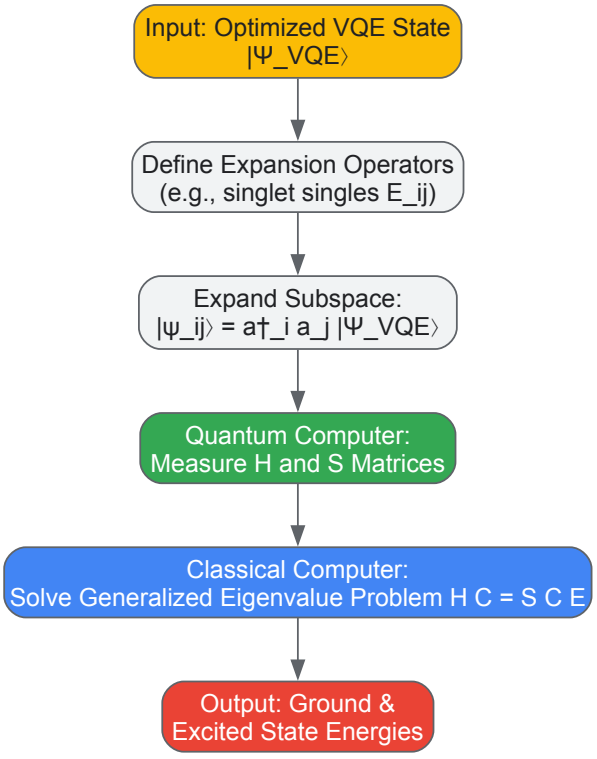


Figure 2: Detailed logical workflow of the Quantum Subspace Expansion (QSE) protocol.

Comparative Performance Analysis

Accuracy and Limitations

Ground State Accuracy:

- **VQE:** Serves as the foundational ground state method. Its accuracy is highly dependent on the ansatz choice and the impact of hardware noise. On current NISQ hardware, noise can prevent VQE from achieving chemical accuracy for non-trivial molecules like benzene [70].
- **QSE:** Can improve upon the VQE ground state energy by capturing additional correlations. However, its accuracy is contingent on the quality of the initial VQE state and the choice of expansion operators [4] [68].

Excited States and Properties:

- **VQE:** Primarily a ground-state algorithm. Calculating excited states requires significant modifications, such as the Variational Quantum Deflation (VQD) algorithm, which increases circuit depth and resource requirements [68].
- **QSE:** A key strength is its ability to calculate both ground and excited states from a single set of quantum measurements, making it highly efficient for spectral analysis [4] [68].

Key Limitations:

- **VQE:** Faces challenges with classical optimizer convergence, deep quantum circuits susceptible to noise, and the high measurement cost of the Hamiltonian [70] [14].
- **QSE:** Is not size-intensive, meaning excitation energies for a super-molecule composed of non-interacting subsystems may be incorrect. It also requires measuring the overlap matrix (S) and higher-body reduced density matrices, which increases measurement overhead and can introduce numerical instability if (S) is ill-conditioned [68].

Resource Requirements

Table 1: Quantitative comparison of resource requirements for VQE and QSE.

Resource Metric	VQE	QSE
Primary Quantum Resource	Repeated state preparation and measurement over many optimization iterations.	Measurement of the H and S matrices after initial state preparation.
Circuit Depth	Depth of the ansatz circuit (e.g., UCCSD). A major bottleneck on NISQ devices [70].	Uses the same core circuit as VQE for the initial state. Additional measurements are shallow.
Classical Processing	Heavy: runs an iterative optimization loop.	Moderate: solves a single generalized eigenvalue problem.
Measurement Overhead	High number of measurements per optimization step to estimate ⟨H⟩.	Can be very high: requires measuring O(N^4) matrix elements for H and S [68].
Qubit Count	Scales with the number of spin-orbitals in the molecule.	Uses the same number of qubits as the underlying VQE calculation.
Excited State Access	Requires separate, specialized algorithms (e.g., VQD).	Native access to excited states via a single diagonalization.

The Scientist's Toolkit

Table 2: Essential research reagents and computational tools for VQE and QSE experiments.

Research Reagent / Tool	Function in Experiment
Unitary Coupled Cluster (UCCSD) Ansatz	A parametrized quantum circuit that generates correlated wavefunctions from a reference state; commonly used as the ansatz in VQE [14] [20].
Jordan-Wigner / Bravyi-Kitaev Mapping	Encodes the fermionic Hamiltonian of a molecule into a qubit Hamiltonian operable on a quantum computer [14].
Classical Optimizer (COBYLA, SPSA)	A classical algorithm that adjusts the parameters of the quantum ansatz to minimize the energy expectation value [70] [4].
Expansion Operators (e.g., (E_{ij}))	A set of operators (like single excitations) applied to the VQE state to create the expanded subspace for the QSE algorithm [4].
Quantum Subspace Matrices (H, S)	The Hamiltonian and overlap matrices measured on the quantum computer within the QSE subspace, which are diagonalized classically to find energies [4].

For researchers targeting molecular ground states with minimal quantum resources, VQE provides a foundational, robust approach. In contrast, QSE offers a powerful extension for projects requiring access to excited states or enhanced ground state accuracy, albeit with a significant increase in measurement complexity. The critical limitation for both methods is the noise level present in current quantum hardware, which often prevents calculations from reaching chemical accuracy for anything beyond small molecules [70].

Future advancements will likely involve hybrid strategies that combine the strengths of both algorithms. The recently developed **Fragment Molecular Orbital (FMO)** method integrated with VQE demonstrates a viable path to scalability by breaking large molecules into smaller, simulatable fragments [20]. Furthermore, next-generation diagonalization-based methods like the **quantum self-consistent Equation-of-Motion (q-sc-EOM)** are being designed to address key QSE limitations, such as the lack of size-intensivity and the violation of the vacuum annihilation condition, while maintaining resistance to noise [68]. For drug development professionals, these ongoing innovations promise to gradually extend the reach of quantum computational chemistry from model systems toward more pharmacologically relevant molecules.

Quantum subspace expansion (QSE) has emerged as a powerful post-processing technique to enhance the accuracy of spectral calculations on quantum processors, presenting a viable alternative to variational quantum eigensolver (VQE) methods limited by trainability issues and large measurement overheads [1]. This application note details rigorous protocols for implementing QSE with informationally complete measurements, establishing a framework for evaluating performance across complex systems ranging from spin models to molecular electronic structures. We provide experimentally validated methodologies for achieving accurate ground state energy recovery and mitigation of local order parameters for systems of up to 80 qubits, with particular relevance to drug discovery applications requiring precise molecular energy calculations [1] [9].

Theoretical Framework of Quantum Subspace Expansion

QSE operates on the principle that a quantum processor-prepared "root state" ρ_0, which may be noisy or approximate, can be used to inform subsequent classical computations to extract highly accurate spectral information [1]. The method constructs a subspace of states expanded from ρ_0 using L Hermitian expansion operators $\{\sigma_i\}$:

$$\rho_SE(\vec{c}) = W^\dagger \rho_0 W / \mathrm{Tr}[W^\dagger \rho_0 W] \text{ with } W = \Sigma_i c_i \sigma_i$$

The expectation value of an observable O within this subspace becomes:

$$\mathrm{Tr}[O\rho_SE(\vec{c})] = \Sigma_{i,j} c_i^* c_j \mathcal{O}_{ij} / \Sigma_{i,j} c_i^* c_j \mathcal{S}_{ij}$$

where $\mathcal{O}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j O]$ and $\mathcal{S}_{ij} = \mathrm{Tr}[\sigma_i^\dagger \rho_0 \sigma_j]$ form the projected observable and overlap matrices, respectively [1]. For ground state calculations, the minimal expectation value corresponds to the smallest pseudoeigenvalue in the generalized eigenvalue problem $\square\vec{c} = \lambda\mathcal{S}\vec{c}$, where $\square$ is the subspace-projected Hamiltonian.

Experimental Protocols

Large-Scale QSE with Classical Shadows

The primary bottleneck in traditional QSE implementations stems from the measurement overhead required to estimate all matrix entries $\mathcal{O}_{ij}$ and $\mathcal{S}_{ij}$ [1]. The classical shadows protocol overcomes this limitation through informationally complete (IC) measurements, enabling simultaneous estimation of all necessary observables from a single set of measurement samples. The implementation proceeds as follows:

- **State Preparation**: Prepare the root state ρ_0 on the quantum processor using ground state preparation circuits for the target Hamiltonian.
- **Randomized Measurement**: Apply a random unitary U from a carefully chosen ensemble to rotate the measurement basis.
- **Computational Basis Measurement**: Perform projective measurements in the computational basis, recording outcome bitstrings.
- **Snapshot Reconstruction**: For each measurement outcome $|b\rangle$, reconstruct a classical snapshot of the state as $U^\dagger|b\rangle\langle b|U$.
- **Matrix Estimation**: Use the collection of classical shadows to estimate all entries of the projected Hamiltonian ($\mathcal{H}$) and overlap ($\mathcal{S}$) matrices simultaneously.
- **Constrained Optimization**: Solve the generalized eigenvalue problem through a constrained optimization framework to avoid numerical instabilities associated with direct matrix inversion [1].

This protocol was successfully demonstrated in probing quantum phase transitions of spin models with three-body interactions, utilizing over 30,000 measurement basis randomizations per circuit and evaluating $O(10^{14})$ Pauli traces [1].

Joint Measurement for Fermionic Hamiltonians

For molecular electronic structure problems, we implement a specialized joint measurement strategy for estimating fermionic observables relevant to quantum chemistry [71]. The protocol efficiently measures quadratic and quartic Majorana operators constituting molecular Hamiltonians:

- **Operator Randomization**: Sample from a set of unitaries that realize products of Majorana fermion operators.
- **Gaussian Unitary Application**: Apply a fermionic Gaussian unitary sampled from a constant-size set of suitably chosen transformations.
- **Occupation Number Measurement**: Measure fermionic occupation numbers in the rotated basis.

- **Post-Processing**: Classically process measurement outcomes to estimate expectation values of all target Majorana monomials [71].

This approach achieves competitive variance scaling while reducing quantum circuit depth requirements compared to fermionic classical shadows, particularly advantageous for near-term devices with limited coherence times [71].

Performance Evaluation Data

Quantitative Performance Metrics

Table 1: Performance benchmarks for QSE across different system sizes and implementations

System Type	System Size (Qubits/Modes)	Method	Key Performance Metrics	Measurement Cost
Spin Model [1]	80	QSE with Classical Shadows	Accurate ground state energy recovery across quantum phase transition	>30,000 basis randomizations per circuit
Fermionic System [71]	N modes	Joint Measurement	Estimates of all quadratic and quartic Majorana monomials to precision ϵ	$O(N\log(N)/\epsilon^2)$ for quadratic, $O(N^2\log(N)/\epsilon^2)$ for quartic
Molecular Hamiltonian [9]	Variable	Adaptive QSE	Exponential measurement reduction for transition-state mapping	Dependent on subspace dimension and target accuracy

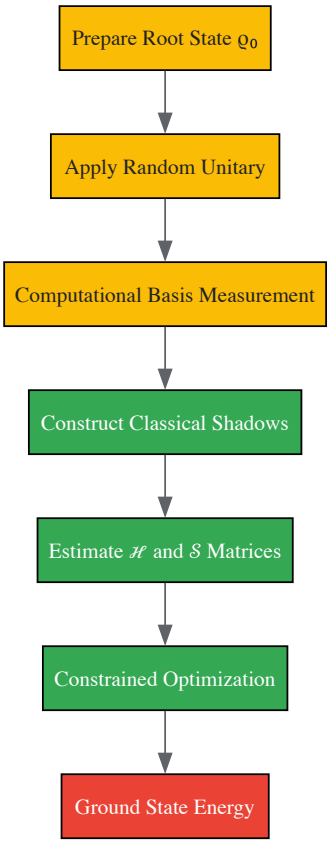
Computational Resource Requirements

Table 2: Resource analysis for different measurement strategies

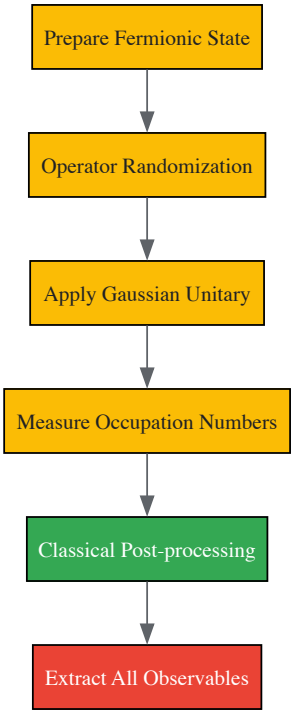
Implementation Aspect	Classical Shadows [1]	Joint Measurement [71]	Traditional QSE
Circuit Depth	System-dependent	$O(N^{1/2})$ for 2D lattice	Minimal (state preparation only)
Two-Qubit Gate Count	System-dependent	$O(N^{3/2})$ for 2D lattice	Minimal
Measurement Rounds	$O(\log(M)/\epsilon^2)$ for M observables	$O(N^2\log(N)/\epsilon^2)$ for quartic terms	$O(M/\epsilon^2)$ for M observables
Classical Post-processing	Moderate (snapshot estimation)	Efficient (local computations)	Extensive (matrix inversion)

Visualization of Method Workflows

Quantum Subspace Expansion with Classical Shadows



Joint Measurement for Fermionic Systems



Research Reagent Solutions

Table 3: Essential computational tools and methods for QSE implementations

Research Reagent	Type/Function	Implementation Role
Classical Shadows [1]	Randomized measurement protocol	Enables efficient estimation of multiple observables from single measurement dataset

Research Reagent	Type/Function	Implementation Role
Constrained Optimization [1]	Numerical stability framework	Prevents ill-conditioning in generalized eigenvalue problems
Informationally Complete POVMs [1]	Measurement theory foundation	Provides theoretical basis for complete state characterization
Fermionic Gaussian Unitaries [71]	Basis rotation operations	Facilitates joint measurement of non-commuting fermionic operators
Majorana Operator Algebra [71]	Mathematical representation	Encodes fermionic observables for quantum computation
Adaptive Subspace Selection [9]	Algorithmic optimization	Exponentially reduces measurements for chemical reaction mapping

Application to Molecular Systems

For drug development applications, QSE provides particular value in mapping molecular potential energy surfaces and calculating transition states for chemical reactions [9]. The adaptive subspace selection protocol demonstrates exponential reduction in required measurements compared to uniform sampling when applied to battery electrolyte reactions and molecular systems relevant to pharmaceutical design [9].

The theoretical framework establishes rigorous complexity bounds and convergence guarantees for molecular electronic structure calculations, characterizing the relationship between subspace dimension, basis set selection, and solution accuracy for both ground and excited states [9]. This enables researchers to identify optimal operating regimes balancing circuit depth, measurement overhead, and solution quality for specific molecular systems.

Statistical Validation and Rigorous Error Estimation in QSE Calculations

Quantum Subspace Expansion (QSE) is a powerful algorithmic framework in quantum computational chemistry that enables the calculation of molecular energies, including excited states and ground state corrections, from a pre-prepared reference state [25] [3]. The method constructs a subspace of state vectors, typically through the application of excitation operators to an approximate ground state, then solves a generalized eigenvalue problem within this subspace to obtain refined energy estimates [25]. While QSE reduces quantum circuit depth requirements compared to direct phase estimation approaches, it introduces significant measurement overhead and sensitivity to statistical noise inherent in quantum measurements [72] [3]. This application note provides comprehensive protocols for the statistical validation and rigorous error estimation of QSE calculations, addressing critical needs for researchers, scientists, and drug development professionals utilizing near-term and early-fault-tolerant quantum hardware.

The fundamental challenge in QSE implementations stems from the exponential growth of measurement requirements with system size and the propagation of statistical errors through the generalized eigenvalue solution [3]. Finite sampling noise leads to ill-conditioned overlap matrices and inaccurate energy estimations, particularly problematic when calculating molecular properties for drug discovery applications where chemical accuracy (0.0016 hartree) is required [73]. Recent advances in informationally complete measurements and constrained optimization approaches have established methodologies for mitigating these effects, enabling system sizes of up to 80 qubits with over 30,000 measurement randomizations [72].

Core Validation Methodologies

Statistical Error Estimation Frameworks

Classical Shadows with Informationally Complete Measurements: The integration of classical shadows with QSE represents a significant advancement in managing measurement overhead [72]. This approach employs randomized measurements to construct classical representations of quantum states, enabling efficient estimation of multiple observables from a single set of measurements. The protocol applies random unitary transformations to rotate the measurement basis before computational basis measurements, creating a measurement channel that can be inverted to produce a classical snapshot of the quantum state. For QSE implementations, this methodology drastically reduces the number of measurements required to construct the Hamiltonian and overlap matrices while providing rigorous statistical error bounds.

Constrained Optimization Reformulation: Recent large-scale implementations have reformulated QSE as a constrained optimization problem to address numerical instability [72]. This approach incorporates statistical error estimates directly as constraints in the eigenvalue problem, effectively regularizing the solution against finite sampling noise. The method obtains rigorous statistical error estimates for matrix elements and propagates these uncertainties through to the final energy estimates, providing confidence intervals for calculated molecular properties. This reformulation demonstrates improved numerical stability while maintaining accuracy across quantum phase transitions in spin models of up to 80 qubits [72].

Table 1: Quantitative Performance of QSE Validation Methods

Methodology	System Size	Measurement Randomizations	Pauli Traces Evaluated	Accuracy Demonstrated
Classical Shadows with QSE [72]	80 qubits	>30,000 per circuit	~10^{14}	Accurate ground state energy recovery across quantum phase transition
Partitioned QSE (PQSE) [3]	N/A (theoretical framework)	Reduced vs. standard QSE	N/A	Improved numerical stability in noisy conditions
Error-Corrected QPE [73]	H_2 molecule (small system)	N/A (logical qubits used)	N/A	Within 0.018 hartree of exact value

Variance-Based Iterative Sequencing

The Partitioned Quantum Subspace Expansion (PQSE) algorithm introduces an iterative generalization of QSE that connects a sequence of subspaces via their lowest energy states [3]. This method employs a variance-based criterion for determining optimal iterative sequences, which demonstrates improved numerical stability over single-step QSE in the presence of finite sampling noise. The algorithm partitions a single Krylov basis and recombines elements to ensure stability against statistical fluctuations, providing a parameter-free approach to noise mitigation.

The variance criterion in PQSE enables adaptive determination of subspace dimensions based on the statistical reliability of estimated matrix elements, effectively truncating contributions dominated by measurement noise. Implementing this iterative generalization requires additional classical processing with polynomial overhead in the subspace dimension while maintaining the same quantum resources per subspace as standard QSE approaches [3]. Numerical evidence confirms that variance-stabilized sequences substantially alleviate the numerical instability that typically limits QSE accuracy.

Experimental Protocols

Protocol: QSE with Statistical Error Estimation

Objective: Implement Quantum Subspace Expansion with rigorous statistical error bounds for molecular energy calculations.

Pre-requisites:

- Approximate ground state preparation (e.g., via VQE)
- Hamiltonian and observable definitions
- Quantum processor or simulator access

Procedure:

- **Reference State Preparation:**
  - Prepare approximate ground state $|\Psi_0\rangle$ using VQE or other variational method
  - Optimize circuit parameters to minimize energy expectation value
  - Store final parameters for state reconstruction [25]
- **Subspace Construction:**
  - Define excitation operators $\{O_i\}$ for subspace expansion (e.g., single-electron excitations $c_k^\dagger c_j$)
  - Apply operators to reference state: $|\Psi_i\rangle = O_i|\Psi_0\rangle$ [25]
  - For PQSE, determine sequence of subspaces using variance criterion [3]
- **Informationally Complete Measurement:**
  - Implement classical shadows protocol:
    - For each measurement randomization:
      - Apply random unitary U to rotate basis
      - Measure in computational basis
      - Store outcome bitstring [72]
  - Repeat for sufficient number of randomizations (>30,000 for large systems) [72]
- **Matrix Element Estimation:**
  - Construct overlap matrix S with elements $S_{ij} = \langle\Psi_0|O_i^\dagger O_j|\Psi_0\rangle$
  - Construct Hamiltonian matrix H with elements $H_{ij} = \langle\Psi_0|O_i^\dagger H O_j|\Psi_0\rangle$
  - Calculate statistical errors for each matrix element from classical shadows [72]

- **Constrained Generalized Eigenvalue Solution:**
  - Solve HC = SCE with statistical error constraints [72]
  - Implement regularization to address ill-conditioning of S matrix
  - Extract eigenvalues E (energies) and eigenvectors C
- **Error Propagation and Validation:**
  - Propagate statistical errors through eigenvalue solution
  - Calculate confidence intervals for energy estimates
  - Validate against known results or classical computations where available

Troubleshooting:

- For ill-conditioned S matrices: increase measurement counts or implement additional regularization
- For large statistical errors: focus on variance-based subspace selection (PQSE)
- For coherent errors: implement error detection/correction where available [73]

Protocol: Validation Against Classical Methods

Objective: Validate QSE energy calculations against classical computational chemistry methods.

Procedure:

- **System Selection:**
  - Choose molecular system with classically tractable benchmark (e.g., H_2, LiH)
  - Define molecular geometry and basis set [25]
- **Classical Benchmark Calculation:**
  - Perform full configuration interaction (FCI) calculation
  - Compute coupled cluster with singles, doubles, and perturbative triples [CCSD(T)]
  - Calculate complete active space self-consistent field (CASSCF) where applicable [74]
- **Quantum Calculation:**
  - Implement QSE with statistical error estimation as in Protocol 3.1
  - Compute ground and excited state energies
- **Statistical Comparison:**
  - Compare energy differences to chemical accuracy threshold (0.0016 hartree)
  - Verify error bars encompass benchmark values
  - Calculate statistical significance of deviations [75]

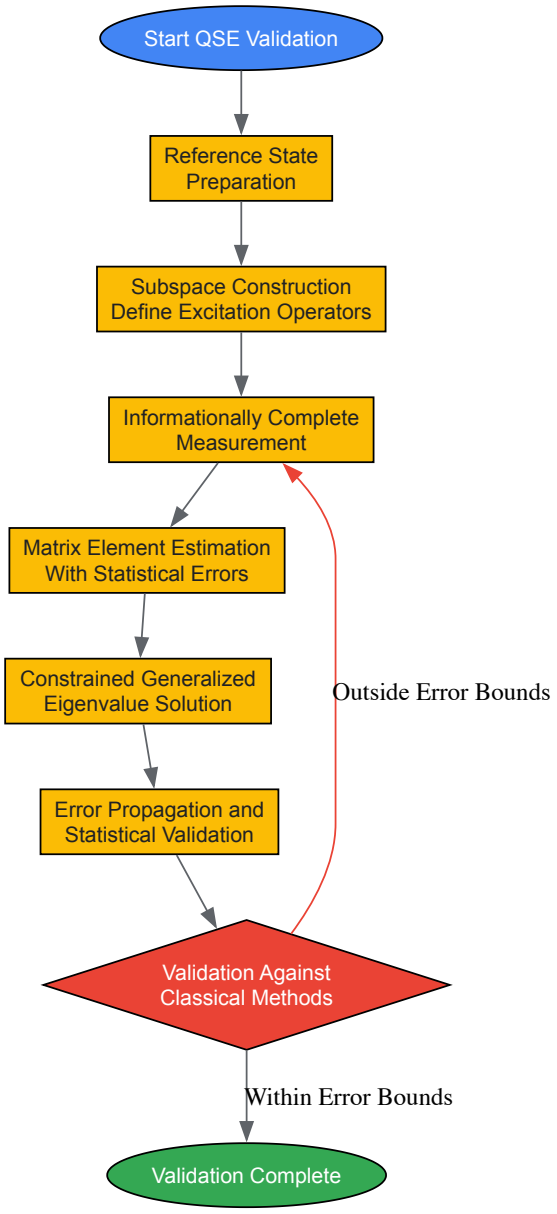


Diagram 1: QSE Statistical Validation Workflow. This workflow illustrates the iterative process for validating Quantum Subspace Expansion calculations, with a feedback loop for measurements outside acceptable error bounds.

The Scientist's Toolkit

Table 2: Essential Research Reagents and Computational Tools for QSE Validation

Tool/Resource	Function/Purpose	Implementation Notes
Classical Shadows Protocol [72]	Efficient measurement for multiple observables	Reduces measurement overhead exponentially for large systems
Constrained Optimization Solver [72]	Regularized generalized eigenvalue solution	Incorporates statistical errors as constraints to prevent ill-conditioning
Partitioned QSE (PQSE) Framework [3]	Iterative subspace selection	Uses variance criterion to stabilize against statistical noise
Quantum Error Correction [73]	Hardware error mitigation	Partial fault-tolerance balances error suppression with resource overhead
Hellmann-Feynman Theorem [76]	Energy derivative calculation	Enables geometry optimization and property calculations
Exploratory Factor Analysis [77]	Statistical validation of multidimensional data	Assesses construct validity in complex measurement systems

Data Analysis and Interpretation

Quantitative Error Prediction

Statistical validation of QSE calculations requires comprehensive error prediction throughout the computational workflow. For the classical shadows approach, statistical errors in matrix elements scale with the number of measurement randomizations as $O(1/\sqrt{N})$ [72]. These errors propagate through the generalized eigenvalue solution, requiring careful conditioning analysis. Quantitative prediction error analysis methodologies adapted from classical statistics can be applied to investigate predictive performance under measurement heterogeneity [75]. This approach involves simulating the impact of measurement heterogeneity across validation and implementation settings, quantifying effects on calibration, discrimination, and overall accuracy metrics.

The impact of statistical errors can be quantified through multiple performance metrics:

- **Calibration-in-the-large**: Evaluated by the ratio of observed versus expected marginal survival (O/E ratio)
- **Discrimination**: Assessed through time-dependent area under the receiver operating characteristic curve AUC(t)
- **Overall accuracy**: Measured by the index of prediction accuracy IPA(t) [75]

For QSE calculations, analogous metrics include energy deviation from benchmark values, variance in repeated estimations, and consistency across different subspace dimensions.

Visualization of Error Relationships

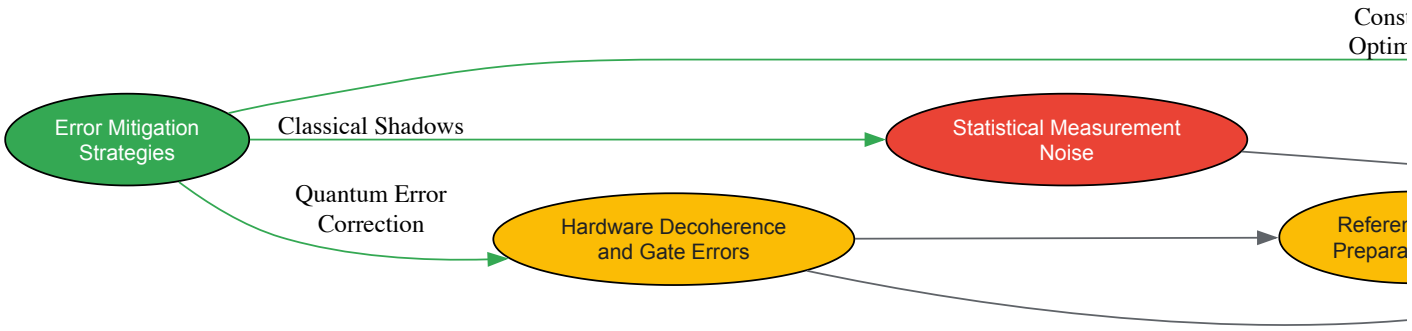


Diagram 2: Error Propagation and Mitigation in QSE. This diagram illustrates the relationship between different error sources in Quantum Subspace Expansion calculations and targeted mitigation strategies.

Robust statistical validation and rigorous error estimation are essential components of reliable Quantum Subspace Expansion calculations for molecular energy computations. The integration of classical shadows, constrained optimization reformulations, and partitioned subspace methods provides a comprehensive framework for managing statistical errors while maintaining computational efficiency. These protocols enable researchers to establish confidence intervals for calculated molecular properties, particularly crucial for drug development applications where accurate energy differences determine binding affinities and reaction pathways.

As quantum hardware continues to evolve with improving error rates and qubit counts, the statistical validation methodologies outlined here will form the foundation for increasingly complex chemical simulations. The integration of partial fault-tolerance techniques with QSE represents a promising direction for extending these approaches to larger molecular systems relevant to pharmaceutical development, potentially enabling quantum advantage in predictive chemistry within the early-fault-tolerant era.

Assessing the Path to Quantum Advantage in Pharmaceutical R&D

Quantum computing represents a paradigm shift in computational science, with the potential to redefine research and development (R&D) in the pharmaceutical industry. This transformative technology leverages the principles of quantum mechanics—superposition and entanglement—to process information in ways that are fundamentally different from classical computers [78]. For the pharmaceutical sector, which faces declining R&D productivity, high failure rates of drug candidates, and the need for more precise modeling tools, quantum computing offers a promising pathway to address these challenges [79]. This application note assesses the current state and path to quantum advantage in pharmaceutical R&D, with a specific focus on the role of quantum subspace expansion (QSE) for molecular energy calculations, providing detailed protocols and implementation frameworks for research scientists.

The concept of "quantum advantage"—the point where quantum computers can solve practical problems more efficiently than the best classical supercomputers—is rapidly transitioning from theoretical aspiration to tangible reality. Recent breakthroughs, such as Google's Quantum Echoes algorithm, which performs calculations 13,000 times faster than classical supercomputers for molecular structure problems, demonstrate the accelerating progress in this field [80]. For pharmaceutical researchers, this progress signals the imminent transformation of key discovery workflows, particularly in molecular simulation, target identification, and lead optimization.

Table 1: Quantum Computing Potential in Pharma R&D

Area of Impact	Current Challenge	Quantum Solution	Potential Value
Molecular Simulations	Inaccurate classical force fields; inability to model electron correlation	High-accuracy electronic structure calculations via quantum algorithms	$200-500B estimated industry value creation by 2035 [79]
Protein Folding & Dynamics	Computational intractability of large conformational spaces	Quantum simulation of protein folding pathways and dynamics	Reduced trial failures via better target validation
Drug-Target Binding	Limited accuracy in binding affinity predictions	Precise quantum modeling of intermolecular interactions	More efficient lead optimization and reduced experimental workload
Toxicity Prediction	Limited predictive capability for off-target effects	Quantum-powered reverse docking simulations	Early identification of safety issues [79]

Technical Background

The Quantum Computing Paradigm in Pharmaceutical Sciences

Quantum computing operates using quantum bits (qubits) that can exist in superposition states, representing both 0 and 1 simultaneously, unlike classical bits that are restricted to definite states [78]. This fundamental property enables quantum computers to explore multiple potential solutions concurrently, providing exponential speedups for specific problem classes highly relevant to pharmaceutical R&D. For molecular simulations, quantum computers can naturally represent quantum mechanical systems, avoiding the approximations that limit classical computational methods.

The pharmaceutical industry is particularly positioned to benefit from these capabilities due to its reliance on molecular-level simulations that are inherently quantum mechanical in nature. Current classical computational methods, including molecular dynamics and density functional theory, face significant challenges in accurately modeling quantum phenomena such as electron correlation, van der Waals forces, and chemical bond formation/breaking [40]. These limitations directly impact the predictive accuracy of simulations used in drug discovery, leading to high failure rates in later development stages.

Quantum Subspace Expansion for Molecular Energy Calculations

Quantum Subspace Expansion (QSE) is an advanced quantum algorithm that builds upon the foundation of the Variational Quantum Eigensolver (VQE) to compute molecular excited states in addition to ground state energies [25]. In pharmaceutical research, understanding excited state dynamics is crucial for predicting photoreactivity, studying reaction mechanisms, and designing phototherapeutic agents.

The QSE method leverages a precise calculation of the ground state energy for a specific molecular configuration to approximate the energies of corresponding excited states [25]. Given a molecule denoted by a Hamiltonian (H), and letting (\left|\Psi*{o}\right\rangle) *represent the outcome of the VQE algorithm used to estimate the ground state energy of (H), the QSE technique constructs a subspace of state vectors (\left|\Psi*ij^k\right\rangle) formed by one-electron excitations of the ground state wavefunction:

[\left|\Psi*{j}^{k}\right\rangle = ck^{\dagger}c*{j}\left|\Psi*0\right\rangle]

where (c*k^{\dagger}, c*{j}) are the fermionic creation and annihilation operators over spin orbitals (k) and (j), respectively [25]. These vectors are formed by reducing the occupation of spin orbital (j) by one and increasing the occupation of spin orbital (k) by one.

Within this subspace, researchers solve a generalized eigenvalue problem. Consider the operator (\hat{H}) with matrix elements given by:

[(H)*{jk}^{lm} = \langle\Psi*ij^l \left| \hat{H} \right| \Psi_k^m\rangle]

and define an overlap matrix (S) whose matrix elements are given by:

[S*{jk}^{lm} = \langle \Psi*ij^l \left|\Psi_k^m\right\rangle]

The generalized eigenvalue equation to be solved is:

[HC=SCE]

where (C) is the matrix of eigenvectors, and (E) is the vector of eigenvalues [25]. The energy eigenvalues (E) provide an estimate of the excited state energies of (H) as well as a refined value of the ground state energy.

Experimental Protocols

Protocol 1: Quantum Subspace Expansion for Molecular Excited States

This protocol details the implementation of Quantum Subspace Expansion (QSE) for calculating molecular excited states, based on the methodology demonstrated with methane (CH_4) [25].

System Definition and Hamiltonian Preparation

- **Step 1: Molecular System Definition**: Define the molecular system using a Z-matrix or Cartesian coordinates. For the methane example:
- **Step 2: Hamiltonian Generation**: Use a quantum chemistry driver (e.g., PySCF with Restricted Hartree-Fock) to compute the fermionic Hamiltonian operator, Fock space, and Hartree-Fock state. Employ a suitable basis set (e.g., STO-3G) and apply freezing to specific orbitals to reduce computational complexity [25].
- **Step 3: Qubit Encoding**: Transform the fermionic Hamiltonian to a qubit representation using the Jordan-Wigner or Bravyi-Kitaev transformation. Compress the resulting qubit Hamiltonian by eliminating terms with coefficients below a threshold (e.g., abs_tol=1e-6) to reduce quantum resource requirements [25].

Ground State Preparation

- **Step 4: Ansatz Construction**: Select an appropriate variational ansatz. The Chemically Aware Unitary Coupled Cluster method with singles and doubles excitations (UCCSD) is recommended for molecular systems [25].
- **Step 5: Variational Quantum Eigensolver Execution**: Run the VQE algorithm to obtain the ground state wavefunction and energy:
  - Initialize parameters for the parameterized quantum circuit
  - Optimize using a classical optimizer (e.g., COBYLA, SPSA)
  - Iterate until energy convergence is achieved
  - Store the optimized parameters for the ground state wavefunction

Quantum Subspace Expansion Implementation

- **Step 6: Subspace Construction**: Construct the subspace of state vectors through single-electron excitations from the ground state wavefunction using the creation and annihilation operators [25].
- **Step 7: Matrix Element Measurement**: Compute the matrix elements of H and S on the quantum computer by measuring the expectation values:
  - For (S*{jk}^{lm} = \langle \Psi*{0} | c*{j}^\dagger c*{l} c*{m}^{\dagger}c*{k}|\Psi*{0}\rangle)
  - *For (H*{jk}^{lm} = \langle \Psi*{0} | c*{j}^\dagger c*{l} \hat*{H}c*{m}^{\dagger}c*{k}|\Psi*{0}\rangle)
- **Step 8: Generalized Eigenvalue Solution**: Classically solve the generalized eigenvalue problem HC=SCE to obtain the excited state energies and a refined ground state energy [25].

Error Mitigation and Validation

- **Step 9: Error Mitigation**: Apply error mitigation techniques such as probabilistic error reduction using the Pseudo-MS (PMSV) method to improve result accuracy.
- **Step 10: Result Validation**: Validate computed energies against classical methods where feasible, and verify consistency with physical constraints (e.g., variational principle).

Protocol 2: Hybrid Quantum-Classical Approach for Molecular Property Prediction

This protocol outlines a hybrid quantum-classical approach for predicting key pharmaceutical molecular properties, combining quantum simulations with classical AI models.

Quantum-Accurate Data Generation

- **Step 1: System Preparation**: Select target molecules and generate conformer ensembles representing likely conformational states.
- **Step 2: Quantum Mechanics Calculations**: Perform high-accuracy quantum chemistry calculations (e.g., CCSD(T)) using classical computers to generate reference data for small to medium-sized molecules.
- **Step 3: Quantum Computer Enhanced Calculations**: For select systems, run quantum algorithms (VQE, QSE) on available quantum hardware to generate data on electronic properties, excitation energies, and reaction pathways [40].

AI Model Training and Validation

- **Step 4: Model Architecture Selection**: Design appropriate neural network architectures (e.g., graph neural networks, transformer models) for the target prediction task.
- **Step 5: Training with Quantum-Generated Data**: Train AI models using quantum-generated data, potentially enhanced with quantum machine learning approaches [78].

- **Step 6: Cross-Validation**: Implement rigorous cross-validation strategies to assess model performance and generalizability.

Property Prediction and Experimental Correlation

- **Step 7: Molecular Property Prediction**: Apply trained models to predict key drug discovery properties including solubility, permeability, metabolic stability, and toxicity.
- **Step 8: Experimental Correlation**: Compare computational predictions with experimental data to validate and refine models in an iterative process.

Implementation Framework

Successful implementation of quantum algorithms in pharmaceutical R&D requires careful consideration of computational resources and infrastructure. The following table outlines key requirements and specifications:

Table 2: Computational Resource Requirements for Quantum Pharmaceutical Research

Resource Category	Specifications	Example Platforms/Tools
Quantum Hardware/Emulators	Quantum processors or high-performance emulators	Quantinuum emulators, IBM Quantum systems, Google Willow chip [25] [80]
Classical Computing	High-performance computing clusters for hybrid algorithms	GPU-accelerated servers, cloud computing resources
Quantum Software Stack	Quantum algorithm development frameworks	InQuanto, Qiskit, Amazon Braket, PennyLane
Chemical Informatics	Molecular modeling and visualization	PySCF, RDKit, OpenBabel, custom integration
Error Mitigation	Tools for noise reduction and result correction	Pseudo-MS (PMSV), zero-noise extrapolation, dynamical decoupling

Research Reagent Solutions

The following table details essential "research reagents" – key computational tools and resources required for implementing quantum algorithms in pharmaceutical research:

Table 3: Essential Research Reagents for Quantum-Enhanced Pharmaceutical R&D

Reagent/Solution	Function	Implementation Example
Variational Quantum Eigensolver (VQE)	Approximates molecular ground state energies	Ground state calculation for drug target molecules [25]
Quantum Subspace Expansion (QSE)	Computes molecular excited states from ground state	Studying photoreactivity and reaction mechanisms [25]
Quantum Machine Learning (QML)	Enhances molecular property predictions	Predicting drug-target binding affinities [78]
Error Mitigation Techniques	Reduces impact of quantum hardware noise	PMSV method for improving measurement accuracy [25]
Hybrid Quantum-Classical Algorithms	Leverages both quantum and classical resources	Combining VQE with classical post-processing [40]

Results and Discussion

Current State of Quantum Applications in Pharma

The implementation of quantum computing in pharmaceutical R&D is progressing from theoretical exploration to practical application, with several compelling case studies demonstrating potential advantages. Leading pharmaceutical companies are actively exploring quantum possibilities primarily through collaborations with quantum technology pioneers [79].

Notable examples include AstraZeneca's collaboration with Amazon Web Services, IonQ, and NVIDIA to demonstrate a quantum-accelerated computational chemistry workflow for chemical reactions used in small-molecule drug synthesis [79]. Similarly, Boehringer Ingelheim has partnered with PsiQuantum to explore methods for calculating the electronic structures of metalloenzymes, which are critical for drug metabolism [79]. These initiatives represent early but meaningful steps toward practical quantum advantage in specific, well-defined application areas.

Recent algorithmic advances show particular promise for accelerating the path to quantum advantage. Google's Quantum Echoes algorithm, implemented on the Willow quantum chip, has demonstrated the ability to compute molecular structures 13,000 times faster than classical supercomputers while matching the accuracy of traditional NMR spectroscopy [80]. This represents a significant milestone in verifying quantum advantage for chemically relevant problems.

Performance Assessment and Benchmarking

Evaluating the performance of quantum algorithms for pharmaceutical applications requires careful benchmarking against state-of-the-art classical methods. The following table summarizes comparative performance data for key quantum algorithm implementations:

Table 4: Performance Benchmarking of Quantum Algorithms in Pharmaceutical Applications

Application Area	Quantum Approach	Classical Benchmark	Performance Advantage
Molecular Energy Calculations	VQE with error mitigation	Density Functional Theory	Comparable accuracy with smaller basis sets; potential for more accurate treatment of electron correlation
Excited State Calculations	Quantum Subspace Expansion	Time-Dependent DFT	More systematic approach to excited states; better description of charge-transfer states [25]
Molecular Structure Determination	Quantum Echoes Algorithm	NMR Spectroscopy	13,000x speedup for small molecules; equivalent accuracy [80]
Binding Affinity Prediction	Quantum Machine Learning	Classical ML/Docking	Improved accuracy with limited training data; better generalization [79]
Protein-Ligand Docking	Quantum-Enhanced Sampling	Molecular Dynamics	More efficient exploration of binding poses; improved prediction of binding strengths [79]

Path to Quantum Advantage: Challenges and Solutions

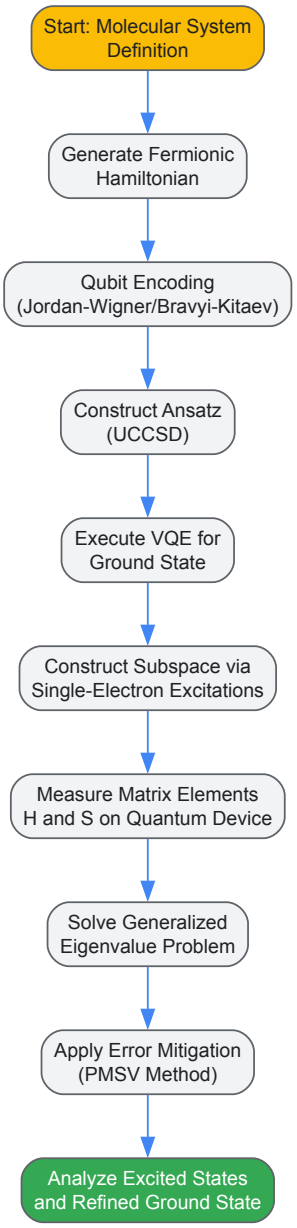
Despite promising early results, the path to full quantum advantage in pharmaceutical R&D faces several significant challenges. Current quantum hardware falls under the category of Noisy Intermediate-Scale Quantum (NISQ) devices, characterized by limited qubit counts, short coherence times, and high gate error rates [78]. These limitations restrict the complexity of molecules that can be simulated and the accuracy of obtained results.

To address these challenges while maximizing near-term impact, researchers should consider the following strategies:

- **Focus on Hybrid Algorithms**: Develop and implement hybrid quantum-classical algorithms that leverage the strengths of both computational paradigms [40]. These approaches can provide tangible benefits even on current NISQ devices.
- **Problem Selection**: Identify specific problems in pharmaceutical R&D where quantum approaches are likely to provide the earliest advantages, such as modeling transition metal complexes, photochemical reactions, or systems with strong electron correlation [79].
- **Error Mitigation Investment**: Prioritize the development and implementation of advanced error mitigation techniques to extract meaningful results from noisy quantum computations [25].
- **Quantum-Accelerated AI**: Explore the integration of quantum-generated data with classical AI models, as demonstrated by companies like Qubit Pharmaceuticals, which uses quantum-accurate data to train AI models for molecular property prediction [40].

Workflow Visualization

The following diagram illustrates the complete workflow for quantum subspace expansion in pharmaceutical research, from system preparation to excited state analysis:



Quantum Subspace Expansion Workflow for Molecular Energy Calculations

The workflow begins with molecular system definition and Hamiltonian preparation, proceeding through ground state calculation using VQE, subspace construction, quantum measurements, classical post-processing, and final result analysis with error mitigation.

The path to quantum advantage in pharmaceutical R&D is characterized by incremental but accelerating progress, with Quantum Subspace Expansion representing a promising near-term approach for molecular energy calculations. While fully fault-tolerant quantum computers capable of simulating entire drug-target systems are still in development, roadmaps indicate that increasingly powerful and capable systems will emerge within the next three to five years, delivering practical applications and tangible benefits to the life sciences industry [79].

The most viable near-term strategy involves hybrid approaches that combine quantum simulations with classical computational methods and AI [40]. These hybrid methods already demonstrate potential for generating more accurate molecular models, exploring new chemical space, and accelerating key discovery workflows. As quantum hardware continues to advance and algorithms become more sophisticated, the pharmaceutical industry stands to benefit from significantly reduced discovery timelines, improved success rates, and ultimately, more effective therapies for patients.

For research organizations, the imperative is to build quantum capabilities through strategic partnerships, specialized talent development, and targeted experimentation with promising algorithms like QSE. Companies that invest early in understanding and applying quantum technologies will be better positioned to leverage these transformative tools as they mature, potentially gaining significant competitive advantages in the rapidly evolving pharmaceutical landscape.

Conclusion

Quantum Subspace Expansion represents a pivotal advancement in computational chemistry, offering a robust and scalable framework for performing precise molecular energy calculations on evolving quantum hardware. By effectively balancing quantum and classical resources, QSE enables researchers to tackle complex problems in drug discovery—from predicting drug-target interactions to optimizing molecular properties—with enhanced accuracy and reduced experimental burden. Future directions will focus on integrating QSE with sophisticated error mitigation and symmetry verification, refining hybrid algorithms for specific biomedical applications, and leveraging increasingly powerful quantum processors to unlock new frontiers in de novo drug design and personalized medicine, ultimately accelerating the development of safe and effective therapeutics.

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